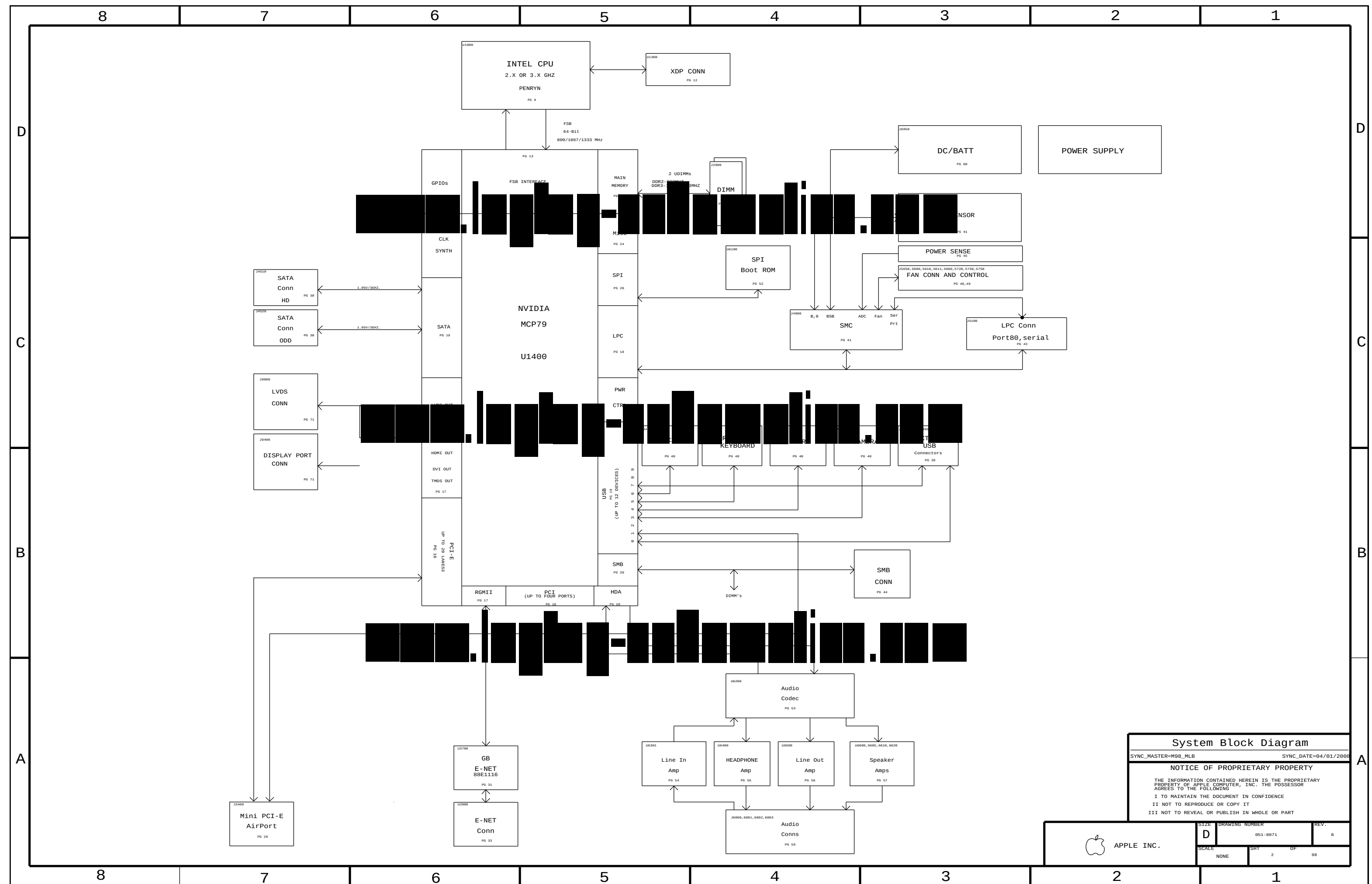
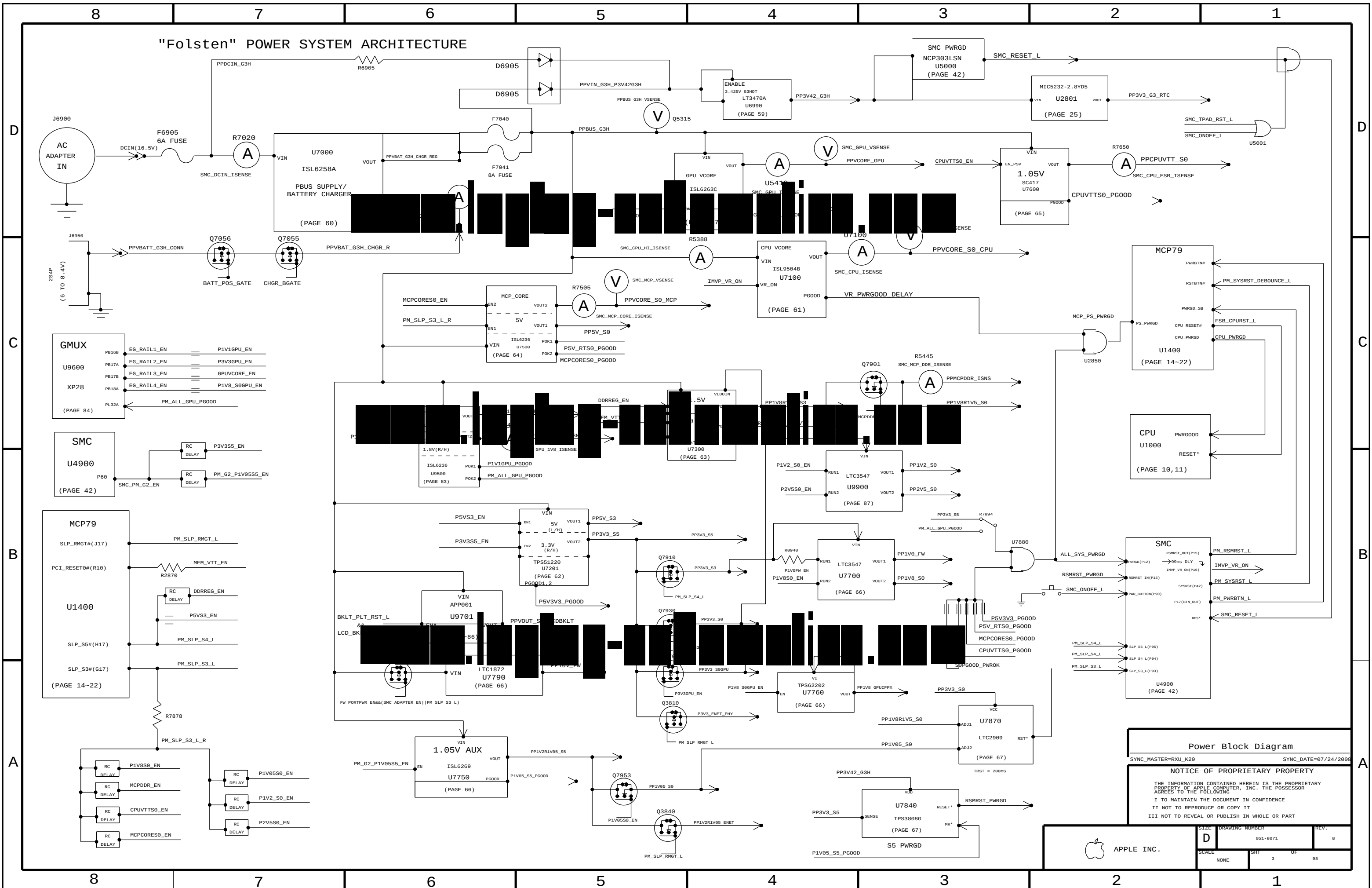


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THE POSSESSOR AGREES TO THE FOLLOWING</div><div>1. TO MAINTAIN THE DOCUMENT IN CONFIDENCE</div><div>2. NOT TO REPRODUCE OR COPY IT</div><div>3. NOT TO REVEAL OR PUBLISH IN WHOLE OR PART</div><div>TITLE</div><div>SCHEM, Folsten, MBP17</div><div>DRAWING NUMBER</div><div>051-8071</div><div>REV.</div><div>B</div><div>SHT 1 OF 98</div></div></div></div> | DRAFTER | /         | DESIGN CK | / | ENG APPD | / | MFG APPD | / | QA APPD | / | DESIGNER | /        | RELEASE | /       | SCALE                 | NONE | MATERIAL/FINISH NOTED AS APPLICABLE |  | SIZE | D |  |  |  |  |  |
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  |   |          |   |         |   |          |          |         |         |                       |      |                                     |  |      |   |  |  |  |  |  |
| ENG APPD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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  |   |          |   |         |   |          |          |         |         |                       |      |                                     |  |      |   |  |  |  |  |  |
| QA APPD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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| RELEASE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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| MATERIAL/FINISH NOTED AS APPLICABLE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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  |   |          |   |         |   |          |          |         |         |                       |      |                                     |  |      |   |  |  |  |  |  |



"Folsten" POWER SYSTEM ARCHITECTURE



**Power Block Diagram**

SYNC\_MASTER=RXU\_K20 SYNC\_DATE=07/24/2008

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PVT:

```
03/24/09
csa.5: Project copied from K20A mlb_pvt.
    Changed CPU APNs for K20 and 3.066GHz CPUs.
    Changed BOM and EEE codes for K20A.

csa.45: Connected =PP1V5_EXP_S0 to J4501.13 for SATA redriver on flex.
03/25/09
csa.9: Added PBUS VS 5V voltage selection resistors for keyboard backlight driver.
03/27/09
csa.90: Added 1000pF cap to the backlight power pin for EMI baseline noise.
03/30/09
csa.5: Changed the bom option to KBDLED_5V per radar# 6723272.
03/31/09
csa.1: Changed rev to 1.0.0
04/09/09
csa.70: No stuff C7099 per radar# 6772695.
04/29/09
Production Release Fab to rev A
csa.5: Changed K20A EFI ROM APN 341S2507 ( BOM change only )
05/05/09
Added 128S0264 (SANYO) as alternate to 128S0257 (KEMET ELEC) per Radar# 6656624.
06/15/09
Added 107S0136 (DALE/VISHAY) as alternate to 107S0132 (CYNTEC) per Radar# 6971400.
For U7871 P/N 353S2718 is made primary. P/N 353S2310 is added back as alternate.
For U6100 Locked Bootrom P/N 341S2506 replaces existing Unlock Bootrom P/N 341S2507.
```

## BOM Variants

| BOM NUMBER | BOM NAME                            | BOM OPTIONS                                       |
|------------|-------------------------------------|---------------------------------------------------|
| 639-0172   | PCBA, BEST, 2.8, 512SAM_VRAM, K20A  | K20A_COMMON, EEE_9EH, CPU_2_80GHZ, FB_512_SAMSUNG |
| 639-0173   | PCBA, BEST, 3.06, 512SAM_VRAM, K20A | K20A_COMMON, EEE_9EK, CPU_3_86GHZ, FB_512_SAMSUNG |
| 639-0174   | PCBA, BEST, 2.8, 512HYN_VRAM, K20A  | K20A_COMMON, EEE_9EL, CPU_2_80GHZ, FB_512_HYNIX   |
| 639-0175   | PCBA, BEST, 3.06, 512HYN_VRAM, K20A | K20A_COMMON, EEE_9EM, CPU_3_86GHZ, FB_512_HYNIX   |

## Alternate Parts

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS:                    |
|-------------|---------------------------|------------|---------|------------------------------|
| 152S0476    | 152S0276                  |            | ALL     | Inductor alternate           |
| 353S1681    | 353S1294                  |            | ALL     | TI alt to National           |
| 138S0603    | 138S0602                  |            | ALL     | Murata alt to Samsung        |
| 152S0684    | 152S0368                  |            | ALL     | Maglayers alt to Dale/Vishay |
| 104S0023    | 104S0018                  |            | ALL     | Cyntec alt to sense resistor |
| 104S0024    | 104S0017                  |            | ALL     | Panasonic alt to FW resistor |
| 341S2367    | 341S2366                  |            | ALL     | Macronix alt to SST          |
| 152S0076    | 152S0782                  |            | ALL     | Maglayer alt to Delta        |
| 157S0058    | 157S0055                  |            | ALL     | Delta alt to TDK Magnetics   |
| 514-0612    | 514-0607                  |            | ALL     | FOXLINK ALT TO FOXCONN KCMV  |
| 514-0613    | 514-0608                  |            | ALL     | FOXLINK ALT TO FOXCONN KCMV  |
| 155S0457    | 155S0329                  |            | ALL     | MAG LAYERS ALT TO MURATA     |
| 128S0264    | 128S0257                  |            | ALL     | SANYO ALT TO KINET ELEC.     |
| 107S0136    | 107S0132                  |            | ALL     | DALE/VISHAY ALT TO CYNTEC.   |
| 353S2310    | 353S2718                  |            | ALL     | INTERSL COMMON TO K24/K19    |

## Folsten BOM GROUPS

| BOM GROUP      | BOM OPTIONS                                                                                                    |
|----------------|----------------------------------------------------------------------------------------------------------------|
| K20A_COMMON    | ALTERNATE, COMMON, K20A_COMMON1, K20A_COMMON2<br>[REDACTED] GPAP [REDACTED]                                    |
| K20A_COMMON1   | ONEWIRE_PU, ISL6258, MEMRESET_HW, MEMRESET_MCP, MCP_B03, MCP_PROD, MCPSEQ_SH [REDACTED]<br>PROD [REDACTED]     |
| K20A_COMMON2   | BOOT_MODE_USER, GPUVID_1P00V, MUXGFX, DPMUX_EN_S0, DP_ESD, [REDACTED], [REDACTED], LD [REDACTED]<br>[REDACTED] |
| K20A_DEBUG     | SMC_DEBUG_YES, XDP, LPCPLUS_NOT, NO_VREFMRGN [REDACTED]                                                        |
| K20A_PROGPARTS | GMUX_PROG, BOOTROM_PROG, SMC_PROG, TPAD_PROG [REDACTED]                                                        |

| BOM GROUP       | BOM OPTIONS              |
|-----------------|--------------------------|
| FB_1024_SAMSUNG | VRAM8, VRAM_1024_SAMSUNG |
| FB_512_SAMSUNG  | VRAM4, VRAM_512_SAMSUNG  |
| FB_512_HYNIX    | VRAM4, VRAM_512_HYNIX    |

Bar Code Labels / EEE #'s

| PART NUMBER | QTY | DESCRIPTION                   | REFERENCE DES | CRIT     |         |
|-------------|-----|-------------------------------|---------------|----------|---------|
| 826-4393    | 1   | LBL,P/N LABEL,PCB,28MM X 6 MM | [EE:9EH]      | CRITICAL | EEE_9EK |
| 826-4393    | 1   | LBL,P/N LABEL,PCB,28MM X 6 MM | [EE:9EK]      | CRITICAL | EEE_9EL |
| 826-4393    | 1   | LBL,P/N LABEL,PCB,28MM X 6 MM | [EE:9EL]      | CRITICAL | EEE_9EM |
| 826-4393    | 1   | LBL,P/N LABEL,PCB,28MM X 6 MM | [EE:9EM]      | CRITICAL | EEE_9EN |

## Module Parts

| PART NUMBER | QTY | DESCRIPTION                                       | REFERENCE DES                                                 | CRITICAL | BOM OPTION        |
|-------------|-----|---------------------------------------------------|---------------------------------------------------------------|----------|-------------------|
| 338S0737    | 1   | IC, ASSP, GPU, NV, G90-G5, VLOWNG, BGA909, LF     | U8000                                                         | CRITICAL |                   |
| 338S0694    | 1   | IC, RTL8251CA-VB-GR, GIGE TRANSCEIVER, 48P, LQFP  | U3700                                                         | CRITICAL |                   |
| 338S0654    | 1   | IC, FW43-E, 2304B PHY/ONC1 LINK/PCI-E, L2         | U4100                                                         | CRITICAL |                   |
| 338S0710    | 1   | IC, MCP78XT-B3, 35X35MM, BGA1437                  | U1400                                                         | CRITICAL | MCP_B03           |
| 338S0563    | 1   | IC, SMC, HSB/2117, 9MMX9MM, TLP                   | U4900                                                         | CRITICAL | SMC_BLANK         |
| 341S2355    | 1   | IC, SMC, DEVELOPMENT, K20                         | U4900                                                         |          |                   |
| 335S0610    | 1   | IC, FLASH, SPI, 32MBIT, 3.3V, 80MHZ, 8-SOP        | U6100                                                         |          |                   |
| 341S2506    | 1   | IC, LOCKED EFI ROM, K20A                          | U6100                                                         |          |                   |
| 341S2384    | 1   | IR, ENCORE II, CY7C6333-LFXC                      | U4800                                                         | CRITICAL |                   |
| 341S2383    | 1   | IC, PSOC +w/USB, 56PIN, MLF, M98                  | U5701                                                         | CRITICAL | TPAD_PROG         |
| 337S3744    | 1   | IC, PDC, SLOAN, PRQ, 3.06, 35M, 1055, ES, 6P, BGA | U1000                                                         | CRITICAL | CPU_3_066GHZ      |
| 337S3682    | 1   | IC, PDC, SLOAN, PRQ, 2.89, 35M, 1055, ES, 6P, BGA | U1000                                                         | CRITICAL | CPU_2_89GHZ       |
| 333S0481    | 4   | IC, SGRAM, GDDR3, 32MX32, 800MHZ, 136 FBGA        | UB480, UB450, UB580, UB550                                    | CRITICAL | VRAM_512_SAMSUNG  |
| 333S0480    | 8   | IC, SGRAM, GDDR3, 32MX32, 800MHZ, 136 FBGA        | UB480, UB450, UB580, UB550, UB330, UB320, UB310, UB300, UB270 | CRITICAL | VRAM_1024_SAMSUNG |
| 333S0506    | 4   | IC, SDRAM, GDDR3, 32MX32, 900MHZ, T1VA, HF        | UB480, UB450, UB580, UB550                                    | CRITICAL | VRAM_512_HYNIX    |

## BOM Configuration

SYNC\_MASTER=K20A\_MLB SYNC\_DATE=04/01/2008

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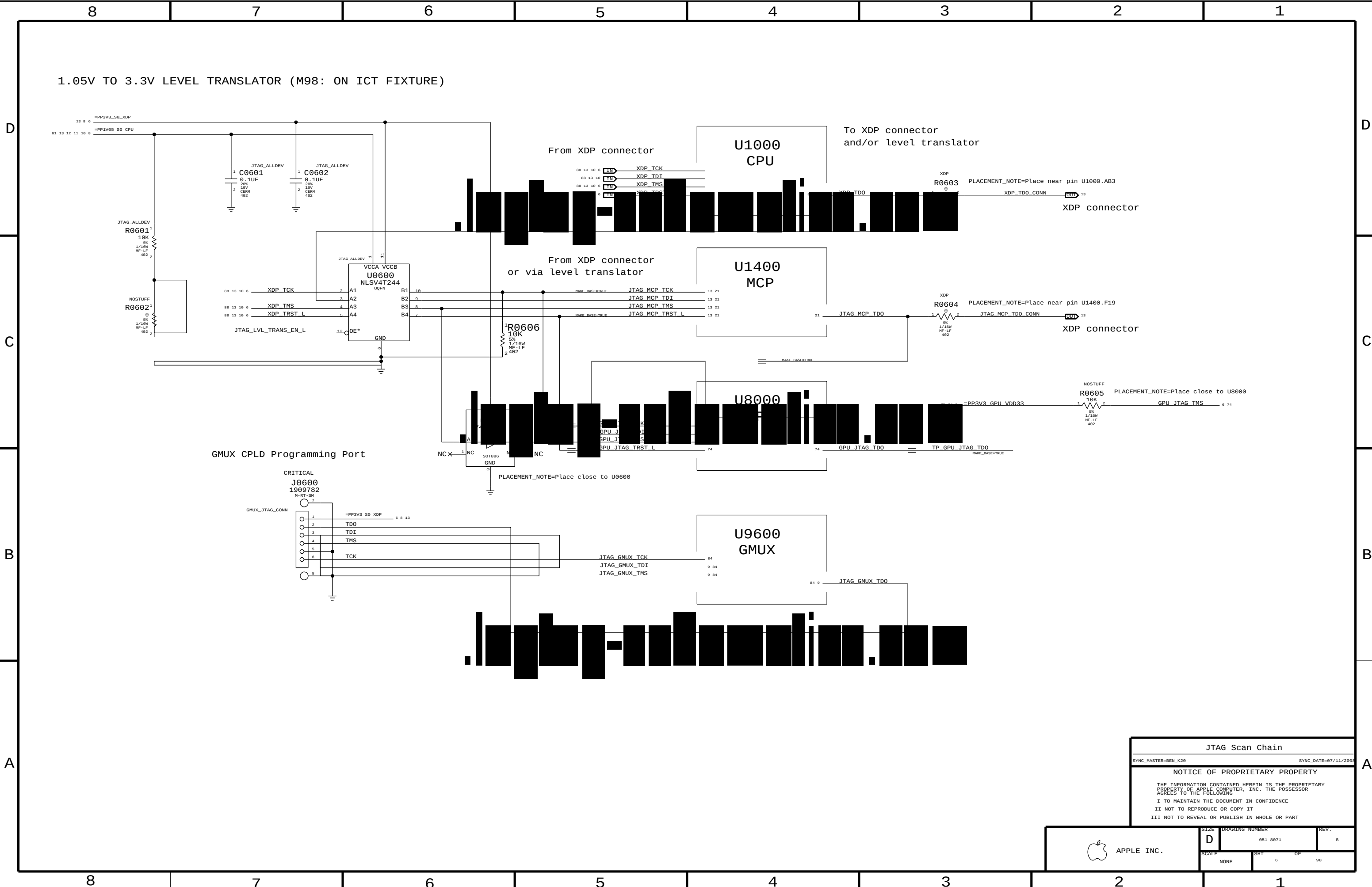
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JTAG Scan Chain

SYNC\_MASTER=BEN\_K20

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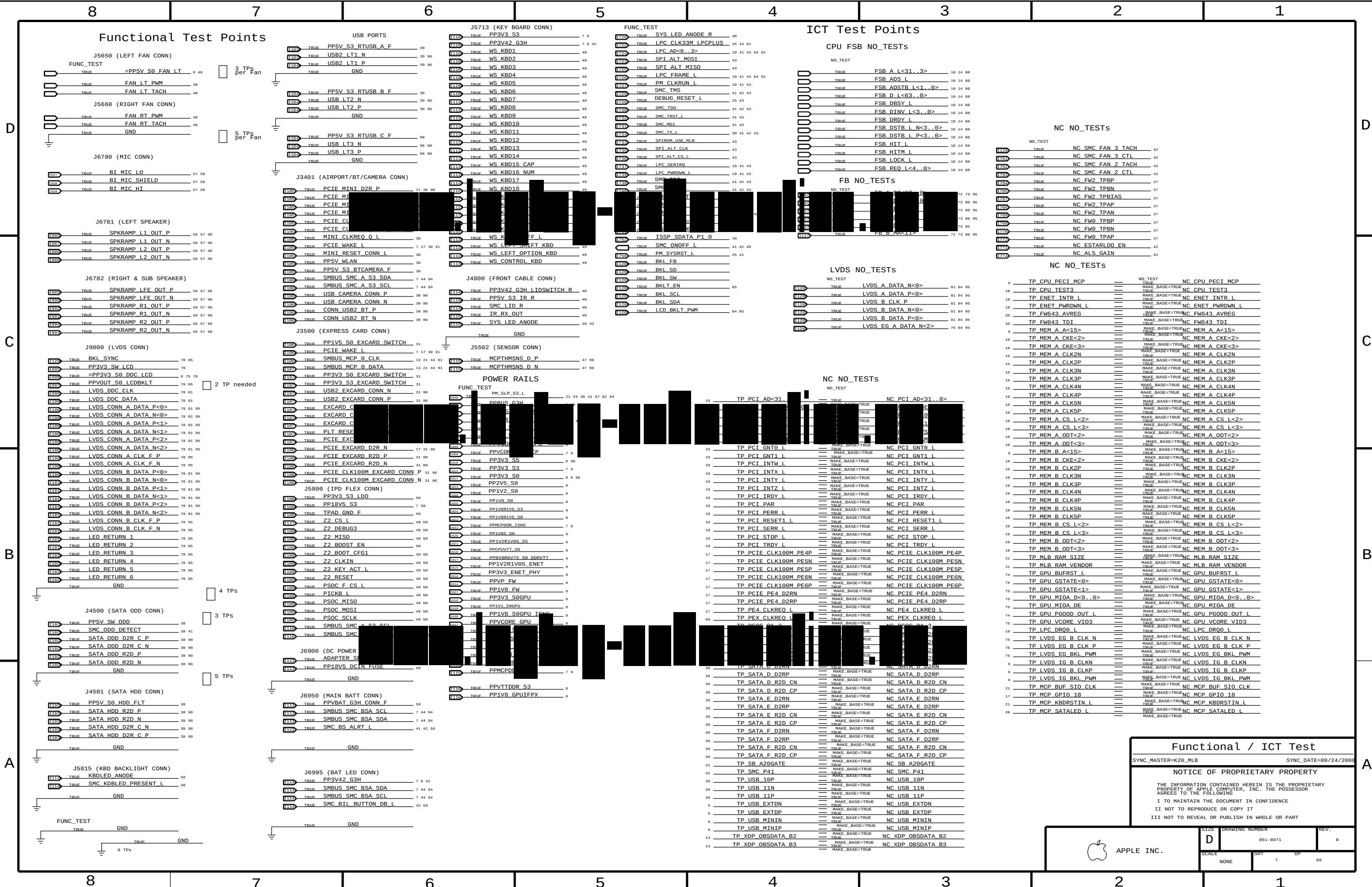
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Functional Test Points

ICT Test Points

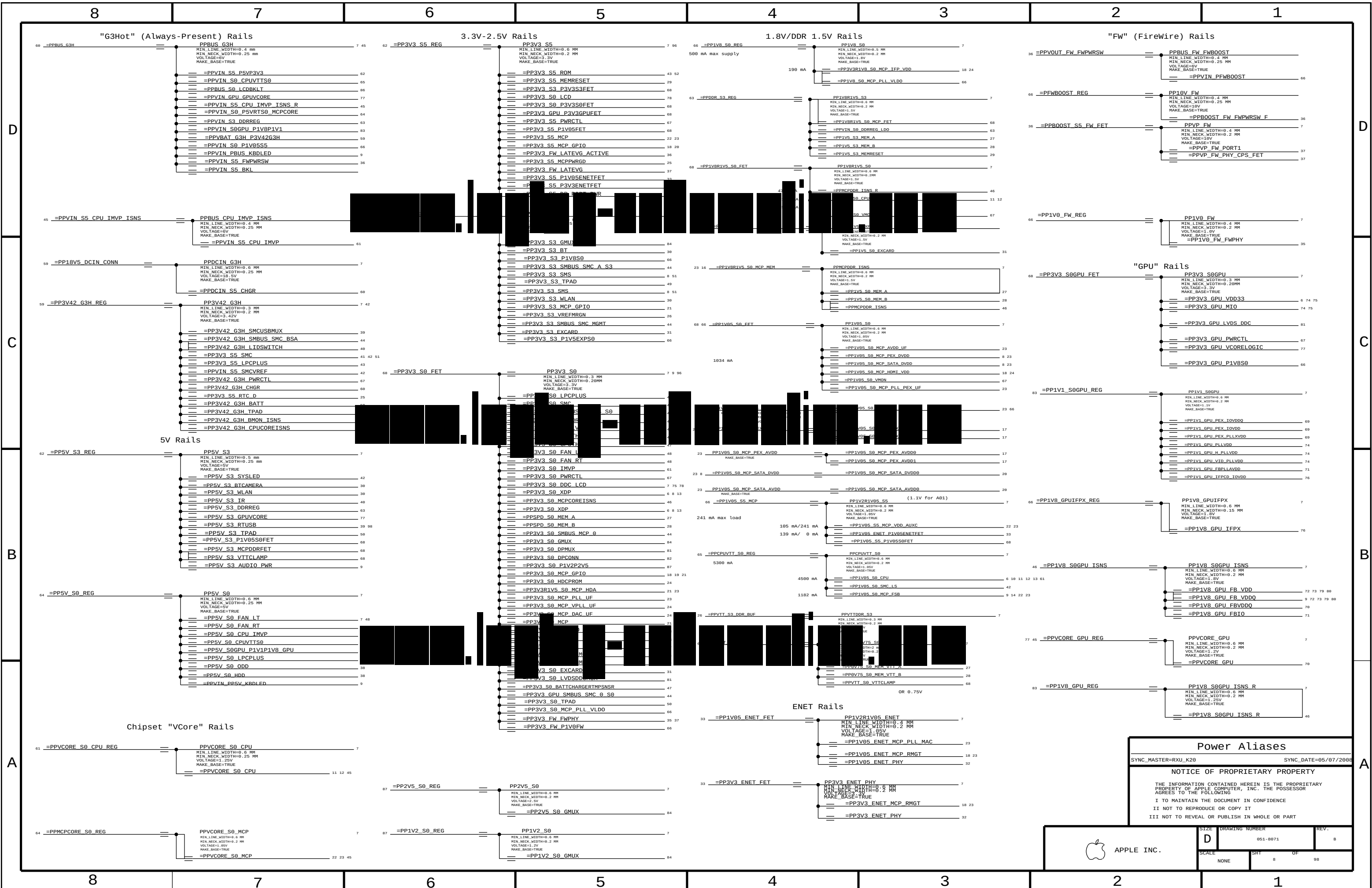
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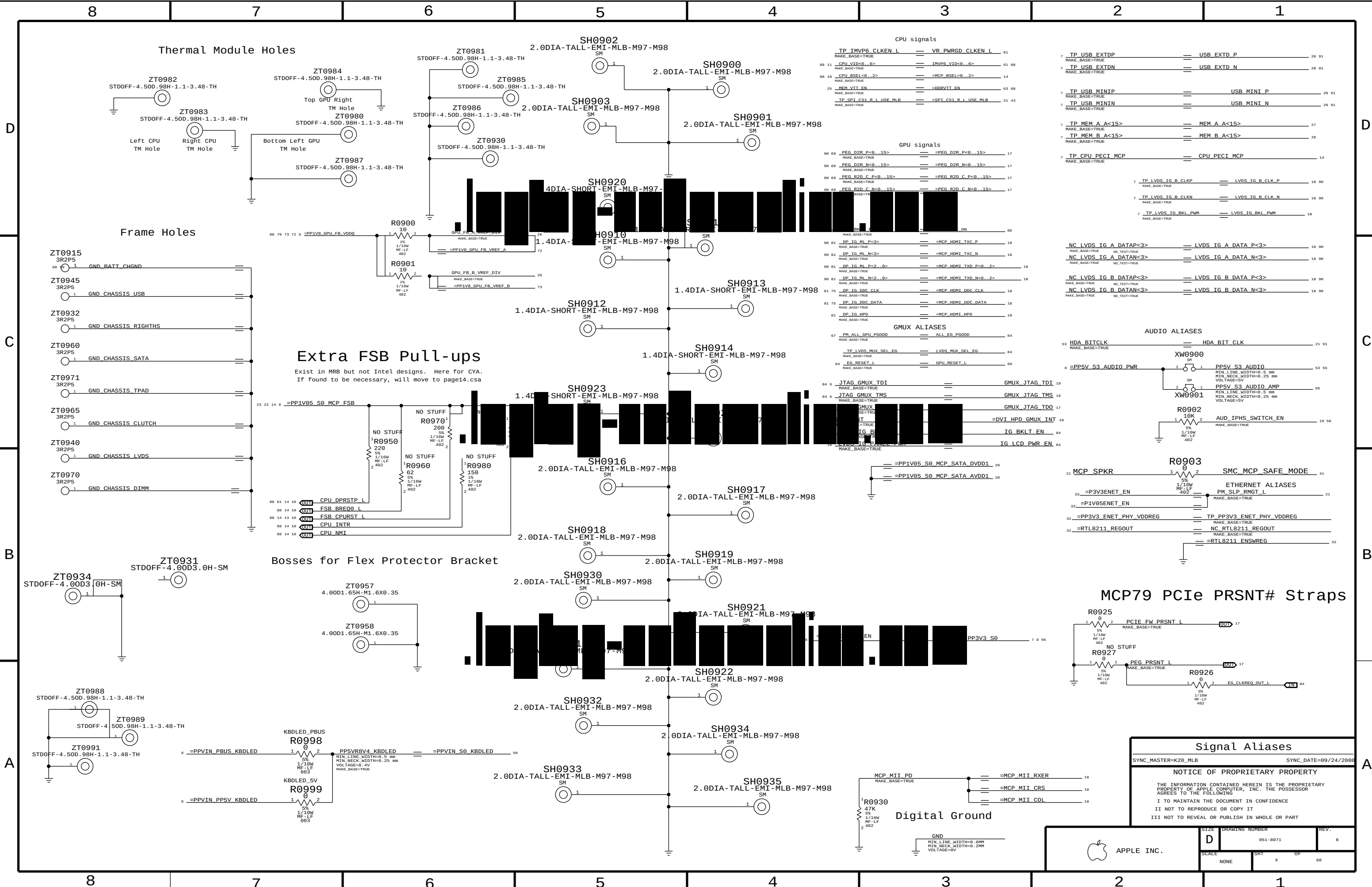
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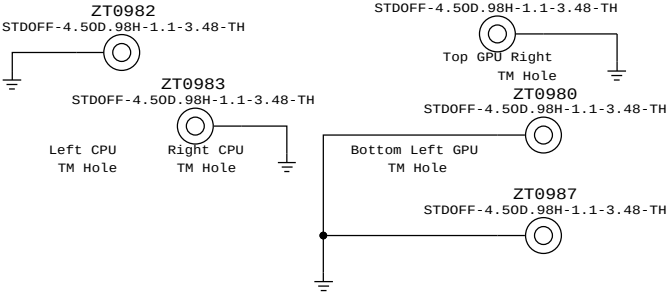
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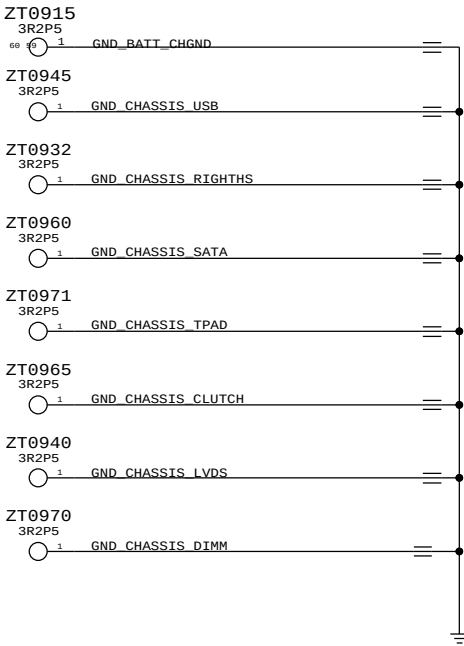




Thermal Module Holes

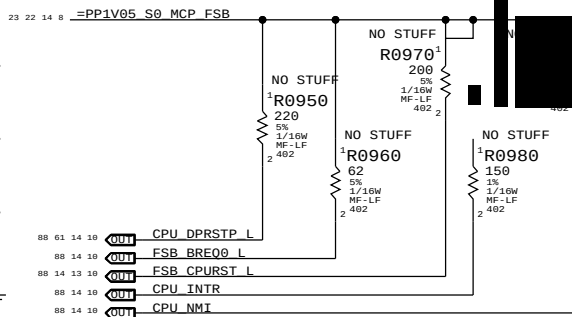


Frame Holes

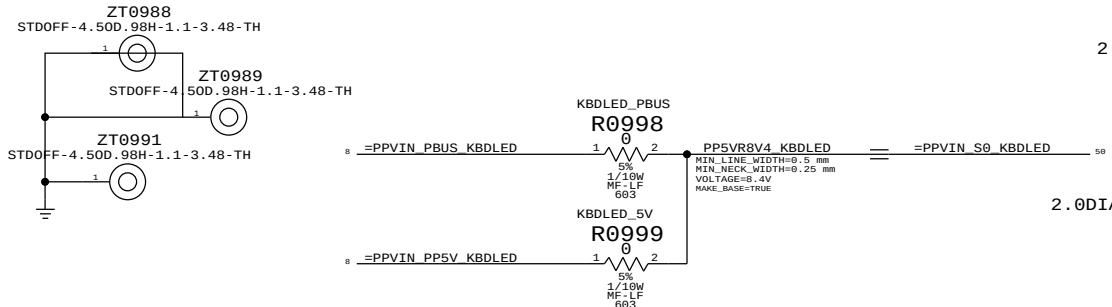
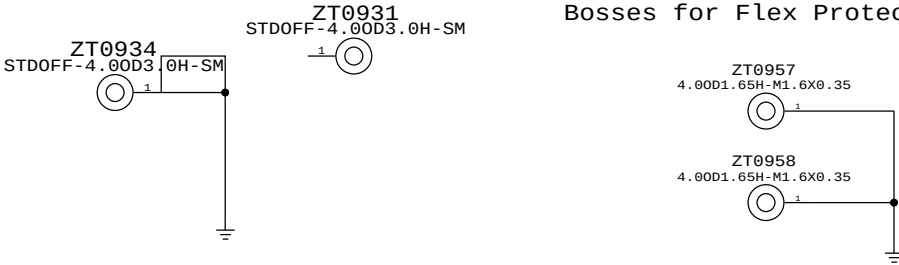


Extra FSB Pull-ups

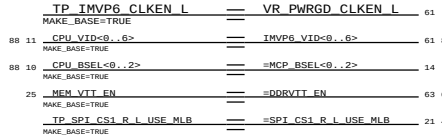
Exist in MRB but not Intel designs. Here for CYA.  
If found to be necessary, will move to page14.csa



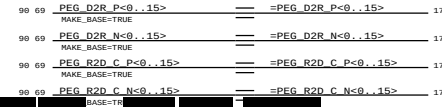
Bosses for Flex Protector Bracket



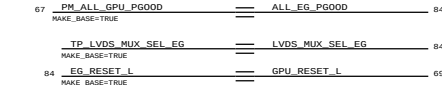
CPU signals



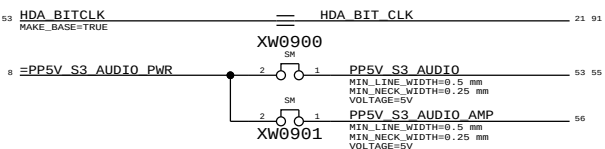
GPU signals



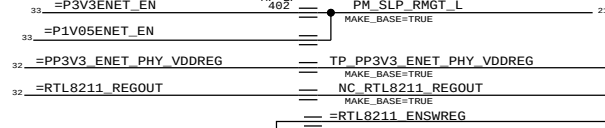
GMUX ALIASES



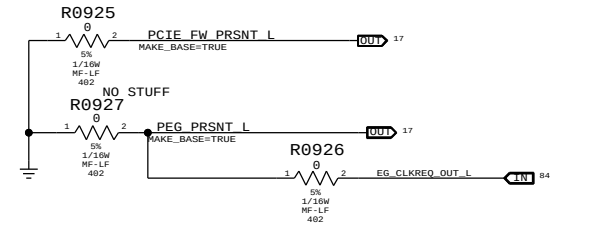
AUDIO ALIASES



ETHERNET ALIASES



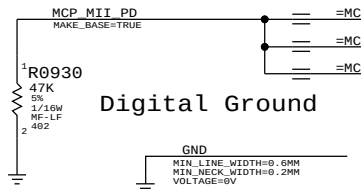
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Signal Aliases

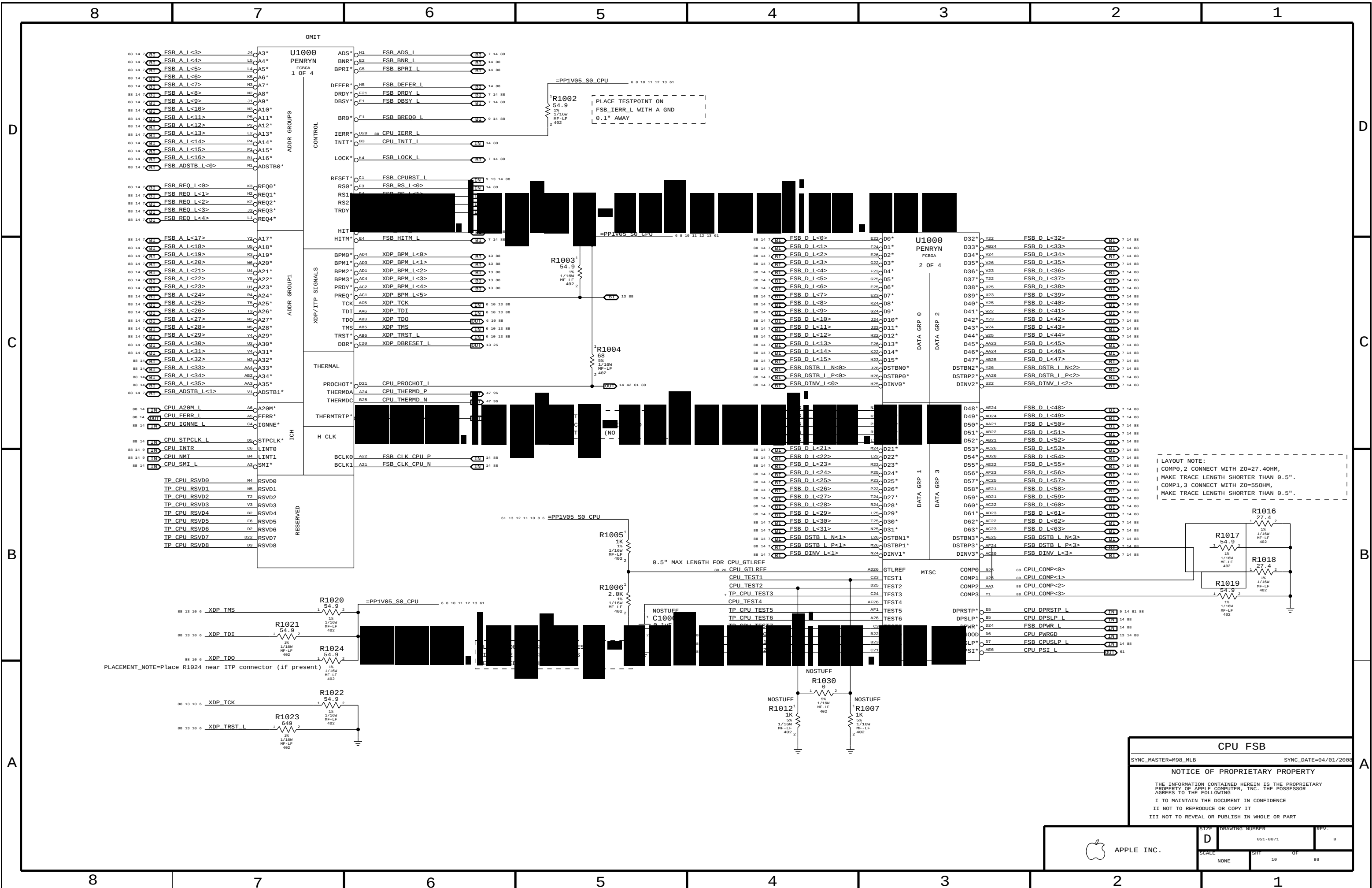
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Digital Ground



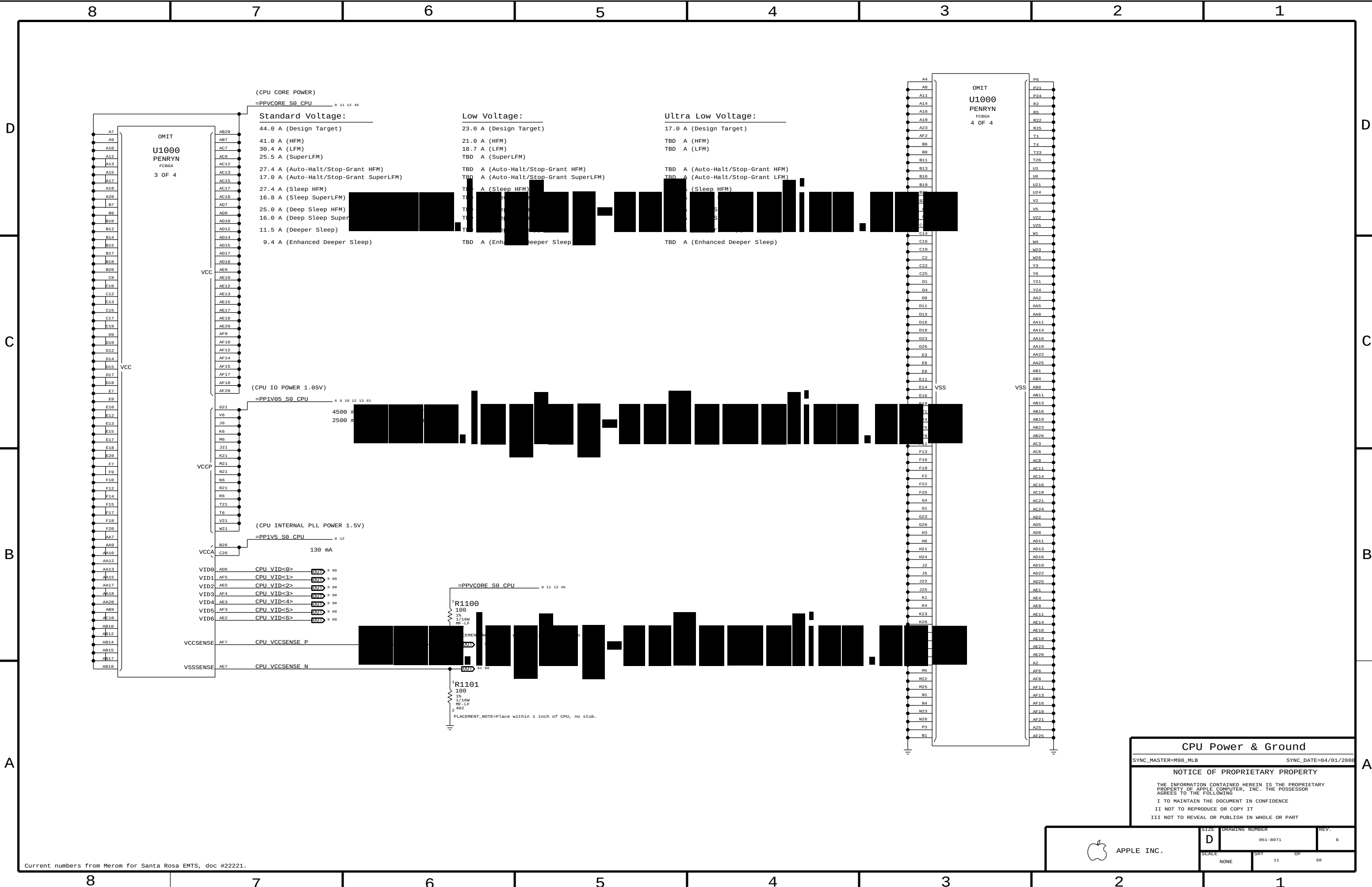
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| SIZE  | DRAWING NUMBER | REV. |
| D     | 051-8071       | B    |
| SCALE | SHT            | OF   |
| NONE  | 9              | 98   |



LAYOUT NOTE:  
COMP0,2 CONNECT WITH Z0=27.40HM,  
MAKE TRACE LENGTH SHORTER THAN 0.5".  
COMP1,3 CONNECT WITH Z0=550HM,  
MAKE TRACE LENGTH SHORTER THAN 0.5".

**CPU FSB**  
SYNC\_MASTER=M98\_MLB SYNC\_DATE=04/01/2008  
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**CPU Power & Ground**

SYNC\_MASTER=M9B\_MLB SYNC\_DATE=04/01/2008

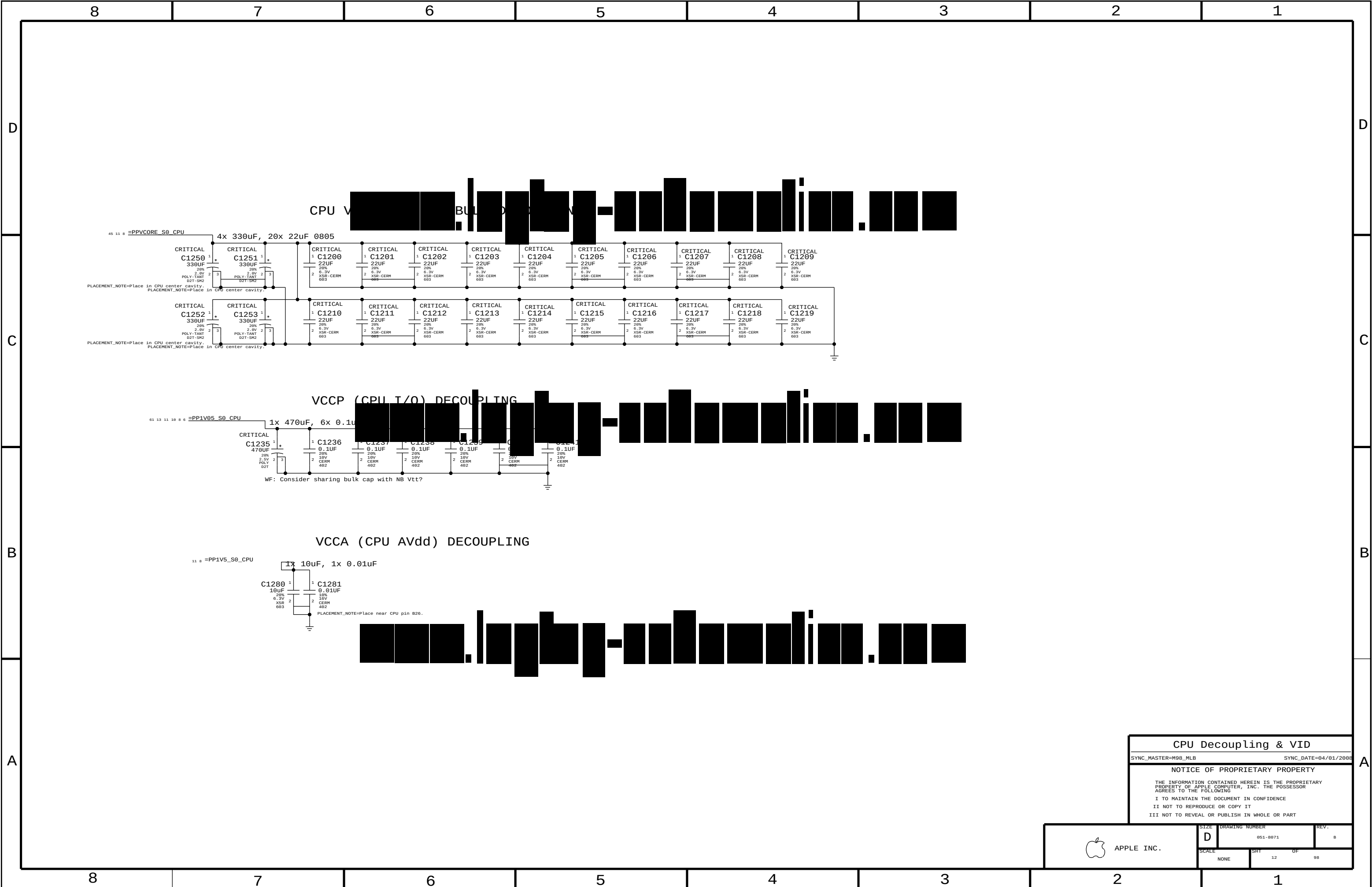
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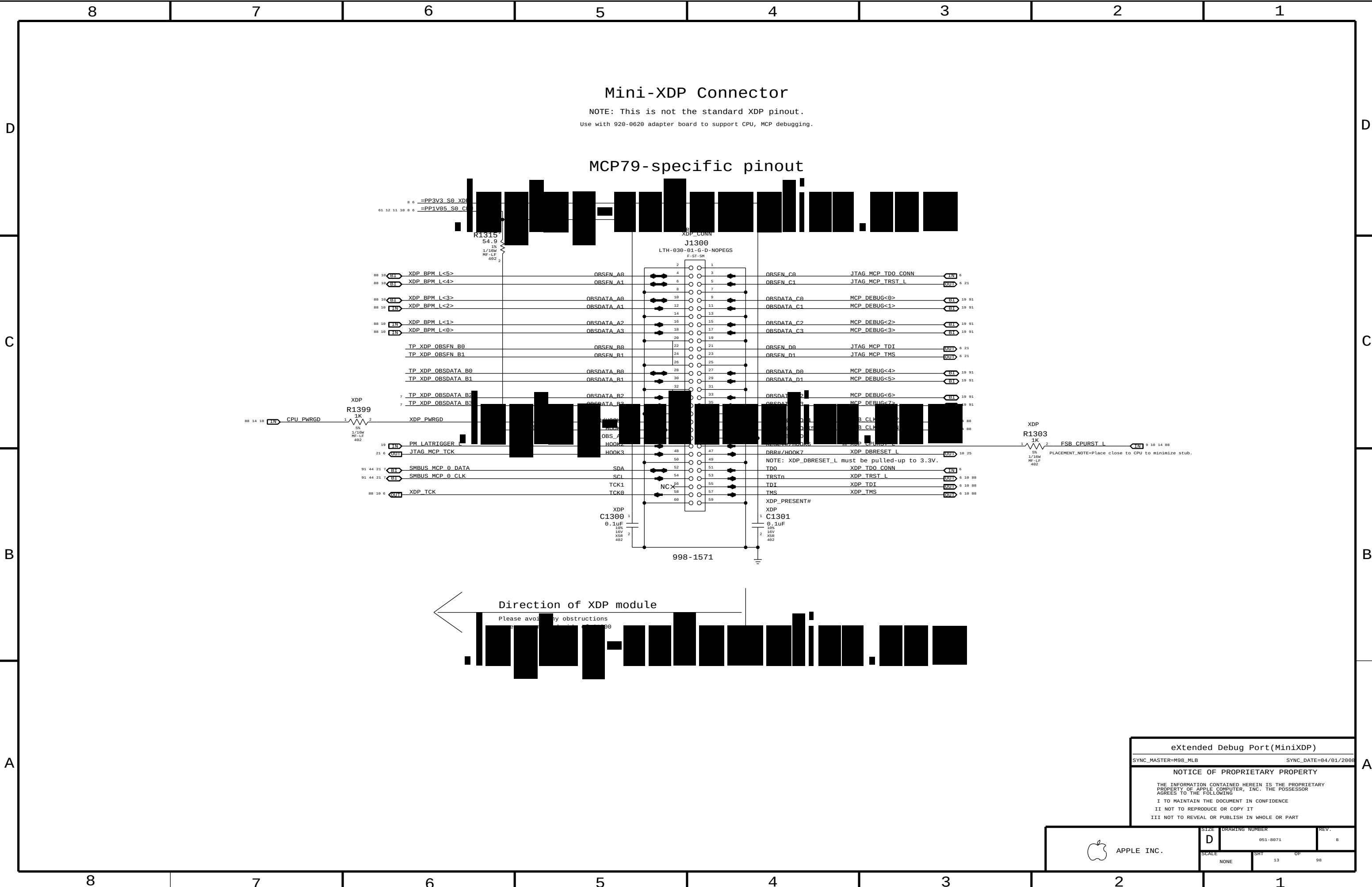
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| CPU Decoupling & VID                                                                                                       |  |                      |
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|            | D    | 051-8071       | B        |
| SCALE      | NONE | SHT            | 12 OF 98 |



Mini-XDP Connector

NOTE: This is not the standard XDP pinout.  
Use with 920-0620 adapter board to support CPU, MCP debugging.

MCP79-specific pinout

eXtended Debug Port(MiniXDP)

SYNC\_MASTER=M98\_MLB

SYNC\_DATE=04/01/2008

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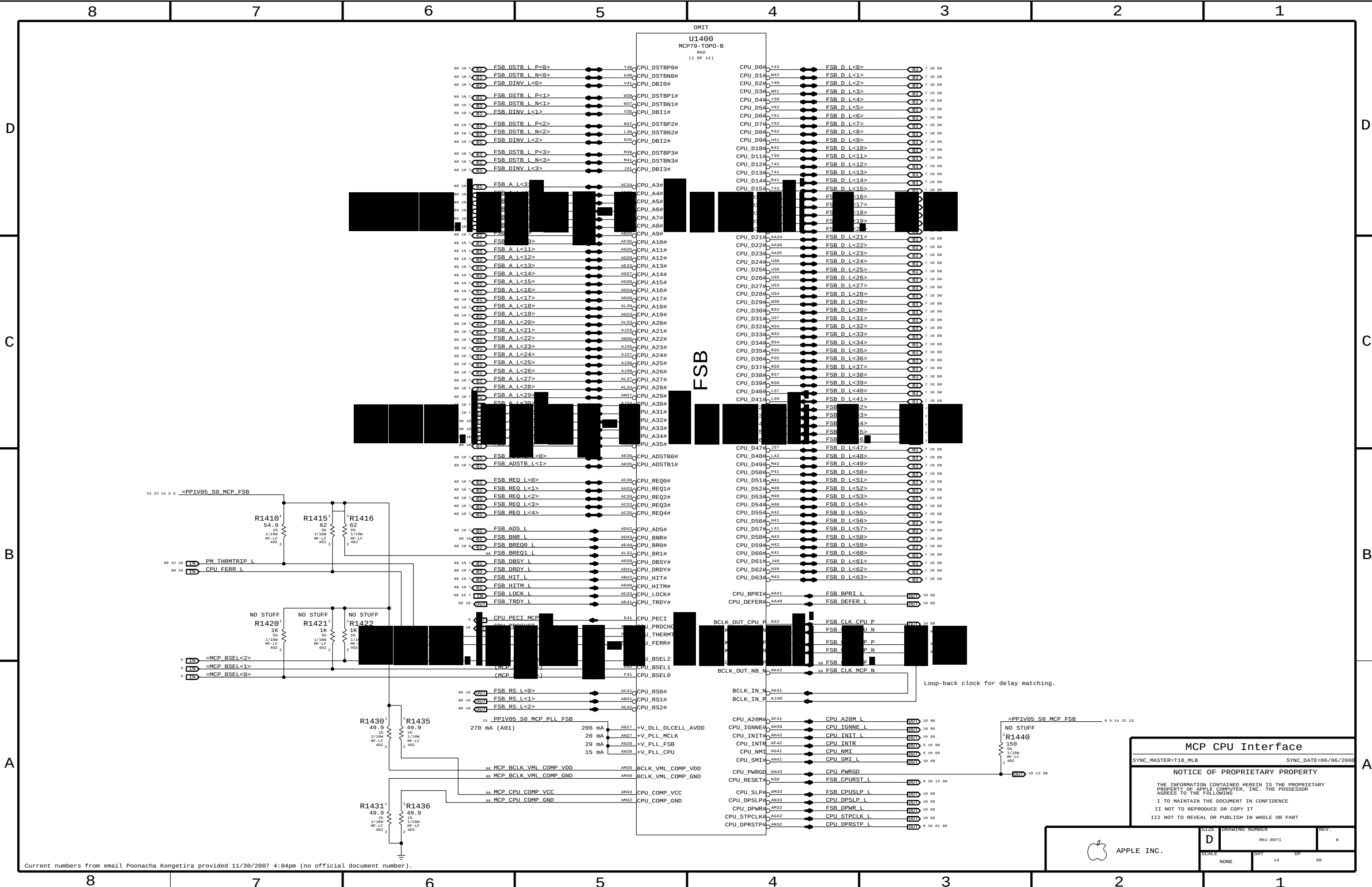
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13

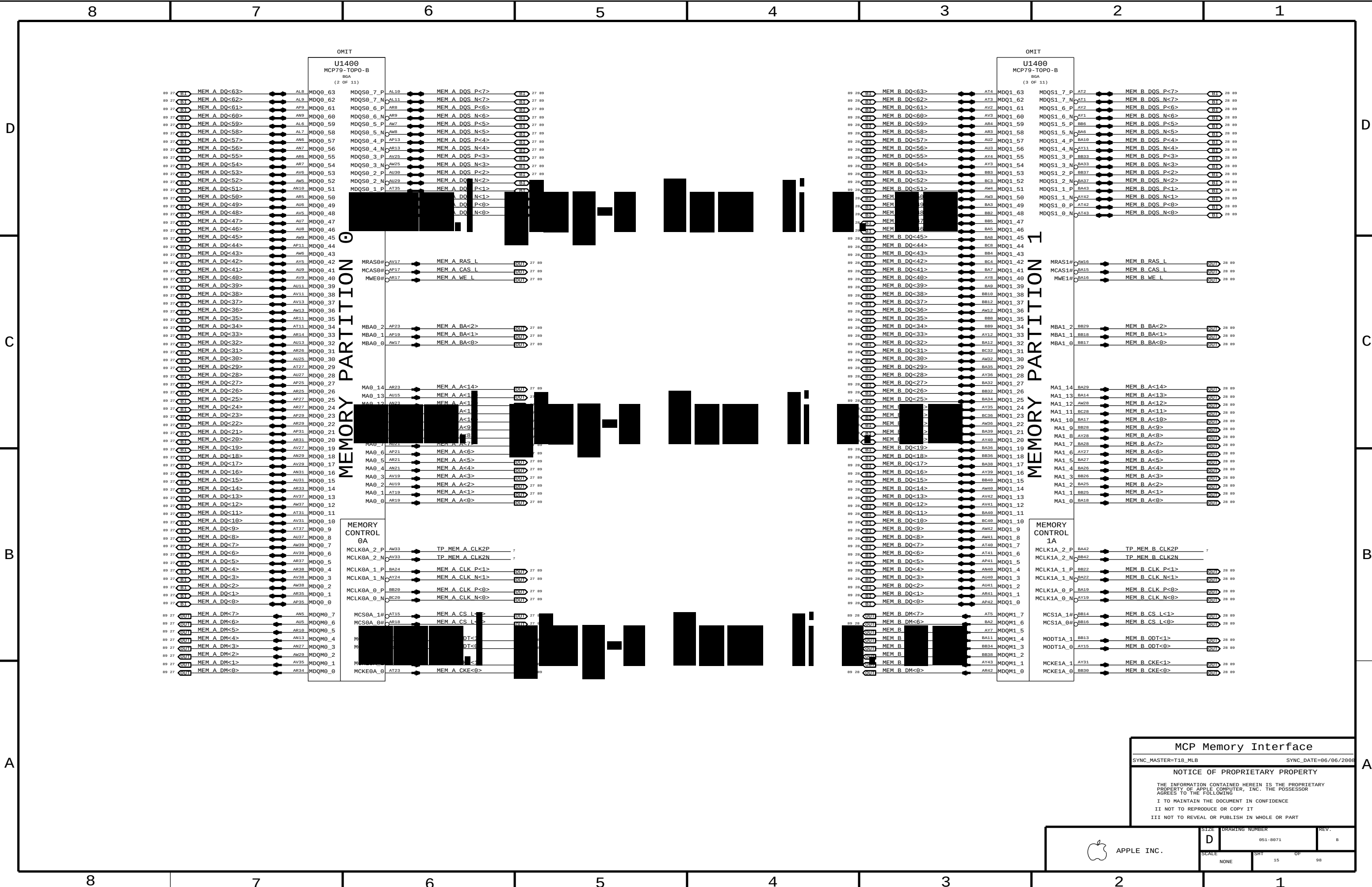
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MCP Memory Interface

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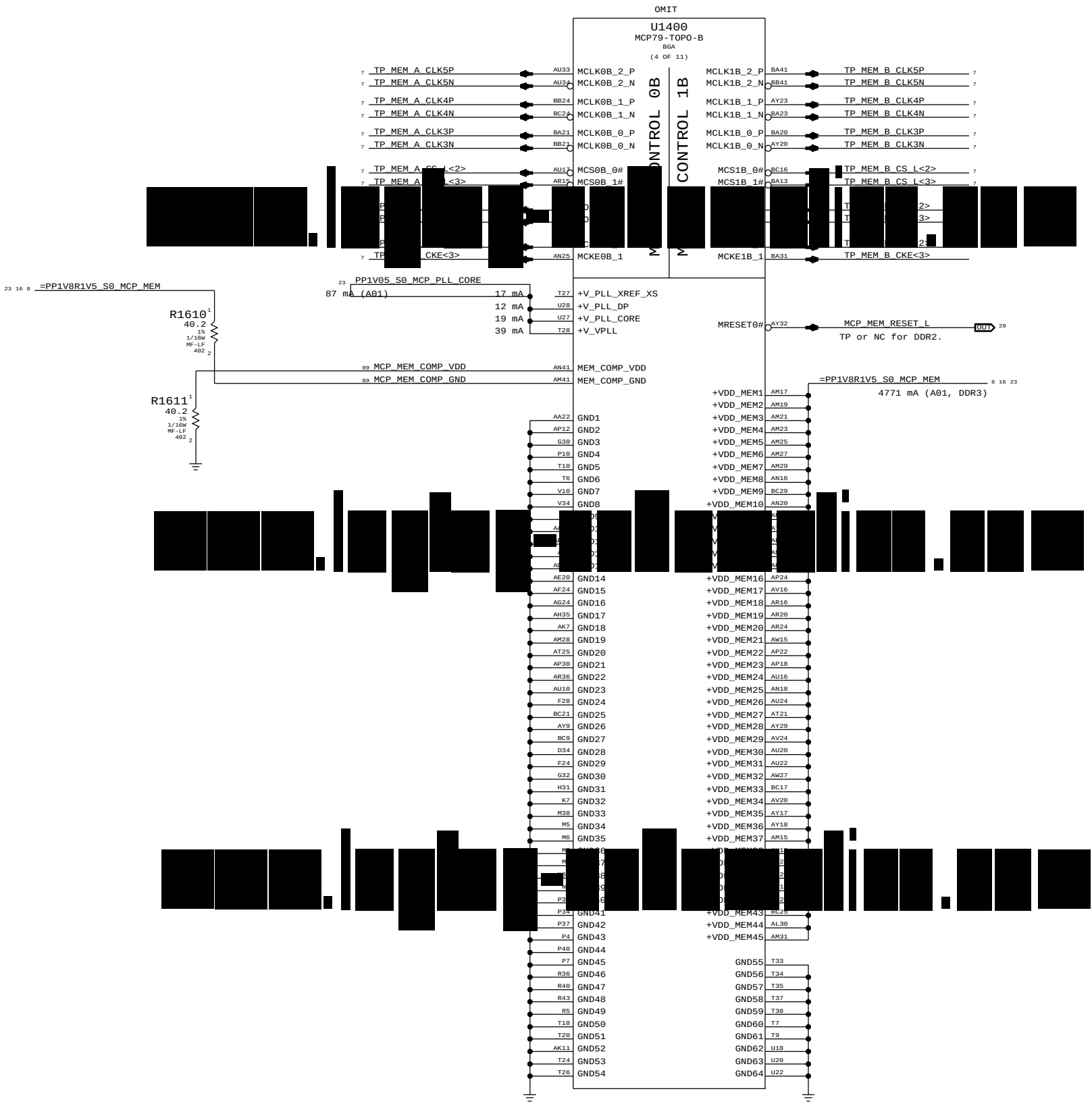
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MCP Memory Misc

SYNC\_MASTER=T18\_MLB

SYNC\_DATE=06/06/2009

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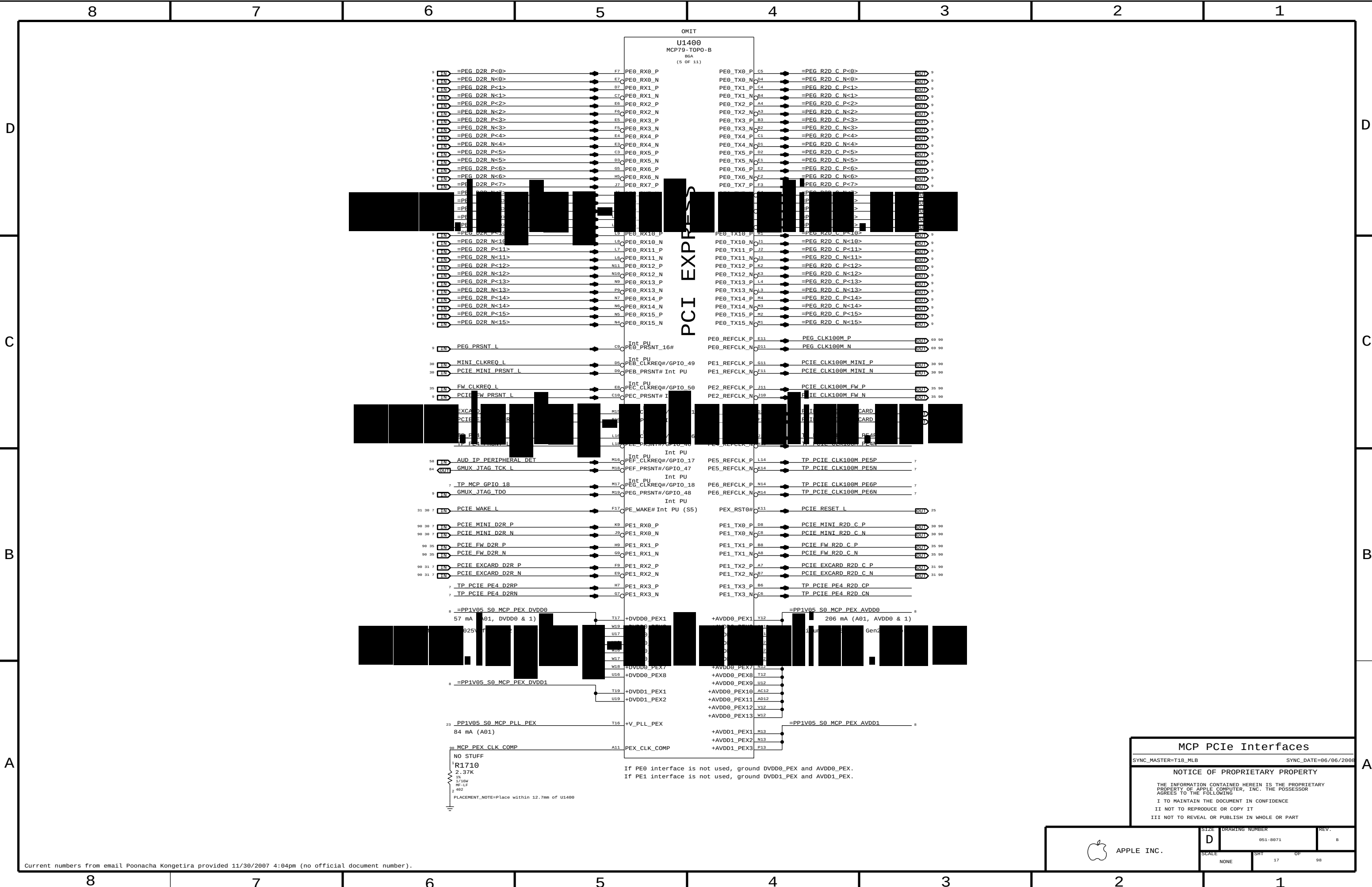
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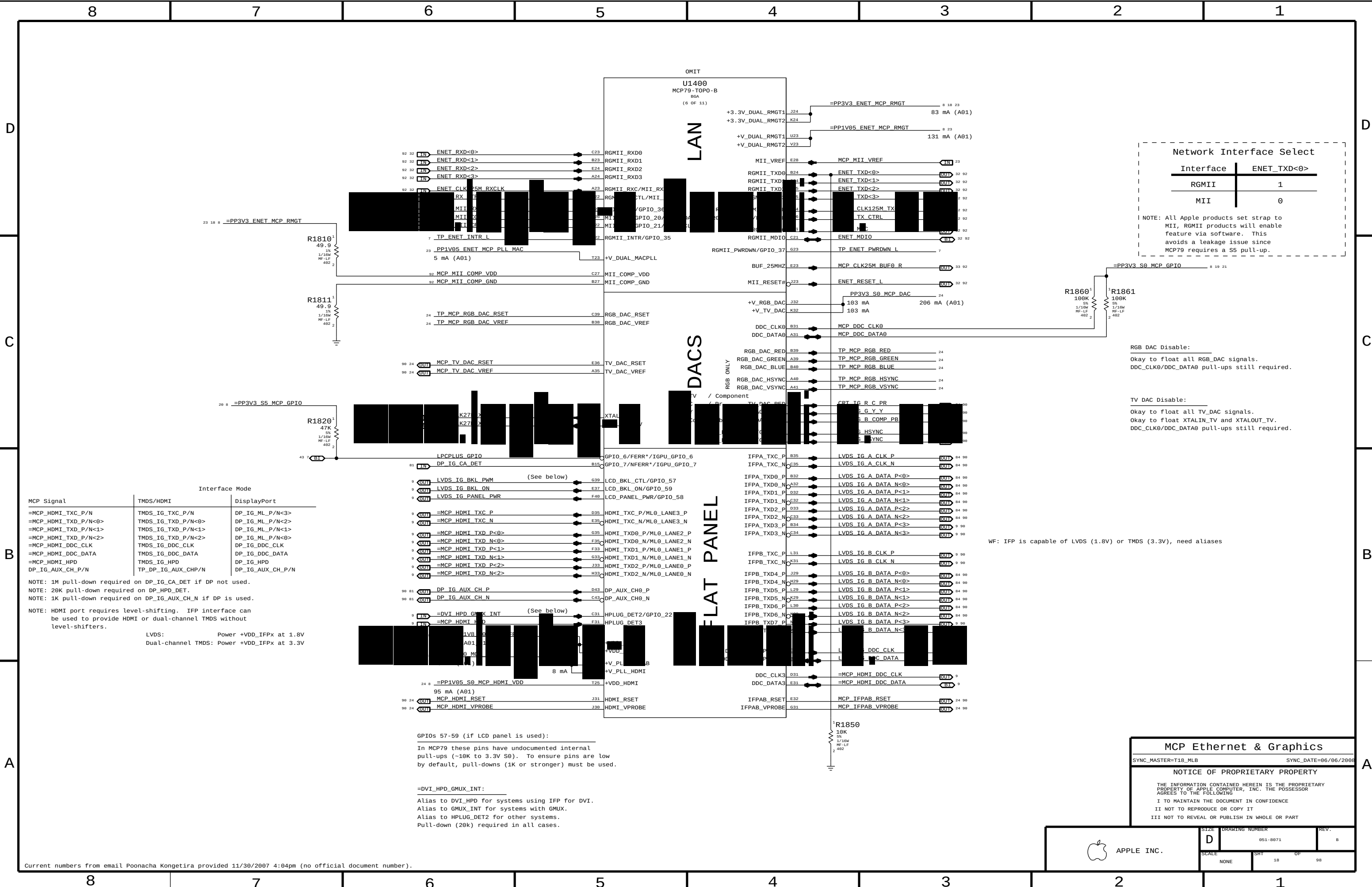
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| MCP PCIe Interfaces                                                                                                        |  |                      |
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| NONE       |                | 17       | 98   |



| Network Interface Select |             |
|--------------------------|-------------|
| Interface                | ENET_TXD<0> |
| RGMII                    | 1           |
| MII                      | 0           |

NOTE: All Apple products set strap to MII, RGMII products will enable feature via software. This avoids a leakage issue since MCP79 requires a S5 pull-up.

RGB DAC Disable: \_\_\_\_\_  
Okay to float all RGB\_DAC signals.  
DDC\_CLK0/DDC\_DATA0 pull-ups still required.

TV DAC Disable: \_\_\_\_\_  
Okay to float all TV\_DAC signals.  
Okay to float XTALIN\_TV and XTALOUT\_TV.  
DDC\_CLK0/DDC\_DATA0 pull-ups still required.

| MCP Signal           | Interface Mode      |                  |
|----------------------|---------------------|------------------|
|                      | TMDS/HDMI           | DisplayPort      |
| =MCP_HDMI_TXC_P/N    | TMDS_IG_TXC_P/N     | DP_IG_ML_P/N<3>  |
| =MCP_HDMI_TXD_P/N<0> | TMDS_IG_TXD_P/N<0>  | DP_IG_ML_P/N<2>  |
| =MCP_HDMI_TXD_P/N<1> | TMDS_IG_TXD_P/N<1>  | DP_IG_ML_P/N<1>  |
| =MCP_HDMI_TXD_P/N<2> | TMDS_IG_TXD_P/N<2>  | DP_IG_ML_P/N<0>  |
| =MCP_HDMI_DDC_CLK    | TMDS_IG_DDC_CLK     | DP_IG_DDC_CLK    |
| =MCP_HDMI_DDC_DATA   | TMDS_IG_DDC_DATA    | DP_IG_DDC_DATA   |
| =MCP_HDMI_HPD        | TMDS_IG_HPD         | DP_IG_HPD        |
| DP_IG_AUX_CH_P/N     | TP_DP_IG_AUX_CH_P/N | DP_IG_AUX_CH_P/N |

NOTE: 1M pull-down required on DP\_IG\_CA\_DET if DP not used.  
NOTE: 20K pull-down required on DP\_HPD\_DET.  
NOTE: 1K pull-down required on DP\_IG\_AUX\_CH\_N if DP is used.

NOTE: HDMI port requires level-shifting. IFP interface can be used to provide HDMI or dual-channel TMDS without level-shifters.

LVDS: Power +VDD\_IFPx at 1.8V  
Dual-channel TMDS: Power +VDD\_IFPx at 3.3V

GPIOs 57-59 (if LCD panel is used):

In MCP79 these pins have undocumented internal pull-ups (~10K to 3.3V S0). To ensure pins are low by default, pull-downs (1K or stronger) must be used.

=DVI\_HPD\_GMUX\_INT:

Alias to DVI\_HPD for systems using IFP for DVI.  
Alias to GMUX\_INT for systems with GMUX.  
Alias to HPLUG\_DET2 for other systems.  
Pull-down (20k) required in all cases.

MCP Ethernet & Graphics

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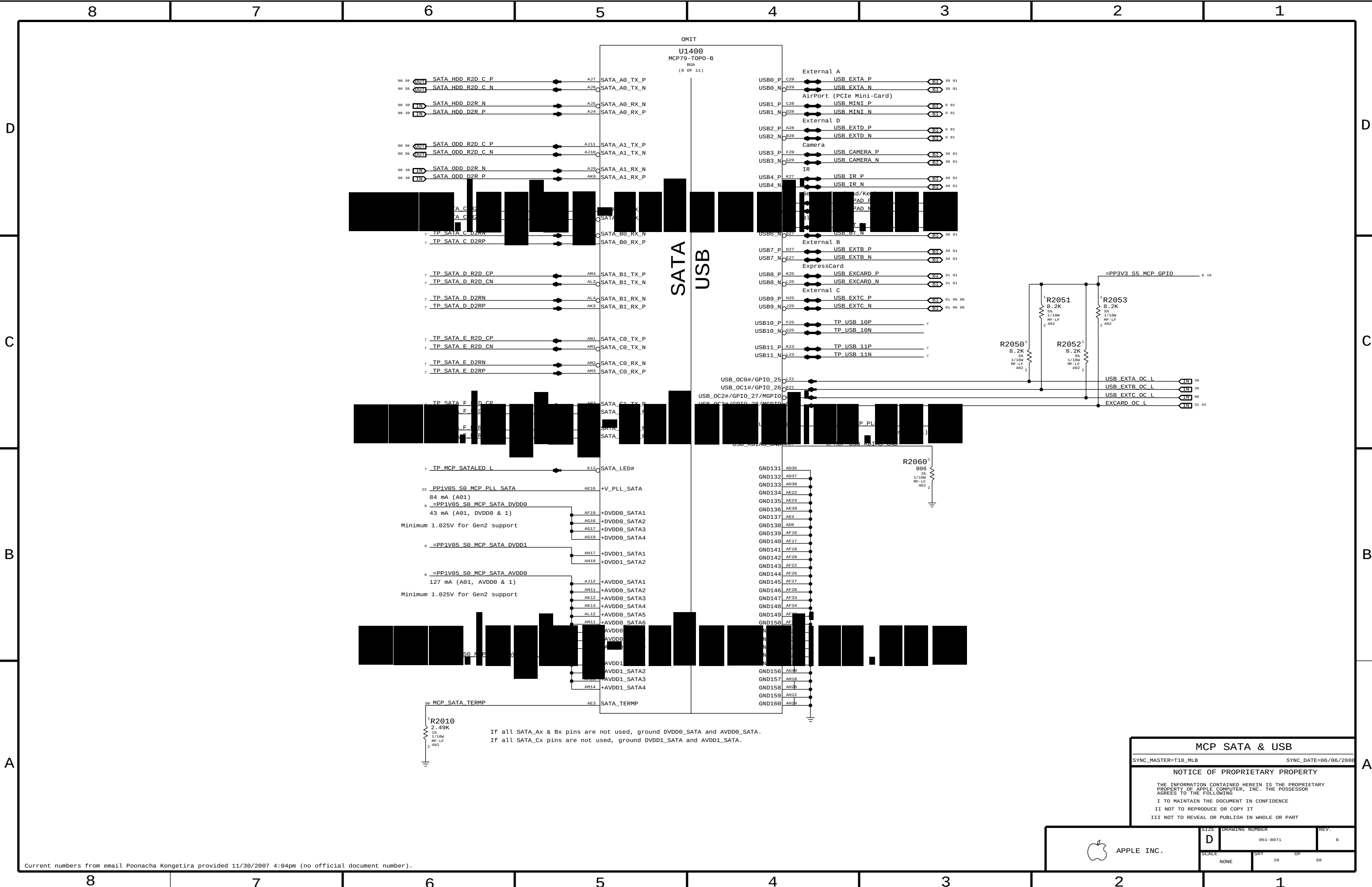
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MCP SATA & USB

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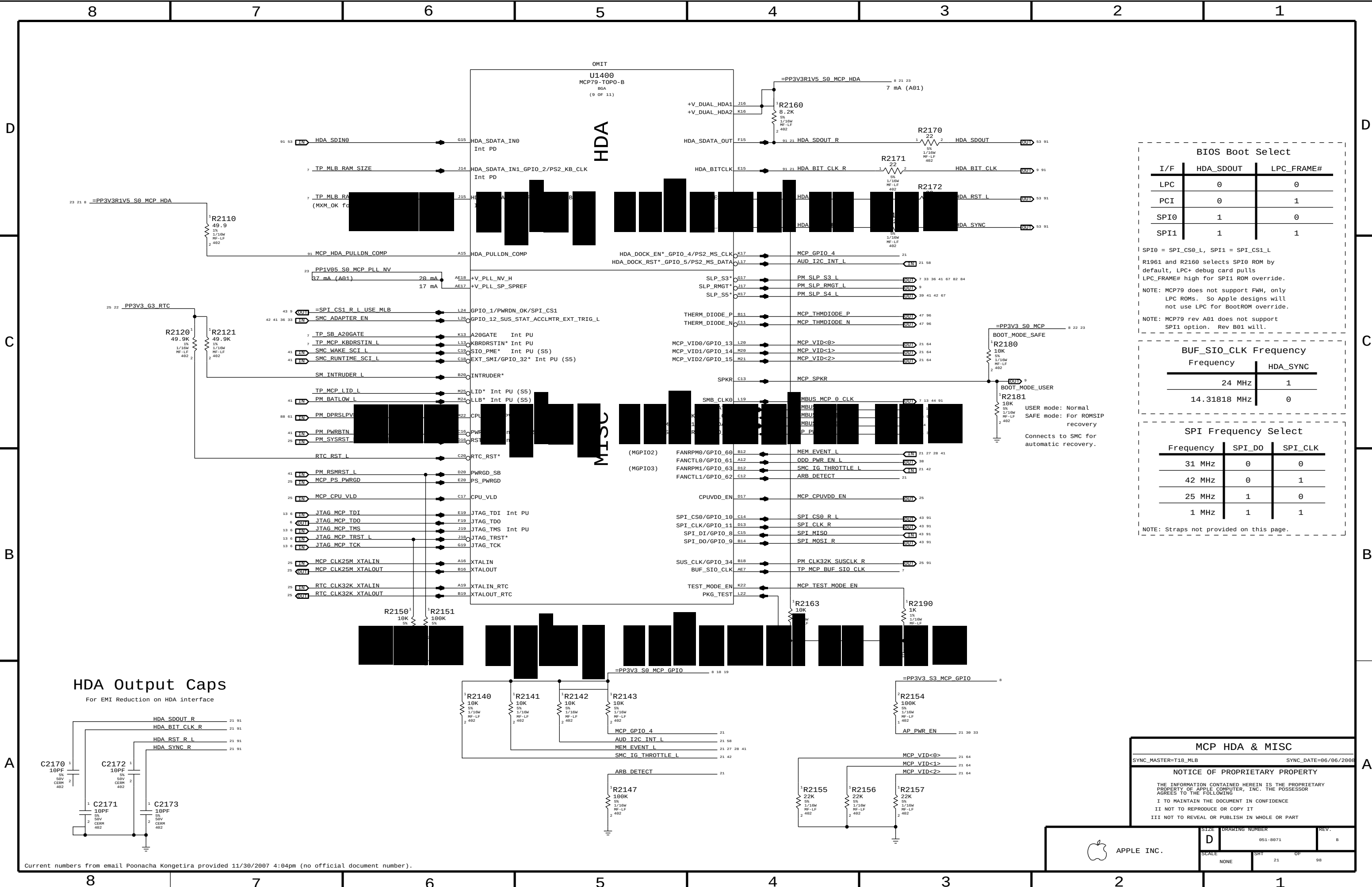
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| BIOS Boot Select |           |            |
|------------------|-----------|------------|
| I/F              | HDA_SDOUT | LPC_FRAME# |
| LPC              | 0         | 0          |
| PCI              | 0         | 1          |
| SPI0             | 1         | 0          |
| SPI1             | 1         | 1          |

SPI0 = SPI\_CS0\_L, SPI1 = SPI\_CS1\_L

R1961 and R2160 selects SPI0 ROM by default, LPC+ debug card pulls LPC\_FRAME# high for SPI1 ROM override.

NOTE: MCP79 does not support FWH, only LPC ROMs. So Apple designs will not use LPC for BootROM override.

NOTE: MCP79 rev A01 does not support SPI1 option. Rev B01 will.

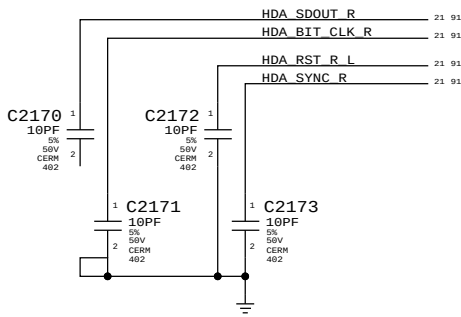
| BUF_SIO_CLK Frequency |          |
|-----------------------|----------|
| Frequency             | HDA_SYNC |
| 24 MHz                | 1        |
| 14.31818 MHz          | 0        |

| SPI Frequency Select |        |         |
|----------------------|--------|---------|
| Frequency            | SPI_D0 | SPI_CLK |
| 31 MHz               | 0      | 0       |
| 42 MHz               | 0      | 1       |
| 25 MHz               | 1      | 0       |
| 1 MHz                | 1      | 1       |

NOTE: Straps not provided on this page.

### HDA Output Caps

For EMI Reduction on HDA interface



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MCP HDA & MISC

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**MCP Power & Ground**

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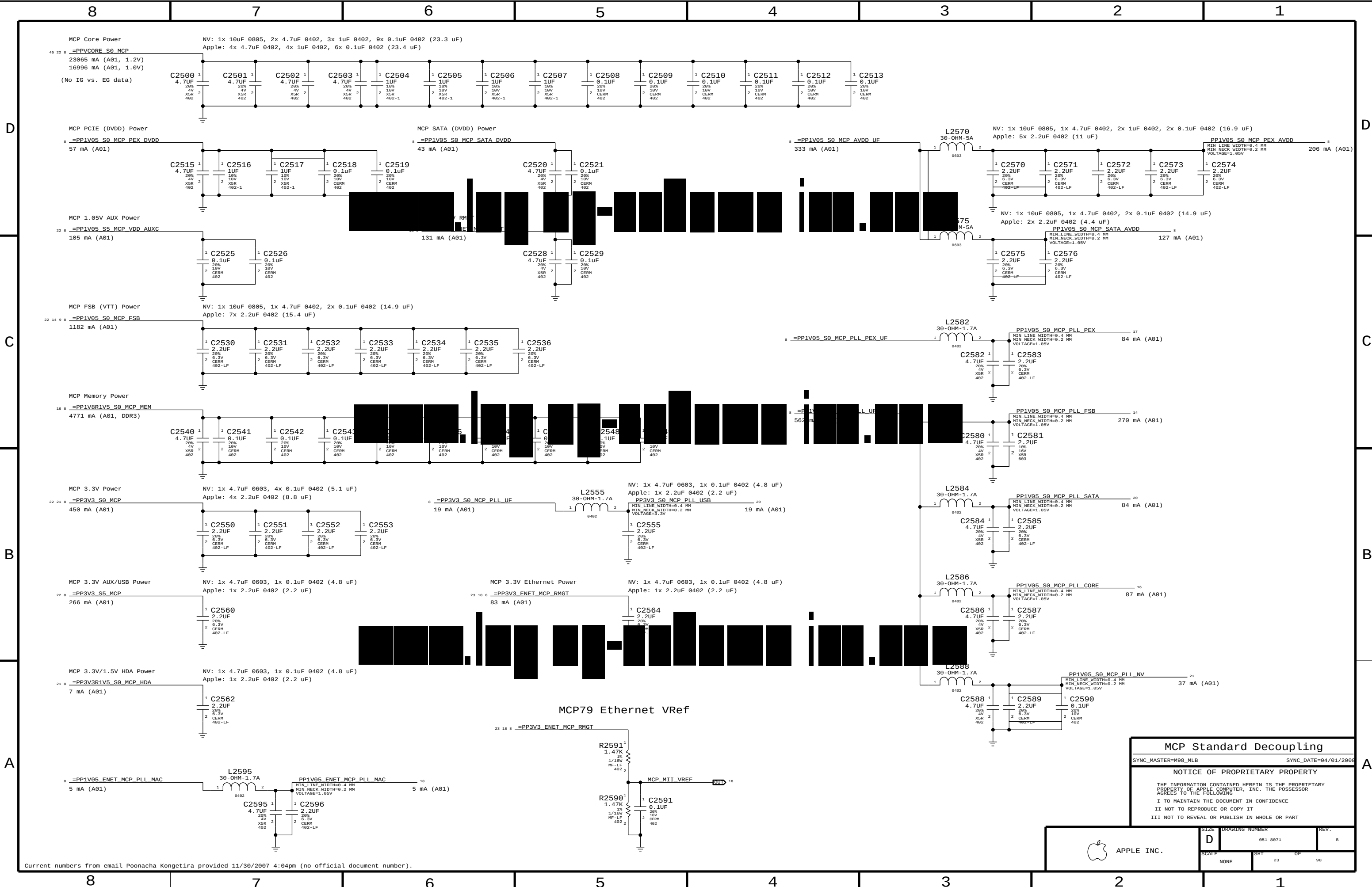
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MCP Standard Decoupling

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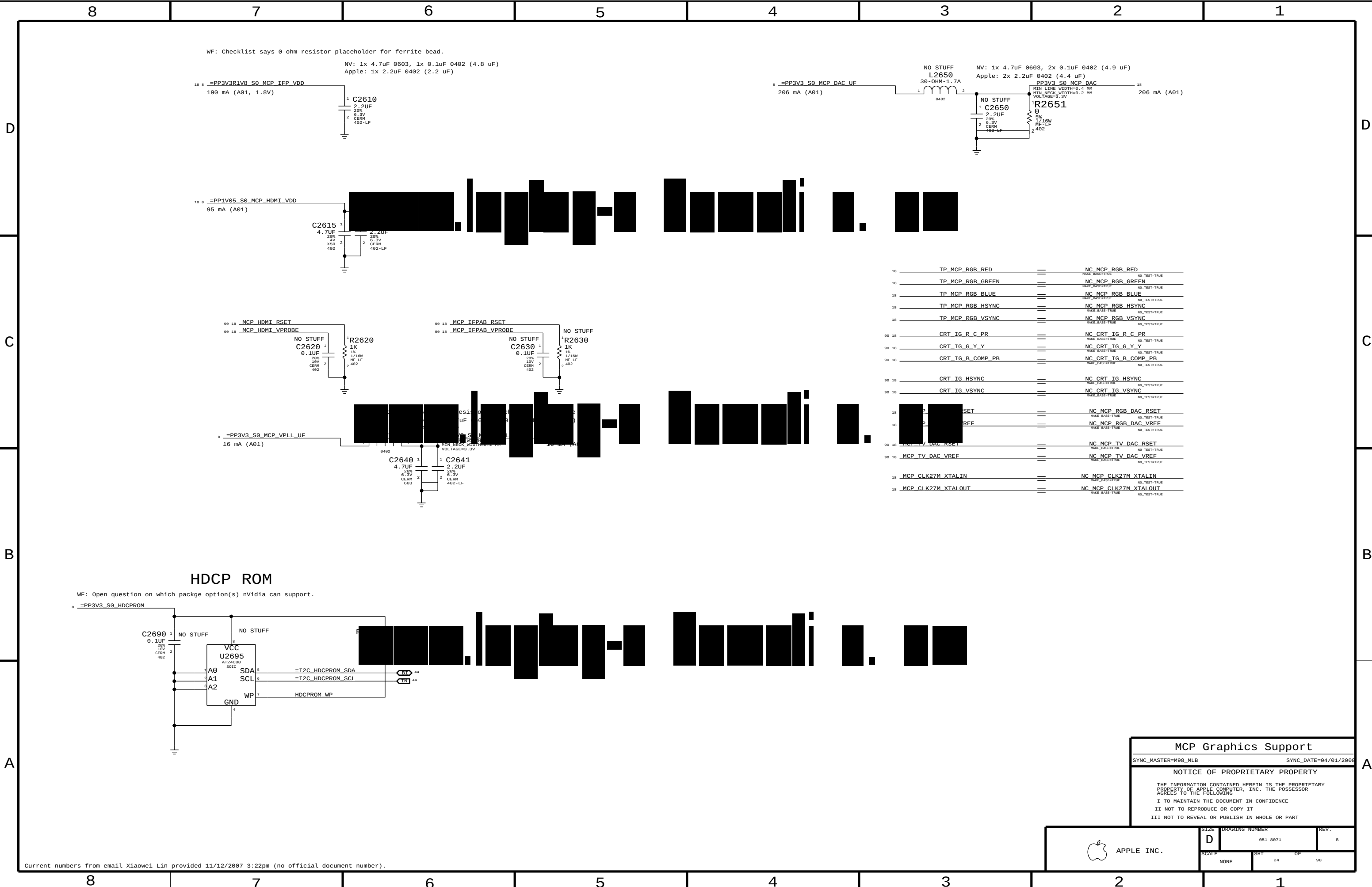
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MCP Graphics Support

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Page Notes

Power aliases required by this page:

- =PP3V3\_S3\_VREFMRGN
- =PP3V3\_S5\_VREFMRGN
- =PPVTT\_S3\_DDR\_BUF

Signal aliases required by this page:

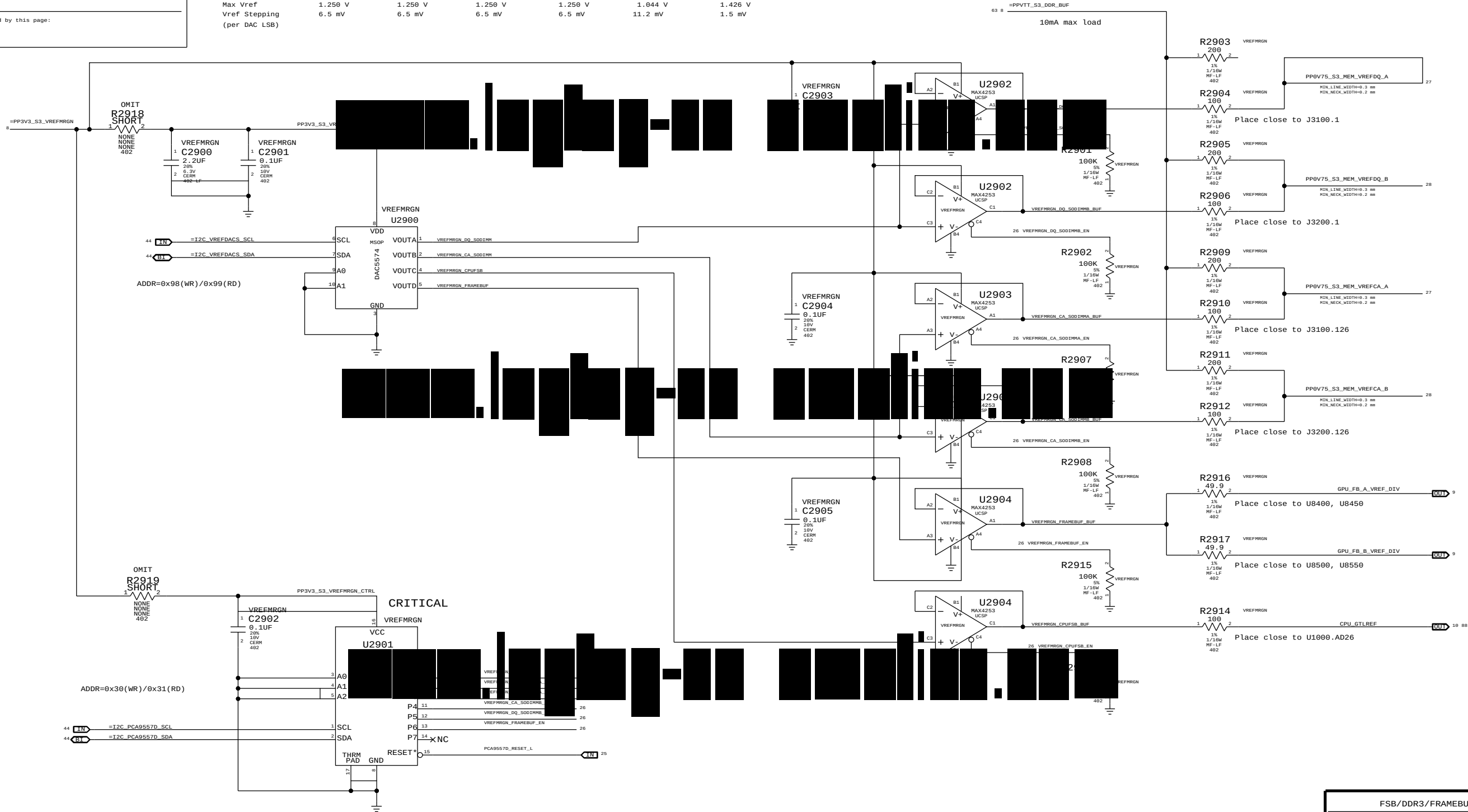
- =I2C\_VREFDAC5\_SCL
- =I2C\_VREFDAC5\_SDA
- =I2C\_PCA9557D\_SCL
- =I2C\_PCA9557D\_SDA

BOM options provided by this page:

- VREFMRGN
- NO\_VREFMRGN

|                             | MEM A VREF DQ |          | MEM A VREF CA |          | MEM B VREF DQ |          | MEM B VREF CA |           | CPU FSB VREF | FRAME BUFFER VREF |
|-----------------------------|---------------|----------|---------------|----------|---------------|----------|---------------|-----------|--------------|-------------------|
| DAC channel                 | A             | B        | A             | B        | A             | B        | C             | D         |              |                   |
| Min DAC code                | 0x00          | 0x00     | 0x00          | 0x00     | 0x00          | 0x00     | 0x00          | 0x00      |              |                   |
| Max DAC code                | 0x87          | 0x87     | 0x87          | 0x87     | 0x87          | 0x87     | 0x55          | 0xFF      |              |                   |
| Max sink I                  | -3.75 mA      | -3.75 mA | -3.75 mA      | -3.75 mA | -3.75 mA      | -3.75 mA | -0.91 mA      | -59.04 mA |              |                   |
| Max source I                | 5 mA          | 5 mA     | 5 mA          | 5 mA     | 5 mA          | 5 mA     | 0.52 mA       | 51.15 mA  |              |                   |
| Nominal Vref                | 0.75 V        | 0.75 V   | 0.75 V        | 0.75 V   | 0.75 V        | 0.75 V   | 0.70 V        | 1.248 V   |              |                   |
| Min Vref                    | 0.375 V       | 0.375 V  | 0.375 V       | 0.375 V  | 0.375 V       | 0.375 V  | 0.091 V       | 1.042 V   |              |                   |
| Max Vref                    | 1.250 V       | 1.250 V  | 1.250 V       | 1.250 V  | 1.250 V       | 1.250 V  | 1.044 V       | 1.426 V   |              |                   |
| Vref Stepping (per DAC LSB) | 6.5 mV        | 6.5 mV   | 6.5 mV        | 6.5 mV   | 6.5 mV        | 6.5 mV   | 11.2 mV       | 1.5 mV    |              |                   |

SO-DIMM A and SO-DIMM B Vref settings should be margined separately (i.e. not simultaneously) due to current limitation of TPS51116 regulator.



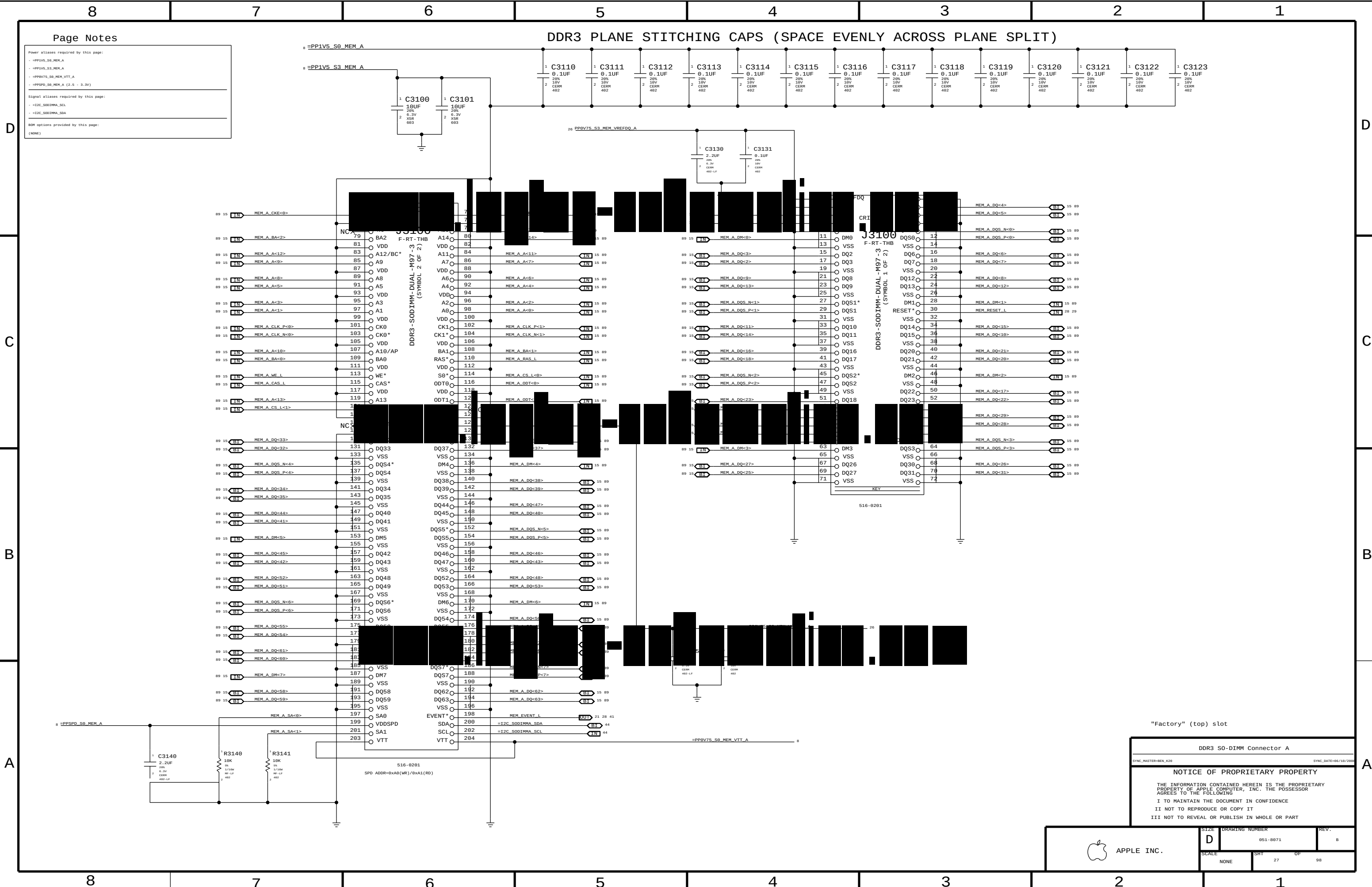
Required zero ohm resistors when no VREF margining circuit stuffed

| PART NUMBER | QTY | DESCRIPTION                        | REFERENCE DES | CRITICAL | BOM OPTION  |
|-------------|-----|------------------------------------|---------------|----------|-------------|
| 116S0004    | 1   | RES, MTL FILM, 0, 5%, 0402, SM, LF | R2903         | CRITICAL | NO_VREFMRGN |
| 116S0004    | 1   | RES, MTL FILM, 0, 5%, 0402, SM, LF | R2905         | CRITICAL | NO_VREFMRGN |
| 116S0004    | 1   | RES, MTL FILM, 0, 5%, 0402, SM, LF | R2909         | CRITICAL | NO_VREFMRGN |
| 116S0004    | 1   | RES, MTL FILM, 0, 5%, 0402, SM, LF | R2911         | CRITICAL | NO_VREFMRGN |

| FSB/DDR3/FRAMEBUF Vref Margining                                                                                           |  |                      |
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Page Notes

Power aliases required by this page:

- PP1V5\_S0\_MEM\_A
- PP1V5\_S3\_MEM\_A
- PP1V5\_S0\_MEM\_VTT\_A
- PPSPD\_S0\_MEM\_A (2.5 - 3.3V)

Signal aliases required by this page:

- I2C\_S0DIMMA\_SCL
- I2C\_S0DIMMA\_SDA

BOM options provided by this page:

(NONE)

"Factory" (top) slot

DDR3 S0-DIMM Connector A

SYNC\_MASTER=SEN\_K20 SYNC\_DATE=06/10/2008

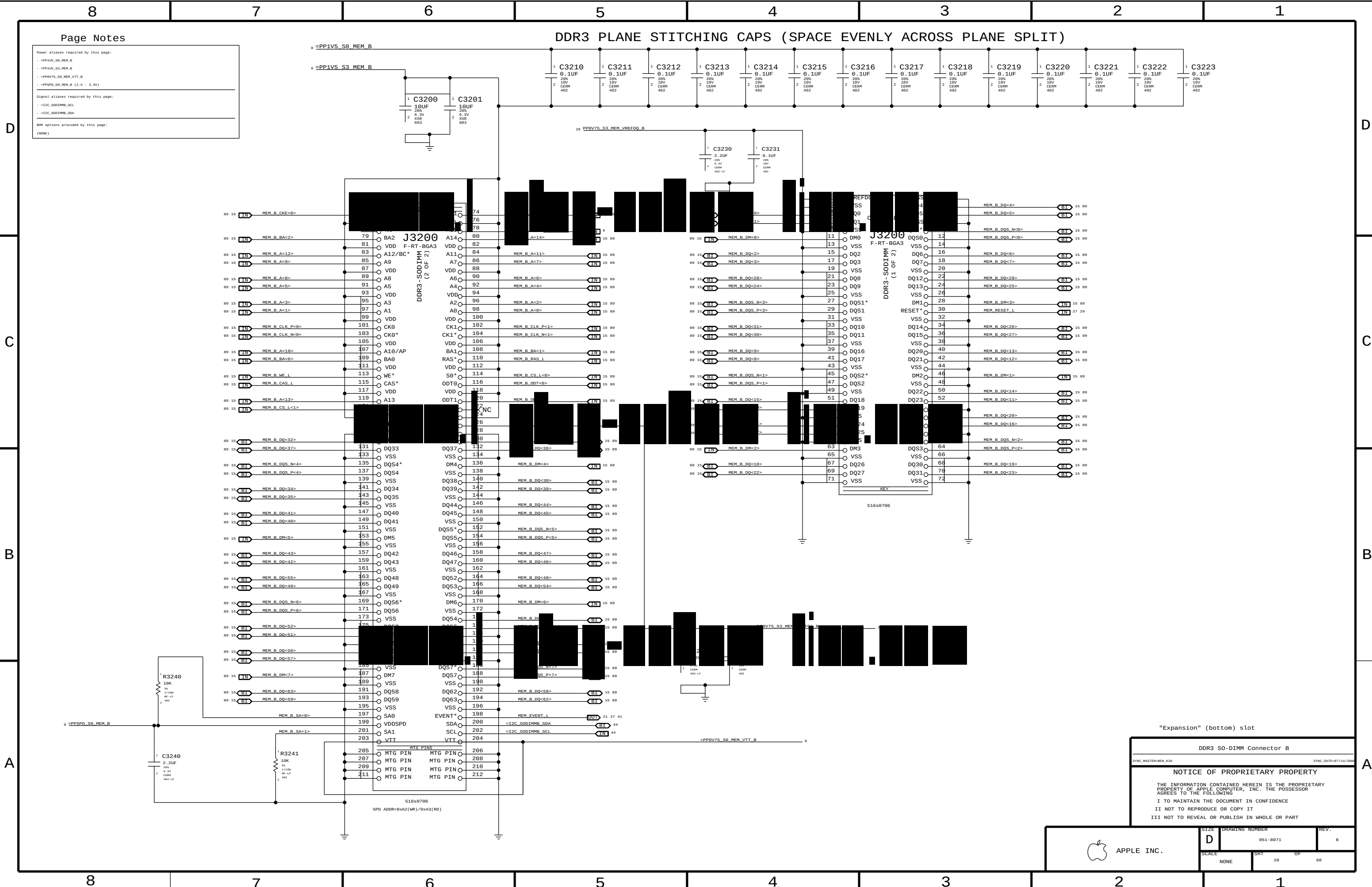
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Page Notes

Power aliases required by this page:  
- =PP1V5\_S0\_MEM\_B  
- =PP1V5\_S3\_MEM\_B  
- =PP0V75\_S0\_MEM\_VTT\_B  
- =PPSPD\_S0\_MEM\_B (2.5 - 3.3V)  
Signal aliases required by this page:  
- =I2C\_S0DIMM\_SCL  
- =I2C\_S0DIMM\_SDA  
BOM notes provided by this page:  
(NONE)

DDR3 PLANE STITCHING CAPS (SPACE EVENLY ACROSS PLANE SPLIT)

"Expansion" (bottom) slot

DDR3 S0-DIMM Connector B

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D 051-8071 B

SCALE: NONE SHY 28 OF 98



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1. **Identify the main topic or subject of the text.**  
 2. **Summarize the key points or findings.**  
 3. **Highlight any specific data or statistics mentioned.**  
 4. **Discuss the implications or significance of the information.**  
 5. **Conclude with a brief statement on the overall message.**

## DDR3 RESET Support

MCP79 cannot control this signal directly since it must be high in sleep and MCP MEM rails are not powered in sleep.



|                     |                      |
|---------------------|----------------------|
| <b>DDR3 Support</b> |                      |
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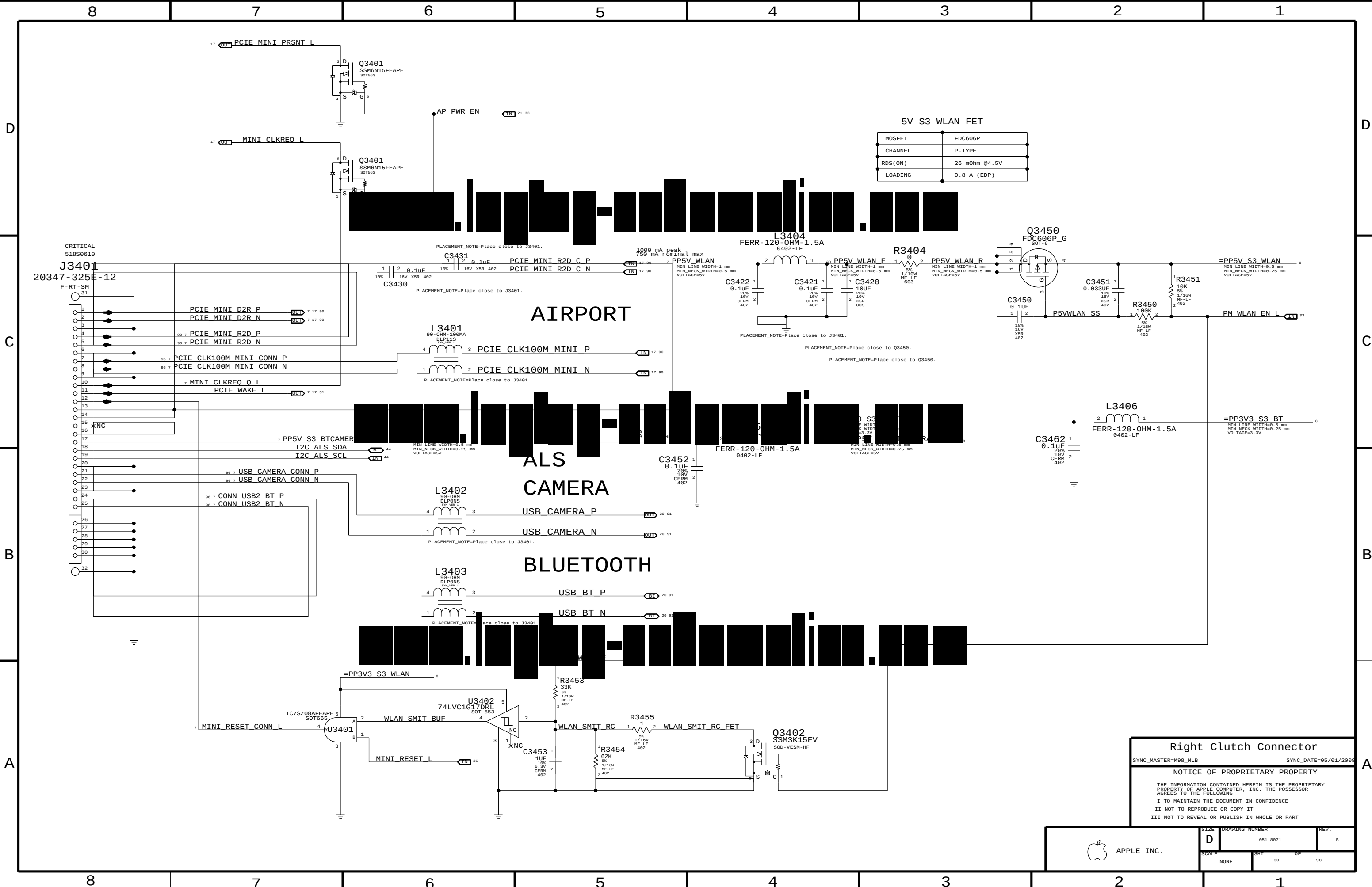
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2



|                |               |
|----------------|---------------|
| 5V S3 WLAN FET |               |
| MOSFET         | FDC606P       |
| CHANNEL        | P-TYPE        |
| RDS(ON)        | 26 mOhm @4.5V |
| LOADING        | 0.8 A (EDP)   |

Right Clutch Connector

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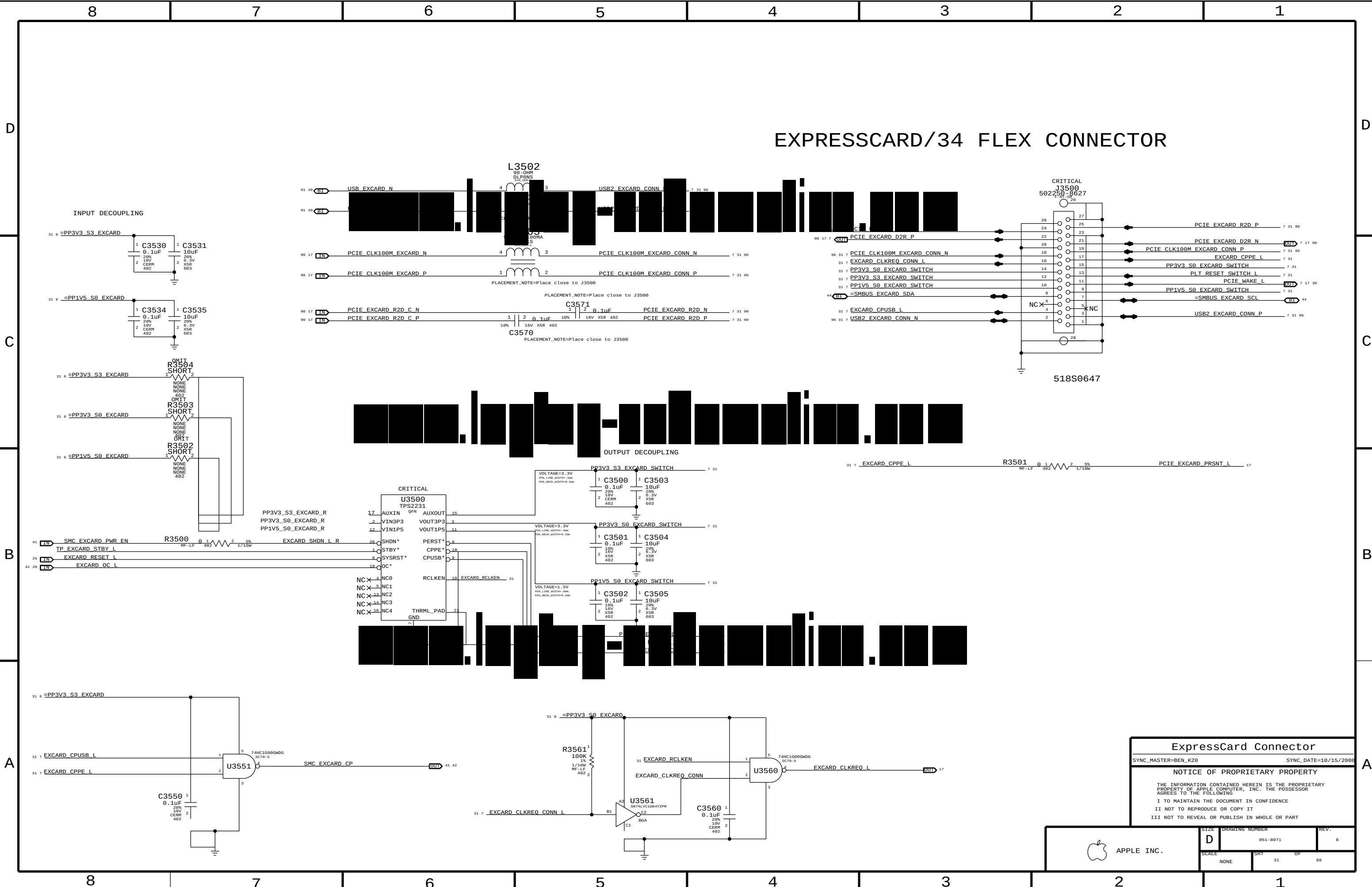
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ExpressCard Connector

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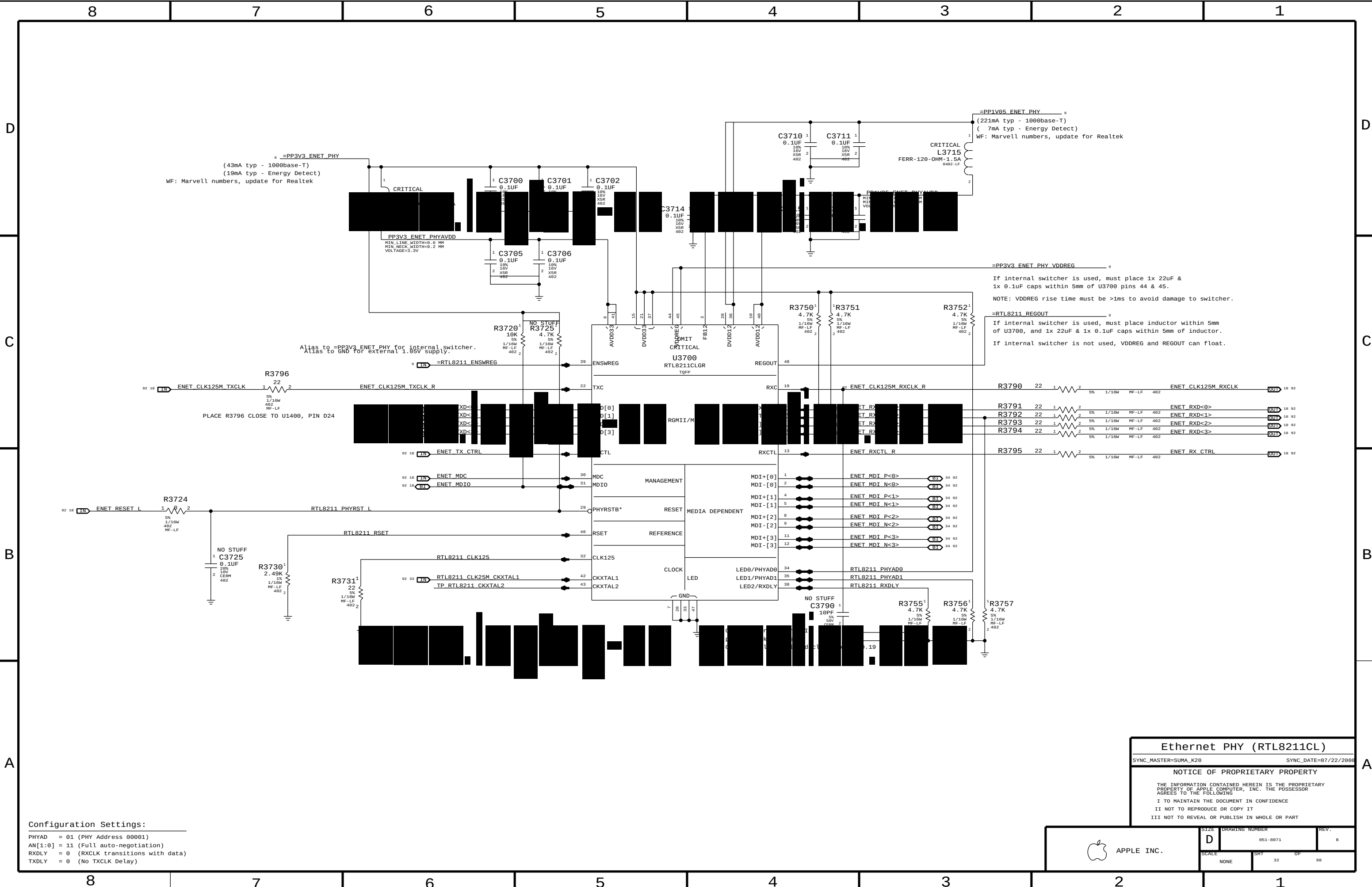
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Configuration Settings:

PHYAD = 01 (PHY Address 00001)  
AN[1:0] = 11 (Full auto-negotiation)  
RXDLY = 0 (RXCLK transitions with data)  
TXDLY = 0 (No TXCLK Delay)

Ethernet PHY (RTL8211CL)

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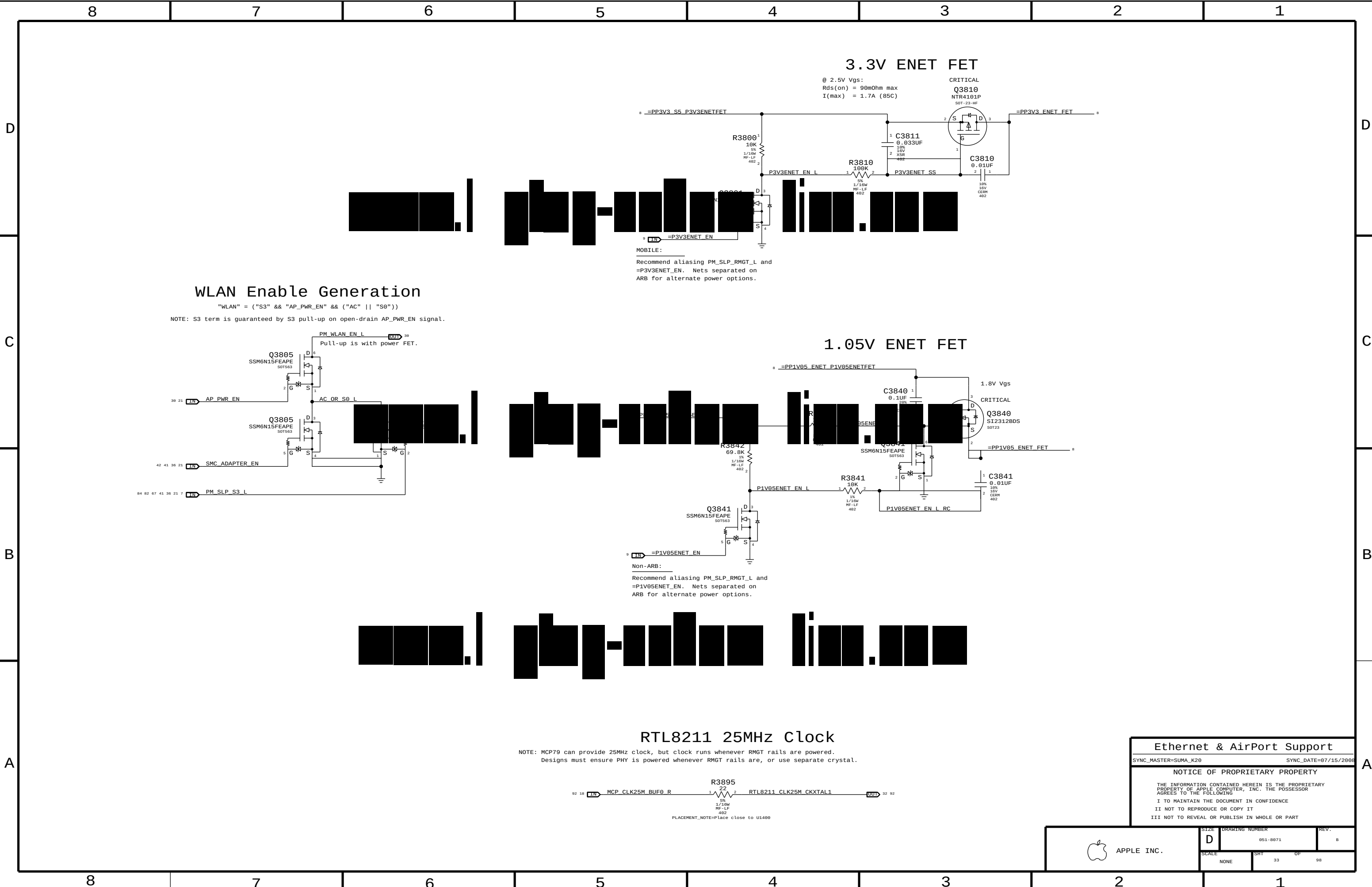
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| APPLE INC. | DRAWING NUMBER |          | REV. |
|            | D              | 051-8071 | B    |
| SCALE      |                | SHT OF   |      |
| NONE       |                | 32 OF 98 |      |





Ethernet & AirPort Support

SYNC\_MASTER=SUMA\_K20

SYNC\_DATE=07/15/2008

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APPLE INC.

SIZE

D

DRAWING NUMBER

051-8071

REV.

B

SCALE

NONE

SHT

33

OF

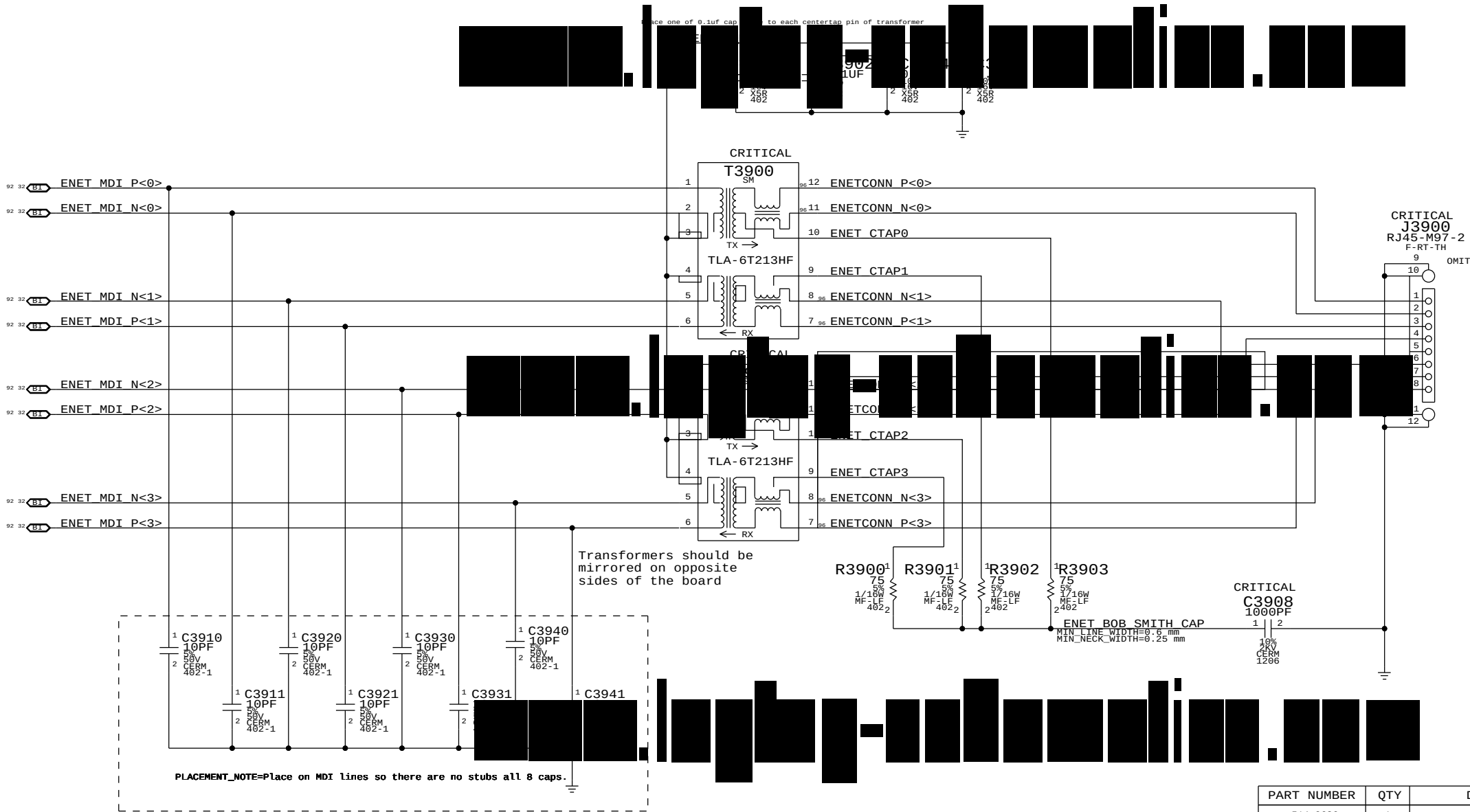
98

Page Notes

Power aliases required by this page:  
(NONE)

Signal aliases required by this page:  
(NONE)

BOM options provided by this page:  
(NONE)



| PART NUMBER | QTY | DESCRIPTION              | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|--------------------------|---------------|----------|------------|
| 514 -0636   | 1   | CONN, RJ45, HB, 10/100TX | J3900         | CRITICAL |            |

Ethernet Connector

SYNC\_MASTER=SUMA\_K20 SYNC\_DATE=07/15/2008

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APPLE INC.

SIZE: DRAWING NUMBER: REV.

D 051-8071 B

SCALE: NONE SHT 34 OF 98



## Page Notes

Power aliases required by this page:

- =PPBUS\_S5\_FWPWRSW (system supply for bus power)
- =PP3V3\_FW\_LATEVG\_ACTIVE
- =PPVP\_FW\_SUMNODE (power passthru summation node)

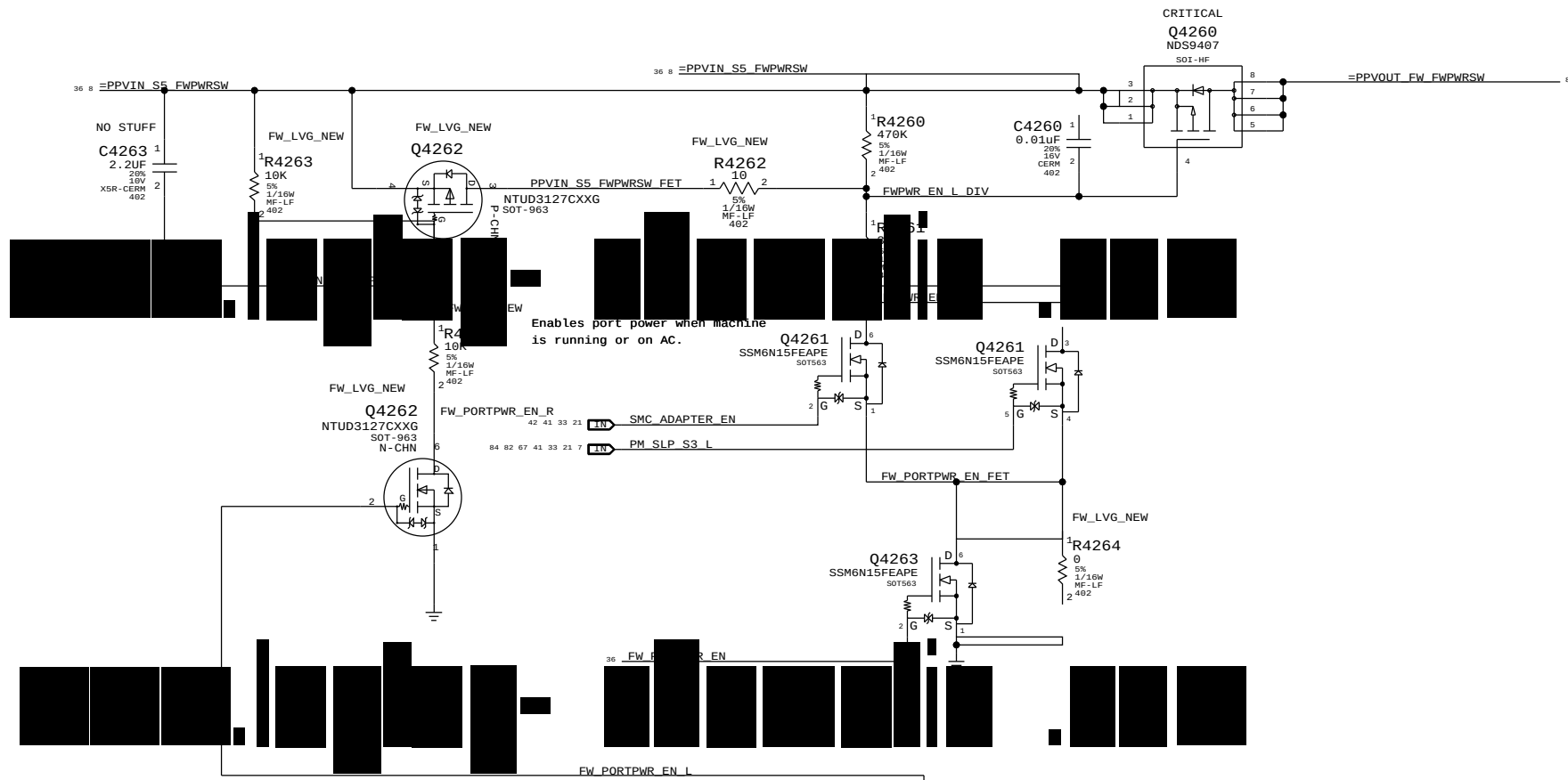
Signal aliases required by this page:

(NONE)

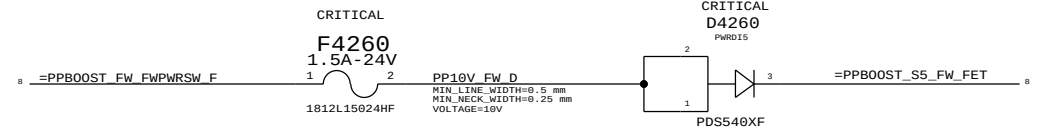
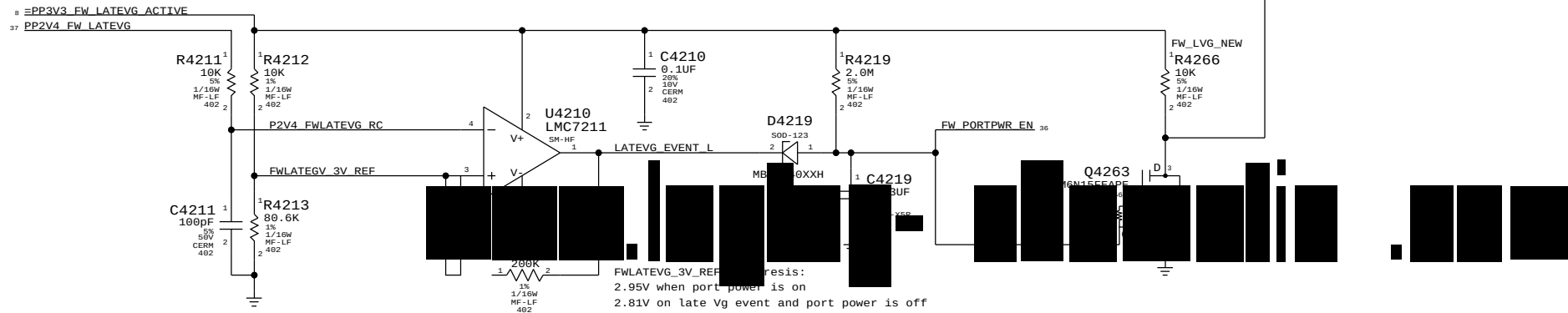
BOM options provided by this page:

- FW\_PORT\_FAULT\_PU

## FireWire Port Power Switch



## Late-VG Event Detection



## FireWire Port Power

SYNC\_MASTER=YWU\_K20 SYNC\_DATE=05/28/2008

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SIZE DRAWING NUMBER REV.

D 051-8071 B

SCALE SRT OF 98

## Page Notes

Power aliases required by this page:

- =PPVP\_FW\_PORT1
- =PP3V3\_FW\_LATEVG

Signal aliases required by this page:

(NONE)

NOTE: This page is expected to contain the necessary aliases to map the FireWire TPA/TPB pairs to their appropriate connectors and/or to properly terminate unused signals.

BOM options provided by this page:

(NONE)

NOTE: FireWire TPA/TPB pairs are NOT constrained on this page. It is assumed that FireWire PHY page will provide the appropriate constraints to apply to entire TPA/TPB XNets.

1394b implementation based on Apple FireWire Design Guide (FWDG 0.6, 5/14/03)

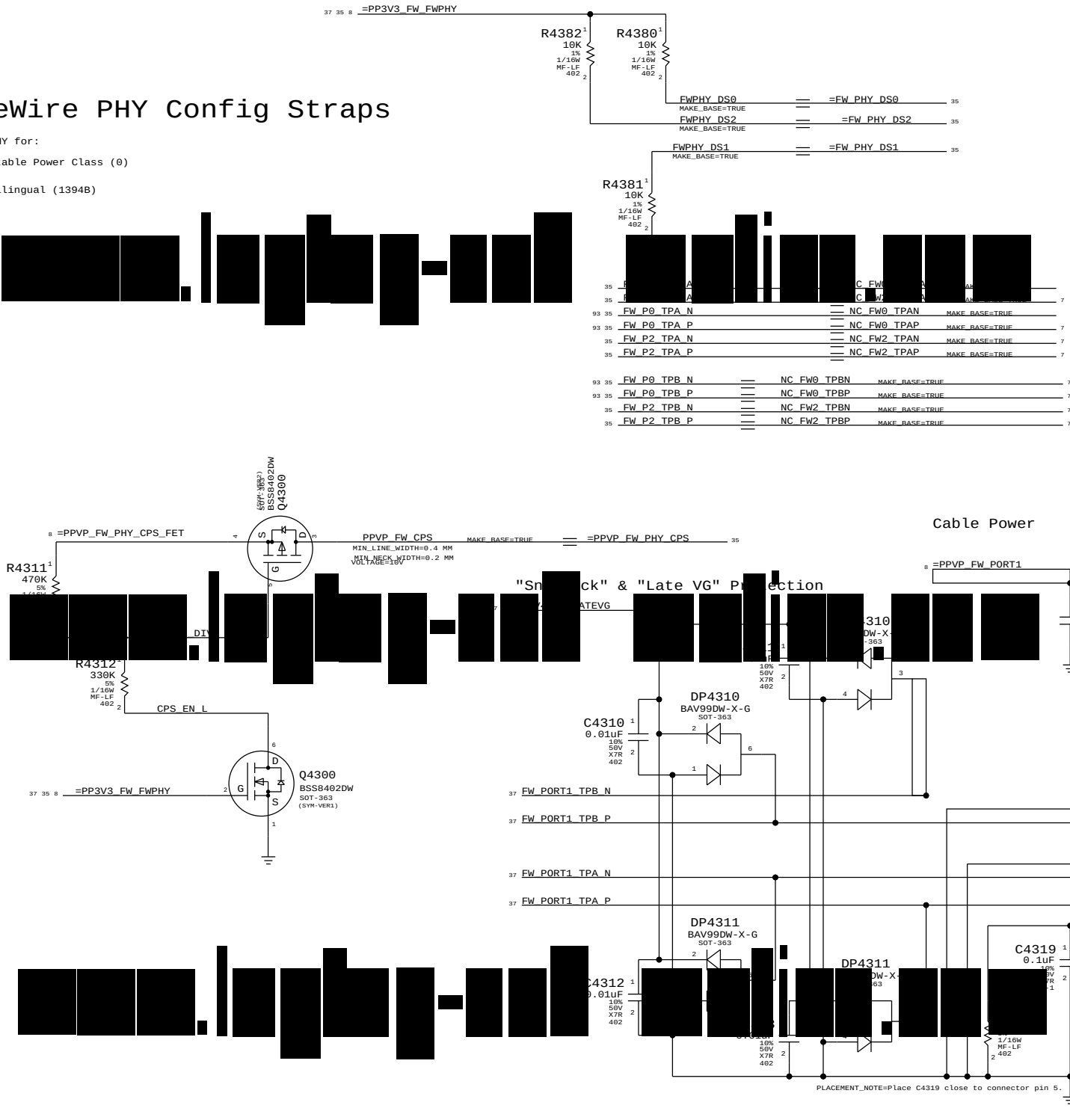
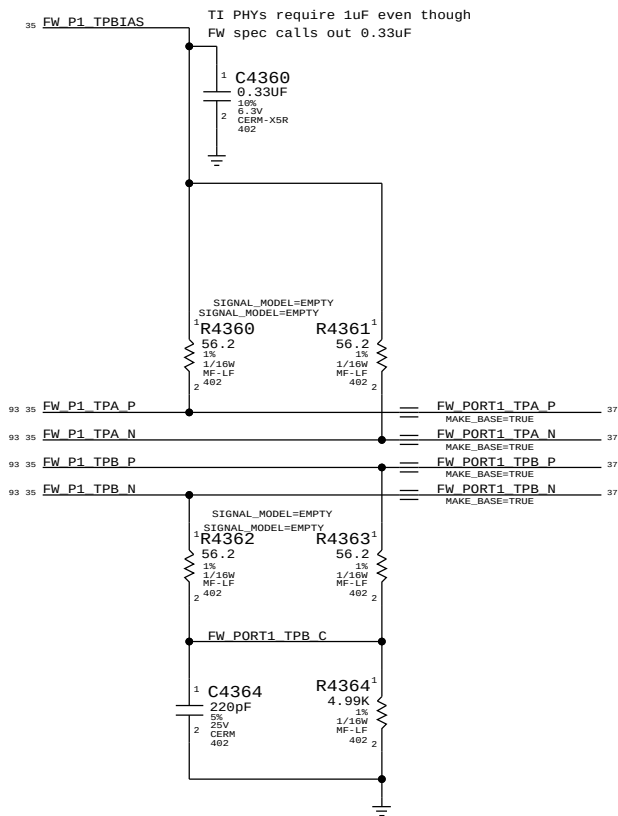
## FireWire PHY Config Straps

Configures PHY for:

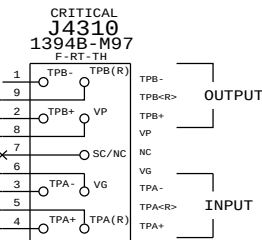
- 1-port Portable Power Class (0)
- Port "1" Bilingual (1394B)

## Termination

Place close to FireWire PHY

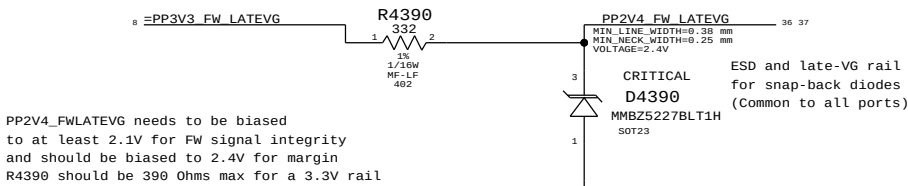


## PORT 1 BILINGUAL



514S0605

## Late-VG Protection Power



## FireWire Ports

SYNC\_MASTER=M9B\_MLB SYNC\_DATE=07/14/2008

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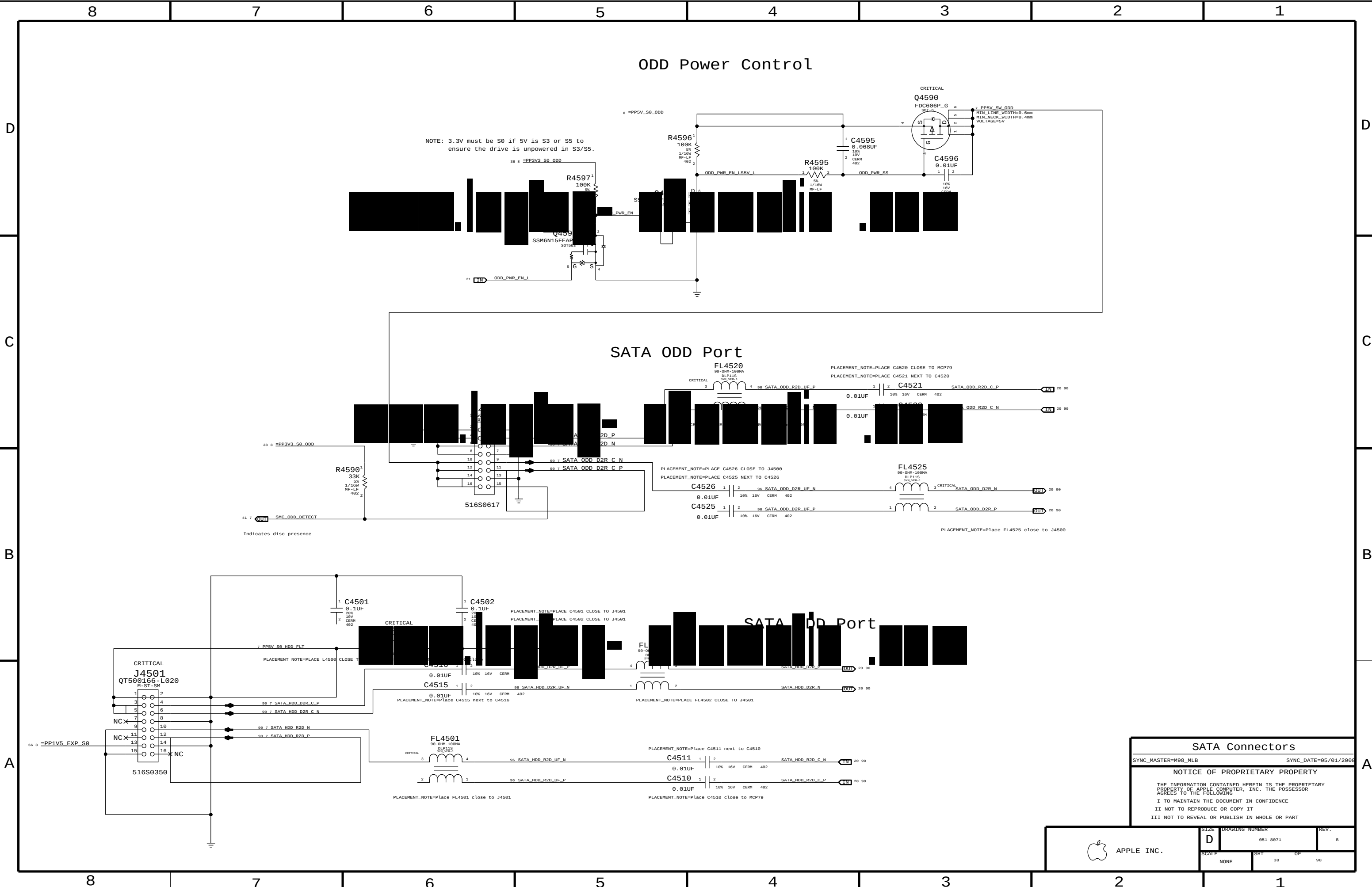
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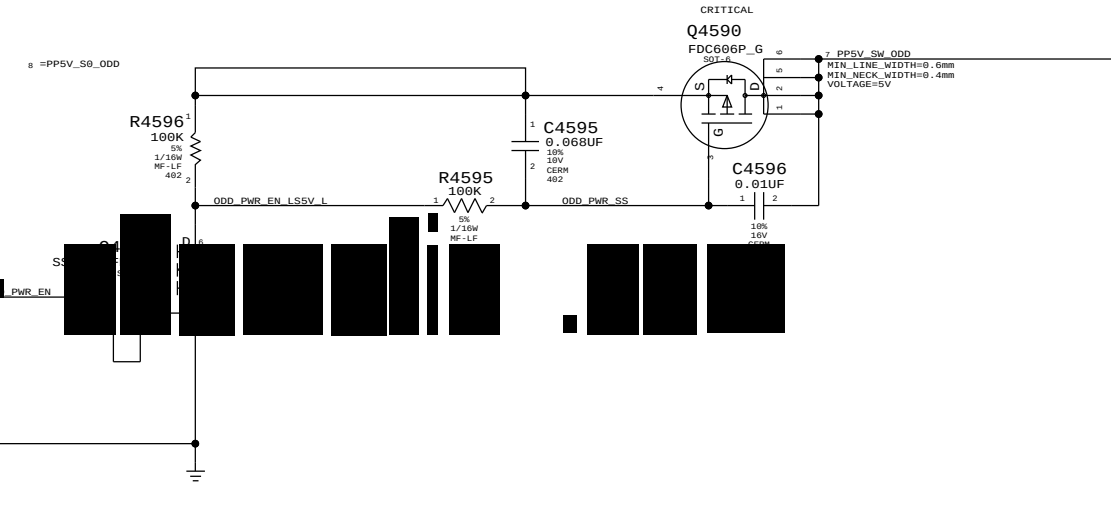
APPLE INC.

SIZE D DRAWING NUMBER 051-8071 REV. B

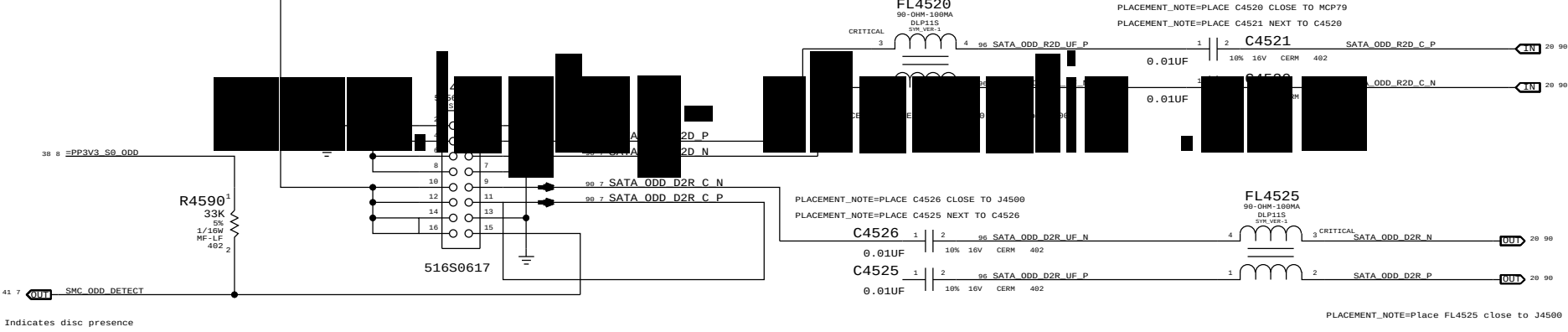
SCALE NONE SHT 37 OF 98



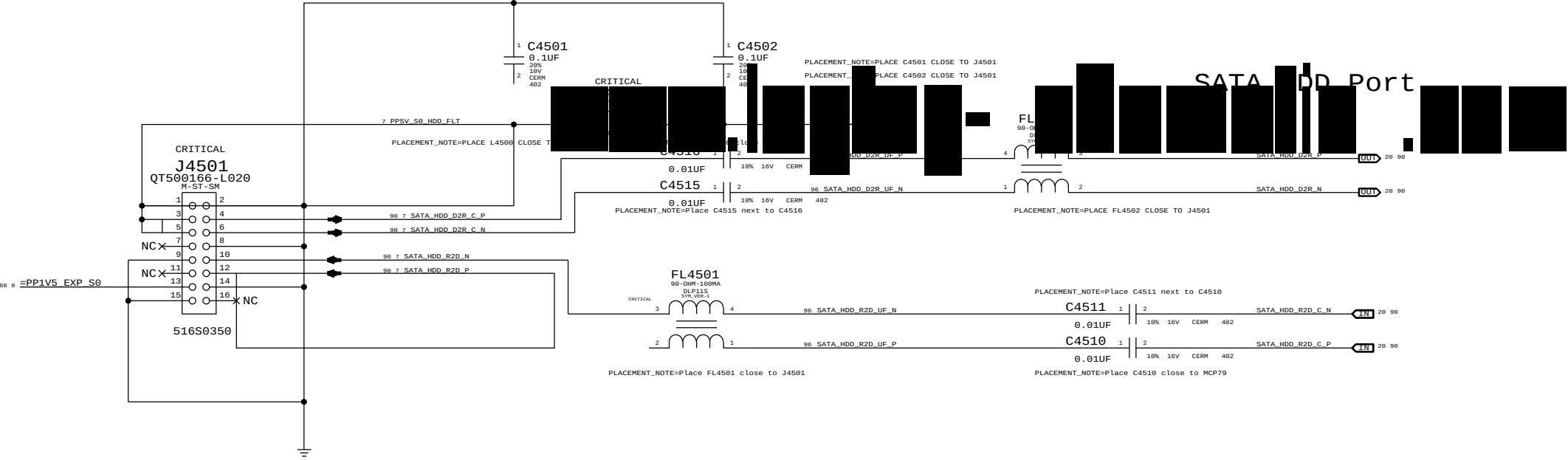
ODD Power Control



SATA ODD Port



SATA HDD Port



SATA Connectors

SYNC\_MASTER=M9B\_MLB      SYNC\_DATE=05/01/2008

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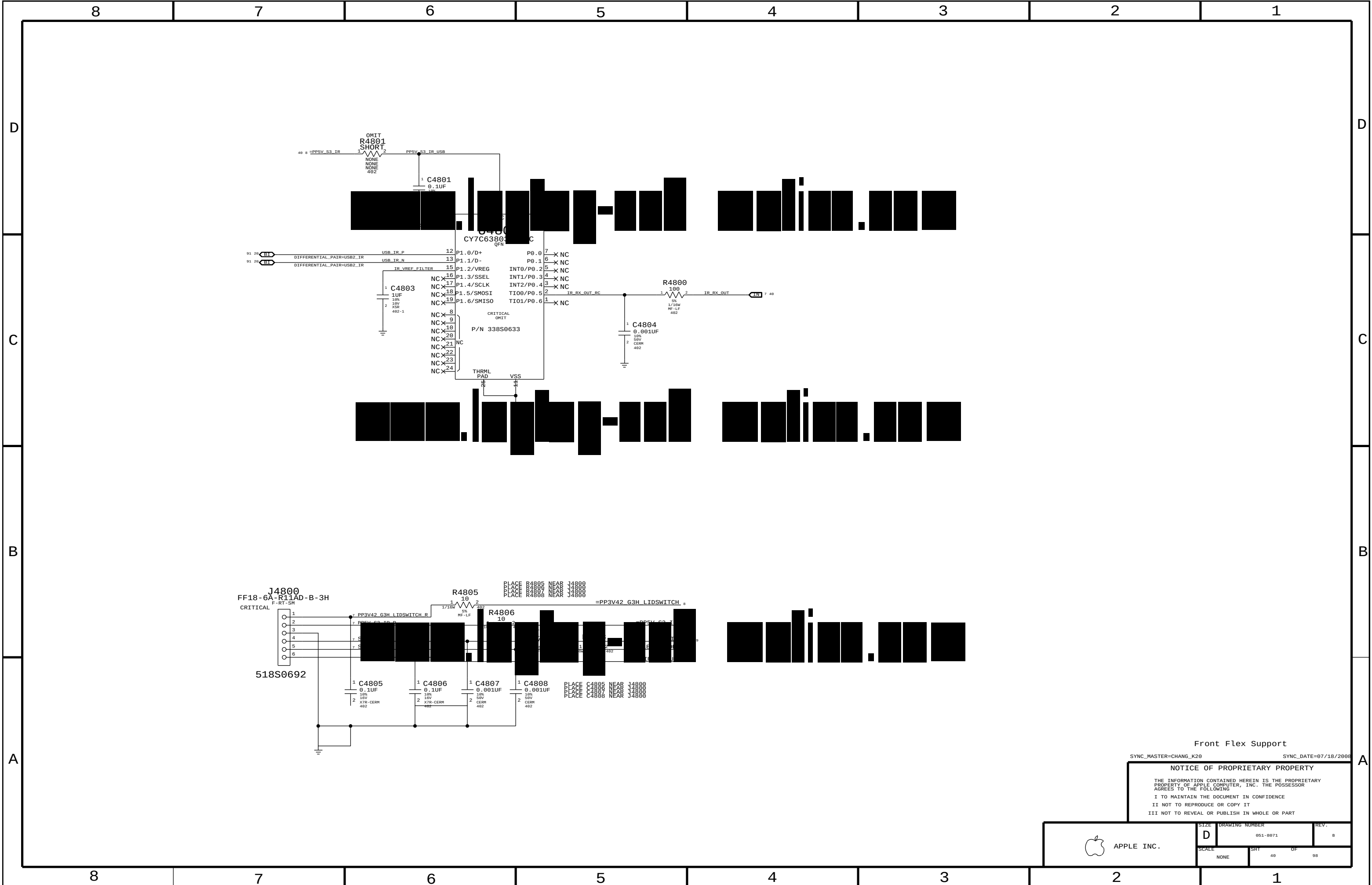
APPLE INC.

SIZE: D      DRAWING NUMBER: 051-8071      REV.: B

SCALE: NONE      SHEET: 38 OF 98







Front Flex Support

SYNC\_MASTER=CHANG\_K20 SYNC\_DATE=07/18/2009

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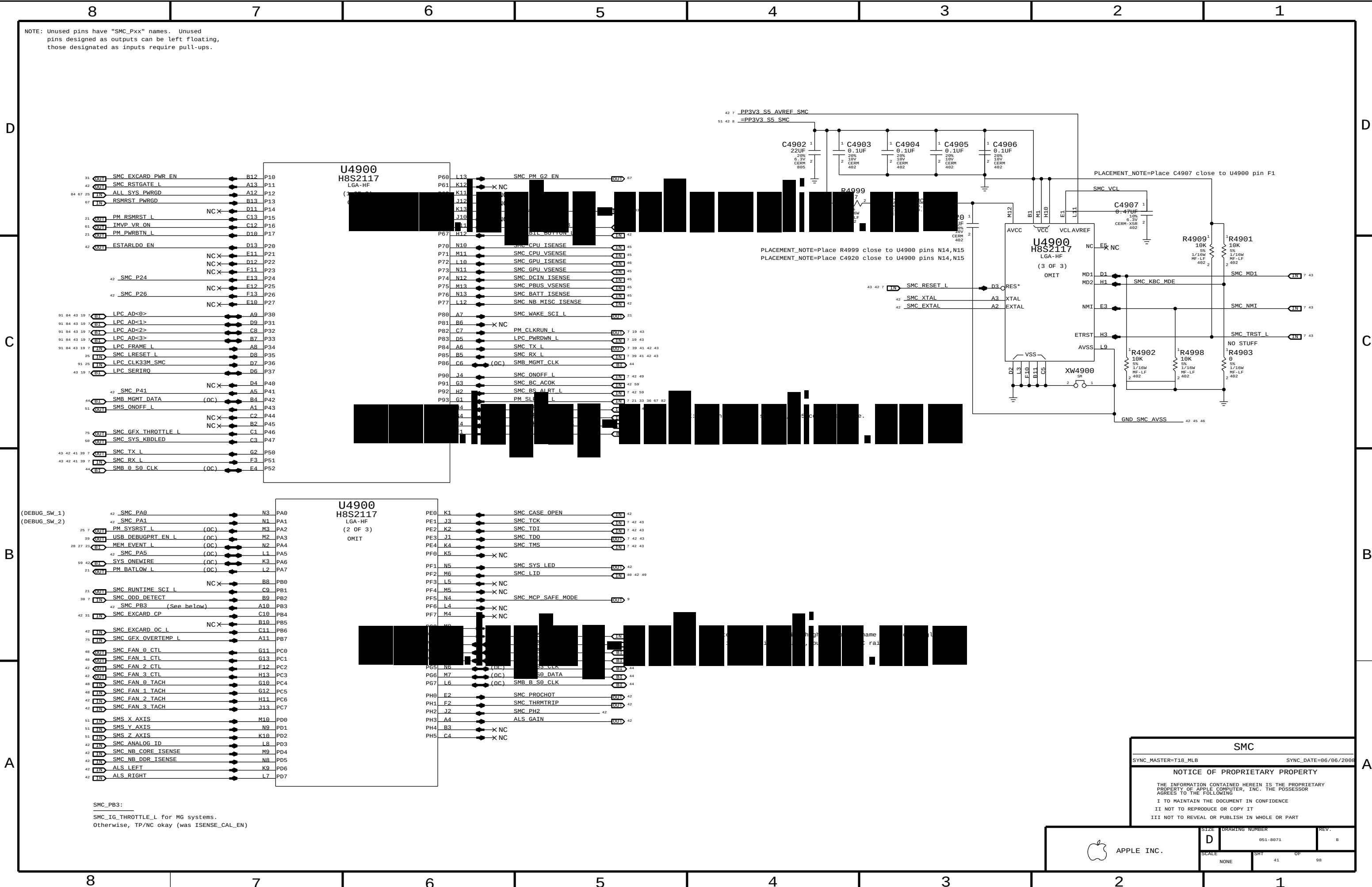
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| SCALE      |                | SHEET OF |      |
| NONE       |                | 40 OF 98 |      |





NOTE: Unused pins have "SMC\_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.

SMC

SYNC\_MASTER=T18\_MLB

SYNC\_DATE=06/06/2009

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APPLE INC.

SIZE: DRAWING NUMBER: 051-8071

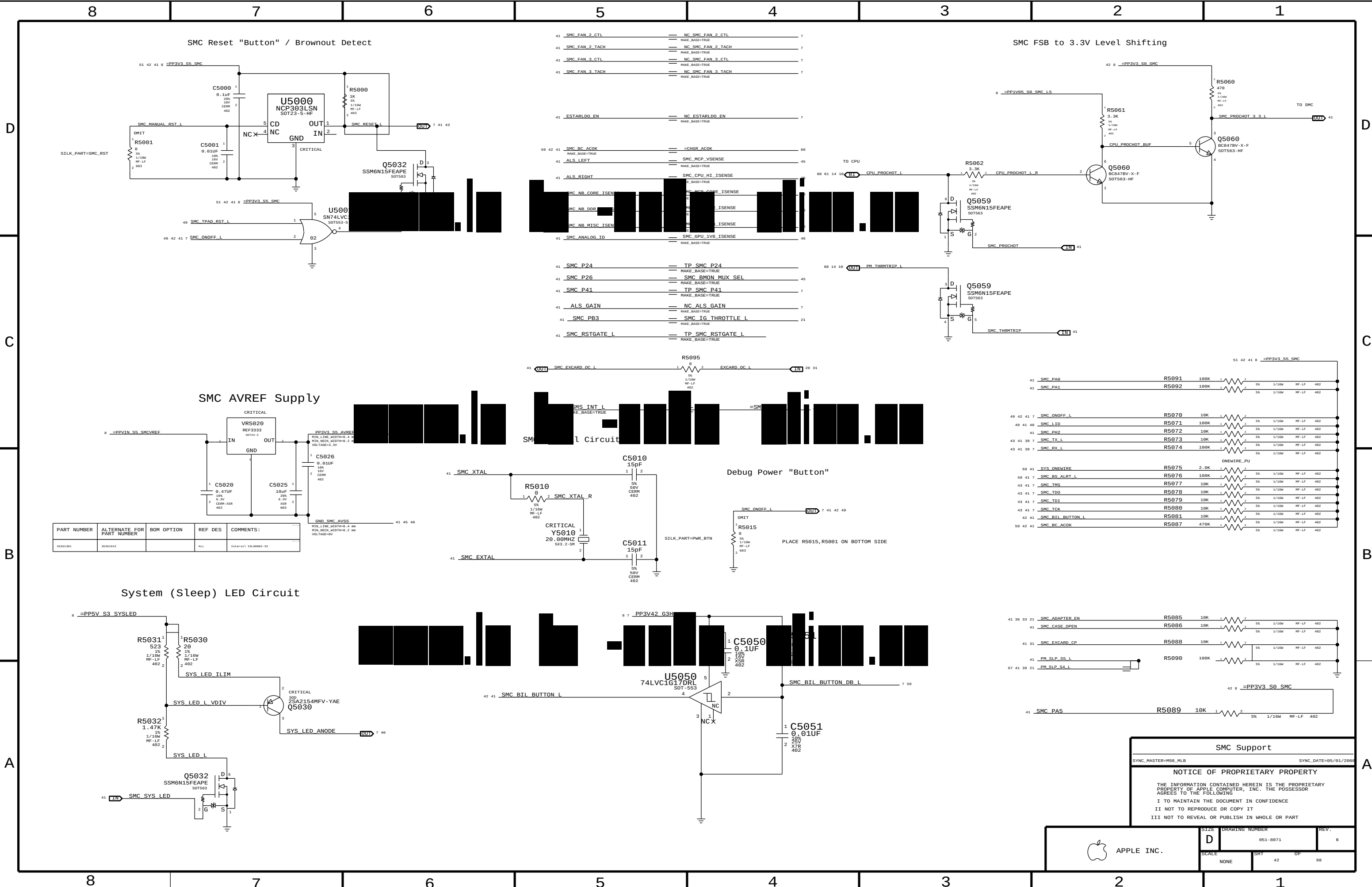
REV: B

SCALE: NONE

SHT: 41

OF: 98

SMC\_PB3:  
SMC\_IG\_THROTTLE\_L for MG systems.  
Otherwise, TP/NC okay (was ISENSE\_CAL\_EN)



SMC Support

SYNC\_MASTER=M98\_MLB SYNC\_DATE=05/01/2008

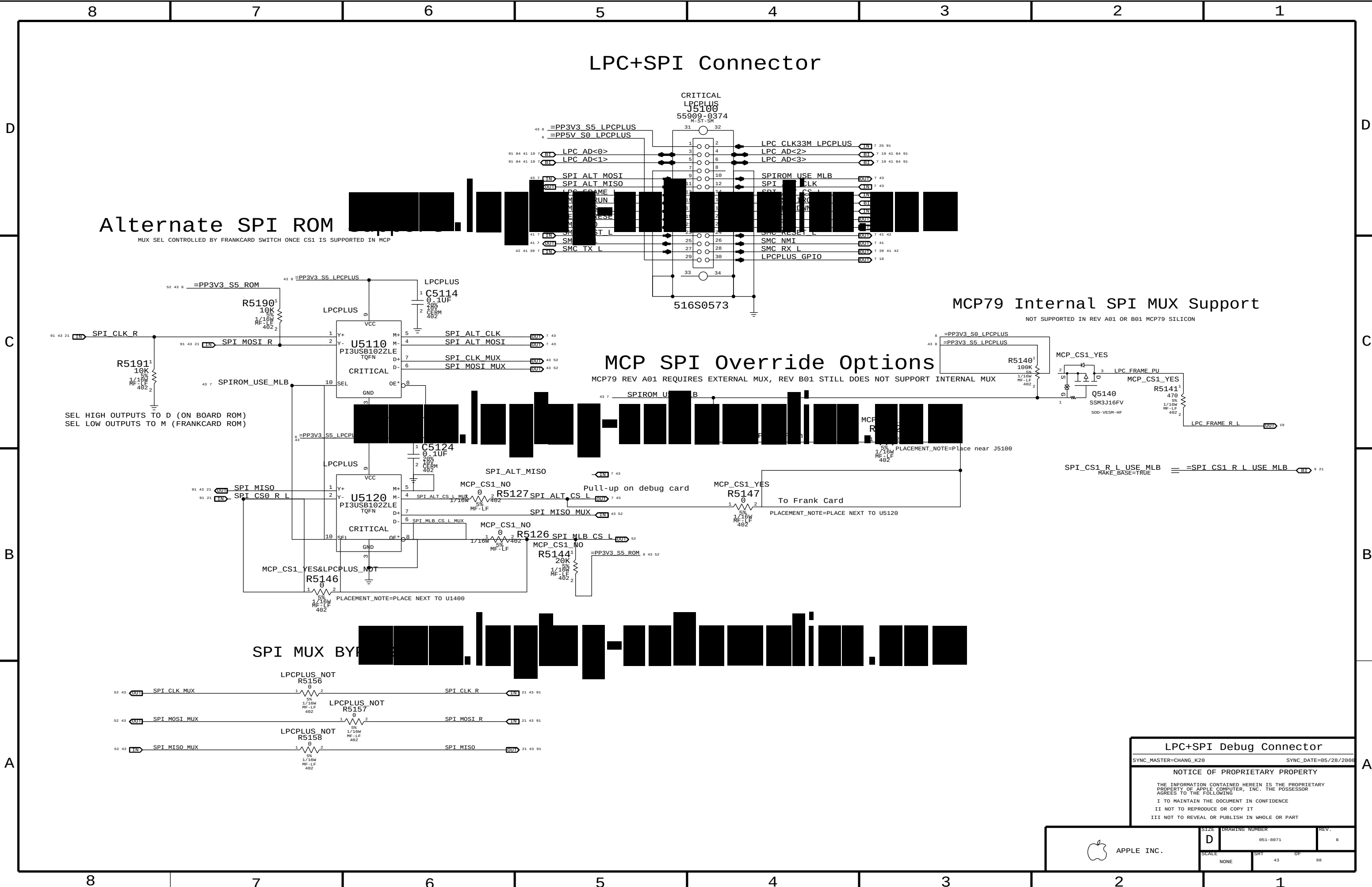
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# LPC+SPI Connector

## Alternate SPI ROM

MUX SEL CONTROLLED BY FRANKCARD SWITCH ONCE CS1 IS SUPPORTED IN MCP

## MCP79 Internal SPI MUX Support

NOT SUPPORTED IN REV A01 OR B01 MCP79 SILICON

## MCP SPI Override Options

MCP79 REV A01 REQUIRES EXTERNAL MUX, REV B01 STILL DOES NOT SUPPORT INTERNAL MUX

## SPI MUX BY

### LPC+SPI Debug Connector

SYNC\_MASTER=CHANG\_K20 SYNC\_DATE=05/28/2008

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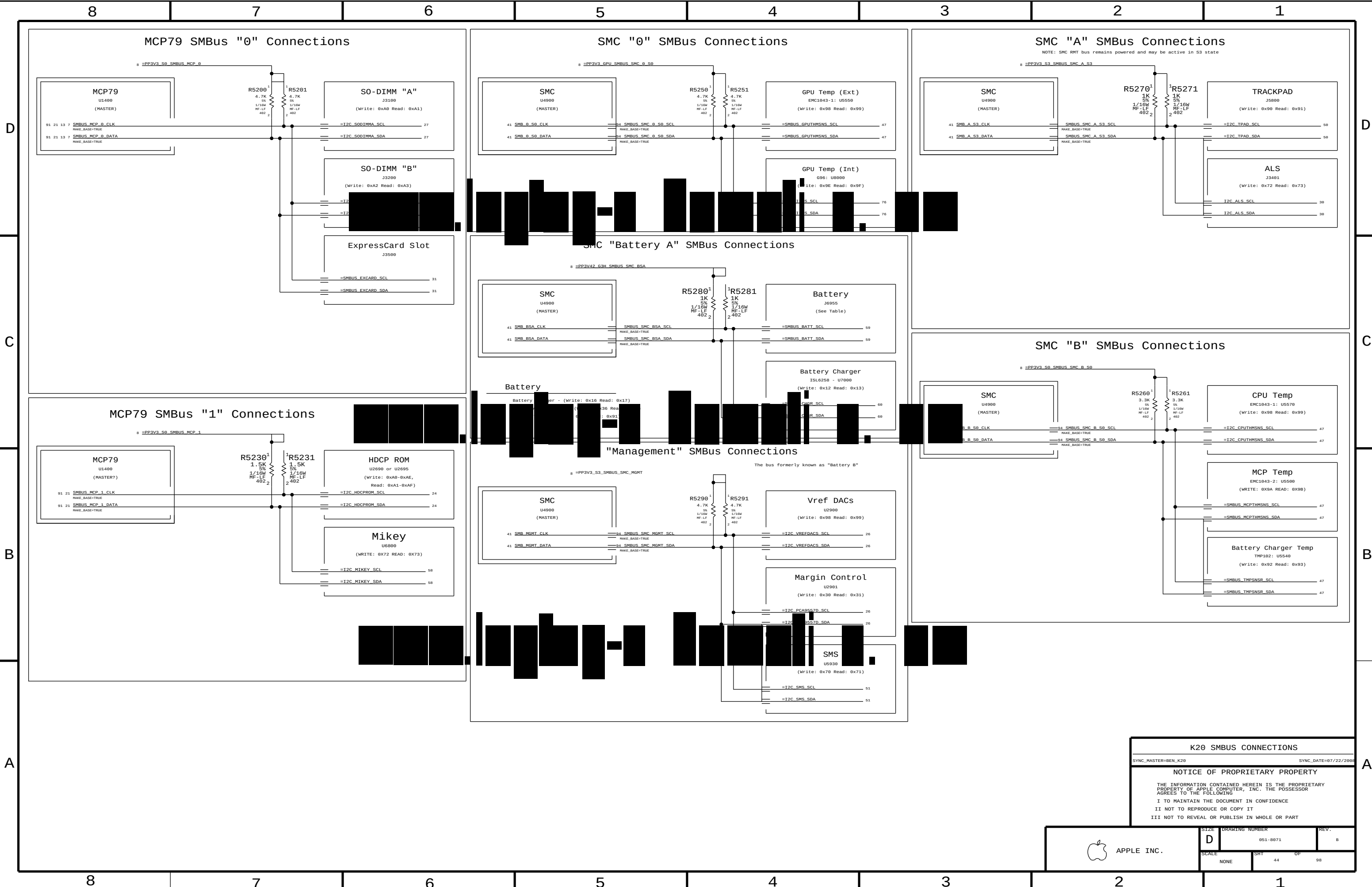
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SIZE: DRAWING NUMBER: REV.

D 051-8071 B

SCALE: NONE SRT 43 OF 98



**K20 SMBUS CONNECTIONS**

SYNC\_MASTER=BEN\_K20 SYNC\_DATE=07/22/2008

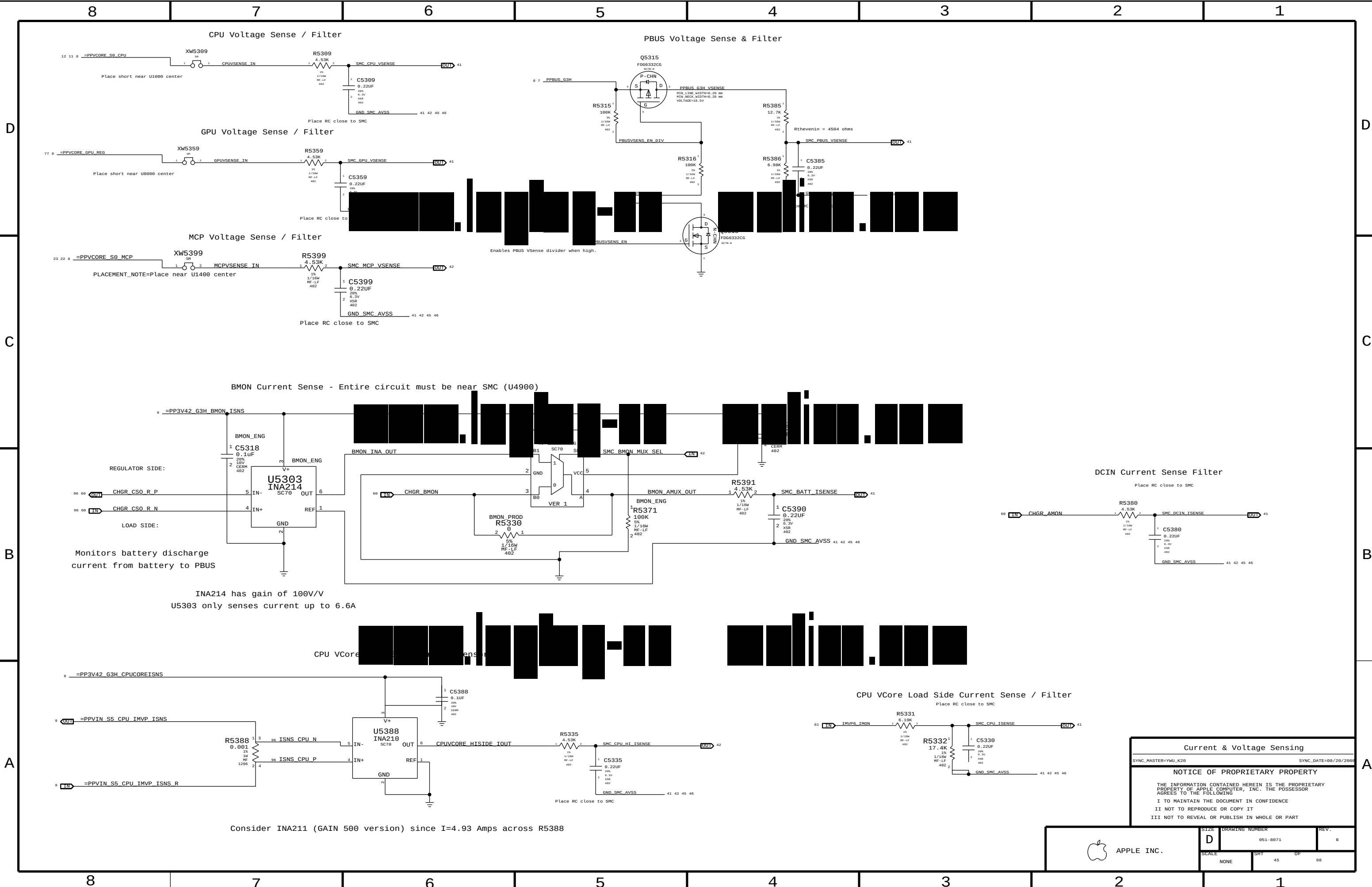
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CPU Voltage Sense / Filter

PBUS Voltage Sense & Filter

GPU Voltage Sense / Filter

MCP Voltage Sense / Filter

BMON Current Sense - Entire circuit must be near SMC (U4900)

DCIN Current Sense Filter

CPU VCore Load Side Current Sense / Filter

Current & Voltage Sensing

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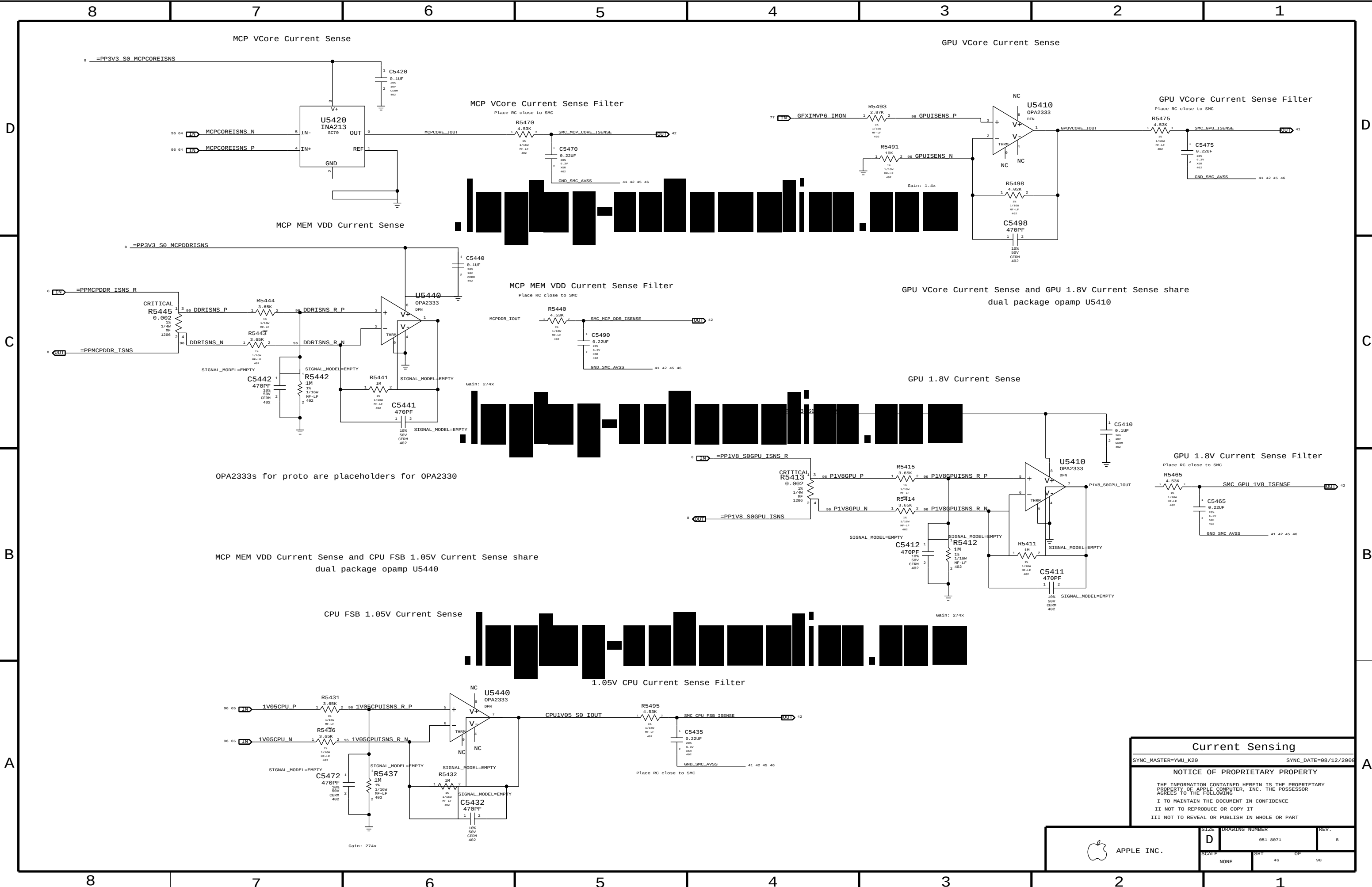
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| NONE           | 45       | 98   |



Current Sensing

SYNC\_MASTER=YWU\_K20 SYNC\_DATE=08/12/2008

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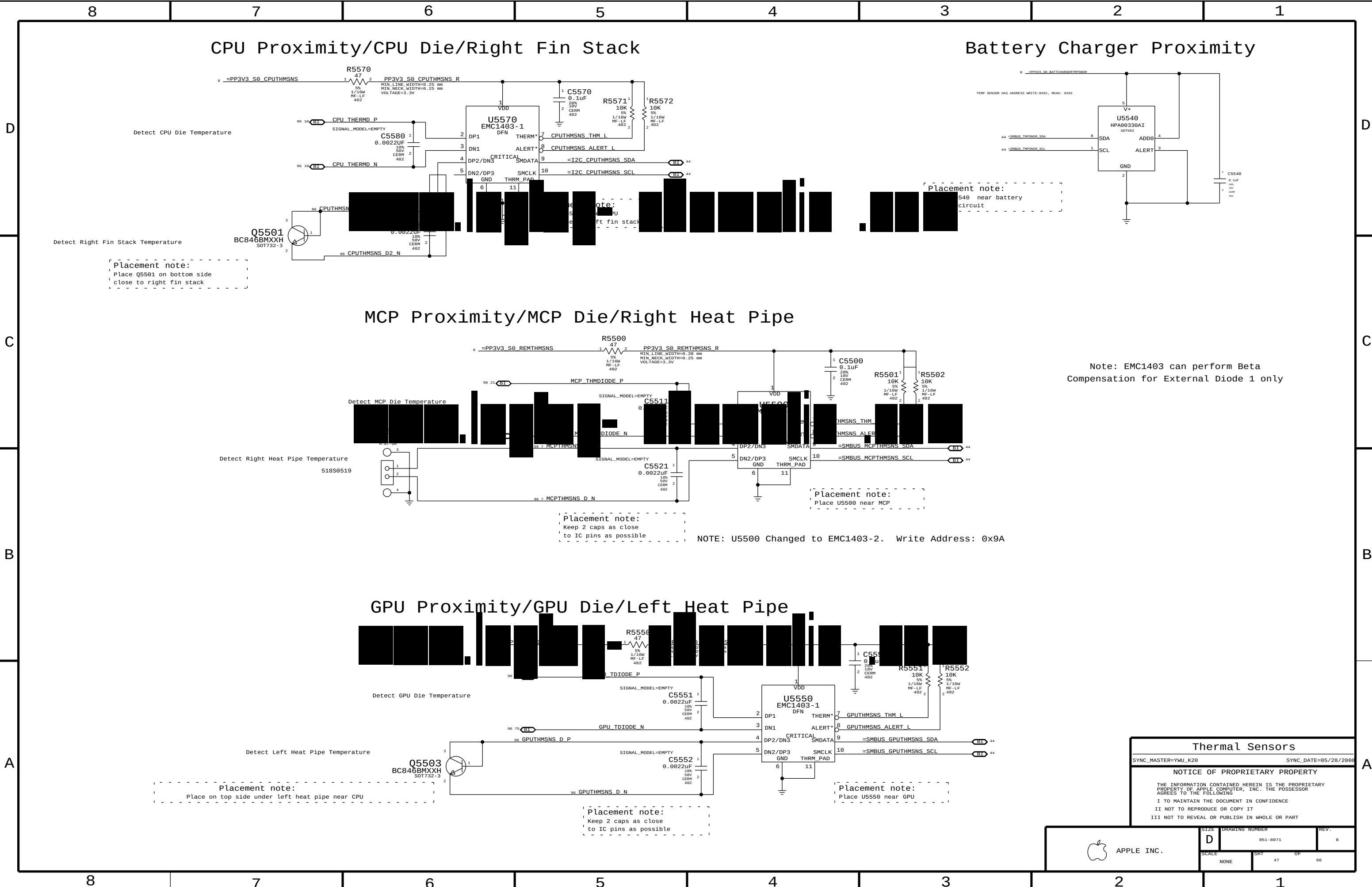
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|-------|----------------|----------|------|
| SCALE | DRAWING NUMBER |          | REV. |
|       | D              | 051-8071 |      |
| NONE  |                | SHY      | OF   |
|       |                | 46       | 98   |







CPU Proximity/CPU Die/Right Fin Stack

Battery Charger Proximity

MCP Proximity/MCP Die/Right Heat Pipe

GPU Proximity/GPU Die/Left Heat Pipe

Note: EMC1403 can perform Beta Compensation for External Diode 1 only

| Thermal Sensors                                                                                                            |  |                      |
|----------------------------------------------------------------------------------------------------------------------------|--|----------------------|
| SYNC_MASTER=YWU_K20                                                                                                        |  | SYNC_DATE=05/28/2008 |
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|----------------|--|------|
| D              |  | B    |
| SCALE          |  | OF   |
| NONE           |  | 98   |



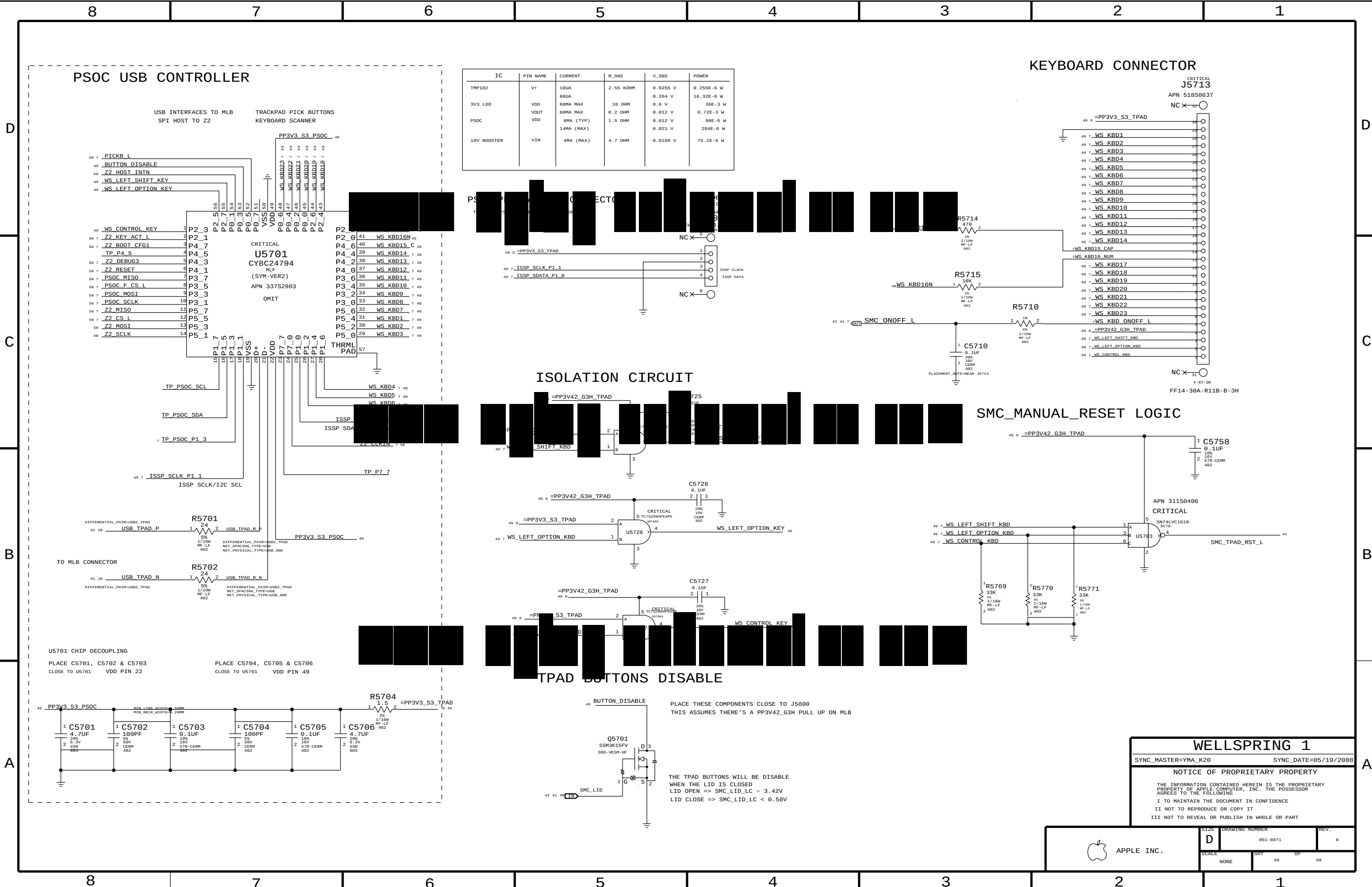
APPLE INC.



| Fan Connectors                                                                                                             |                |                      |
|----------------------------------------------------------------------------------------------------------------------------|----------------|----------------------|
| SYNC_MASTER=M9B_MLB                                                                                                        |                | SYNC_DATE=04/01/2008 |
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| SIZE                                                                                                                       | DRAWING NUMBER | REV.                 |
| D                                                                                                                          | 051-8071       | B                    |
| SCALE                                                                                                                      | SHT OF         |                      |
| NONE                                                                                                                       | 48 OF 98       |                      |







| IC          | PIN NAME | CURRENT                 | R_SNS     | V_SNS    | POWER               |
|-------------|----------|-------------------------|-----------|----------|---------------------|
| TMP102      | V+       | 10UA                    | 2.55 KOHM | 0.0255 V | 0.255E-6 W          |
| 3V3 LDO     | VDD      | 98UA                    | 10 OHM    | 0.204 V  | 16.32E-6 W          |
|             | VOUT     | 68MA MAX                | 0.2 OHM   | 0.012 V  | 0.72E-3 W           |
| PSOC        | VDD      | 8MA (TYP)<br>14MA (MAX) | 1.5 OHM   | 0.012 V  | 96E-6 W<br>294E-6 W |
| 18V BOOSTER | VIN      | 4MA (MAX)               | 4.7 OHM   | 0.0188 V | 75.2E-6 W           |

KEYBOARD CONNECTOR

CRITICAL  
J5713  
APN 518S0637  
NC X 32

ISOLATION CIRCUIT

SMC\_MANUAL\_RESET LOGIC

TPAD BUTTONS DISABLE

WELLSPRING 1

SYNC\_MASTER=YMA\_K20 SYNC\_DATE=05/19/2008

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SIZE DRAWING NUMBER

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REV.

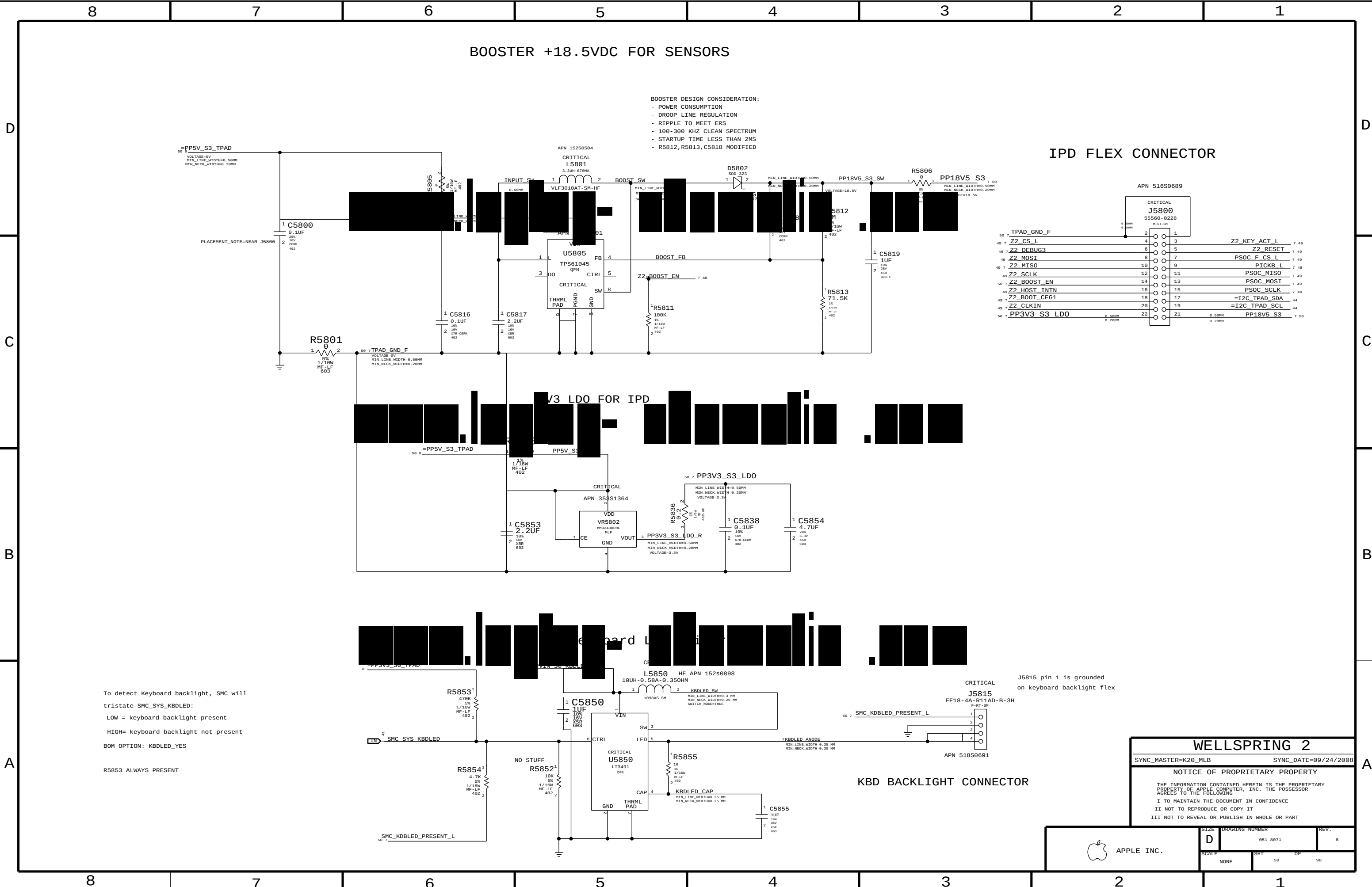
B

SCALE

SHT

OF

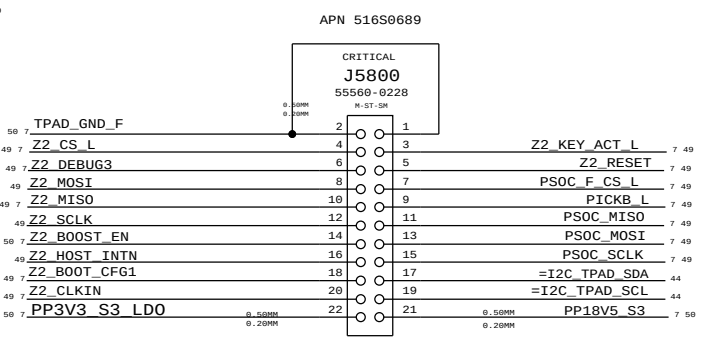
98



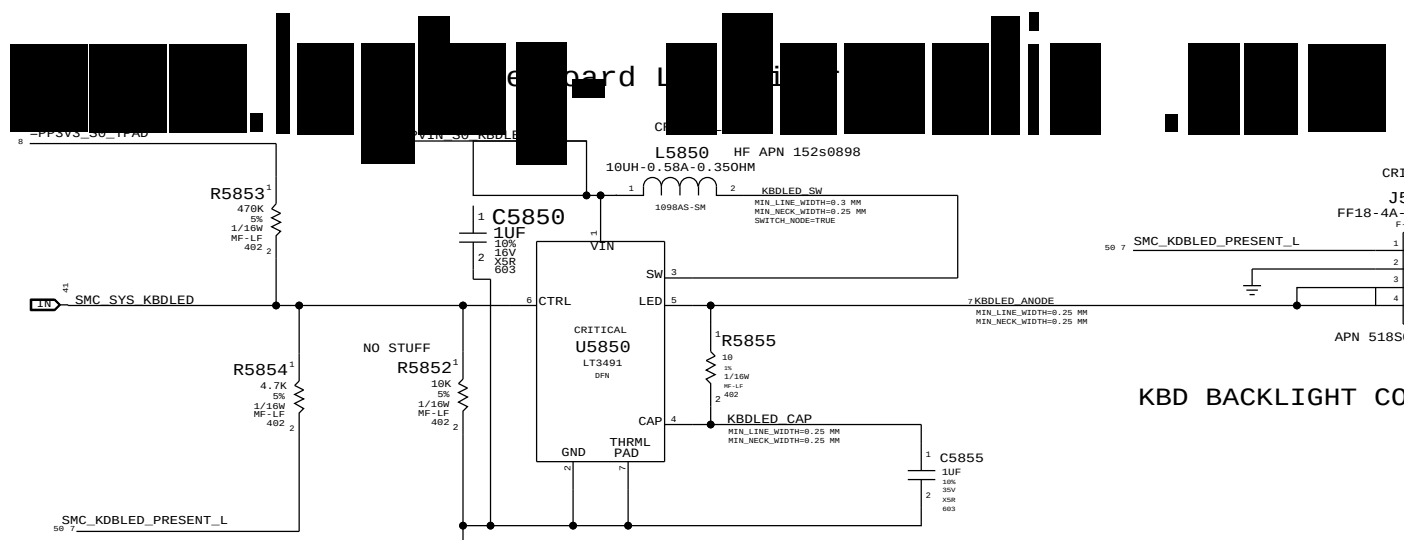
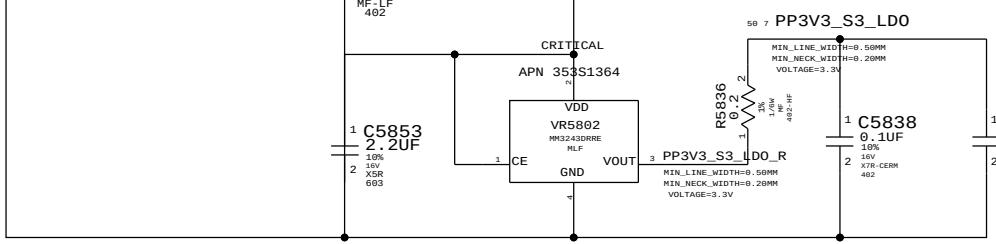
BOOSTER DESIGN CONSIDERATION:

- POWER CONSUMPTION
- DROOP LINE REGULATION
- RIPPLE TO MEET ERS
- 100-300 KHZ CLEAN SPECTRUM
- STARTUP TIME LESS THAN 2MS
- R5812, R5813, C5818 MODIFIED

IPD FLEX CONNECTOR



PP3V3\_LDO FOR IPD



To detect Keyboard backlight, SMC will tristate SMC\_SYS\_KBDLED:  
LOW = keyboard backlight present  
HIGH= keyboard backlight not present  
BOM OPTION: KBDLED\_YES  
R5853 ALWAYS PRESENT

J5815 pin 1 is grounded on keyboard backlight flex

KBD BACKLIGHT CONNECTOR

WELLSPRING 2

SYNC\_MASTER=K20\_MLB      SYNC\_DATE=09/24/2008

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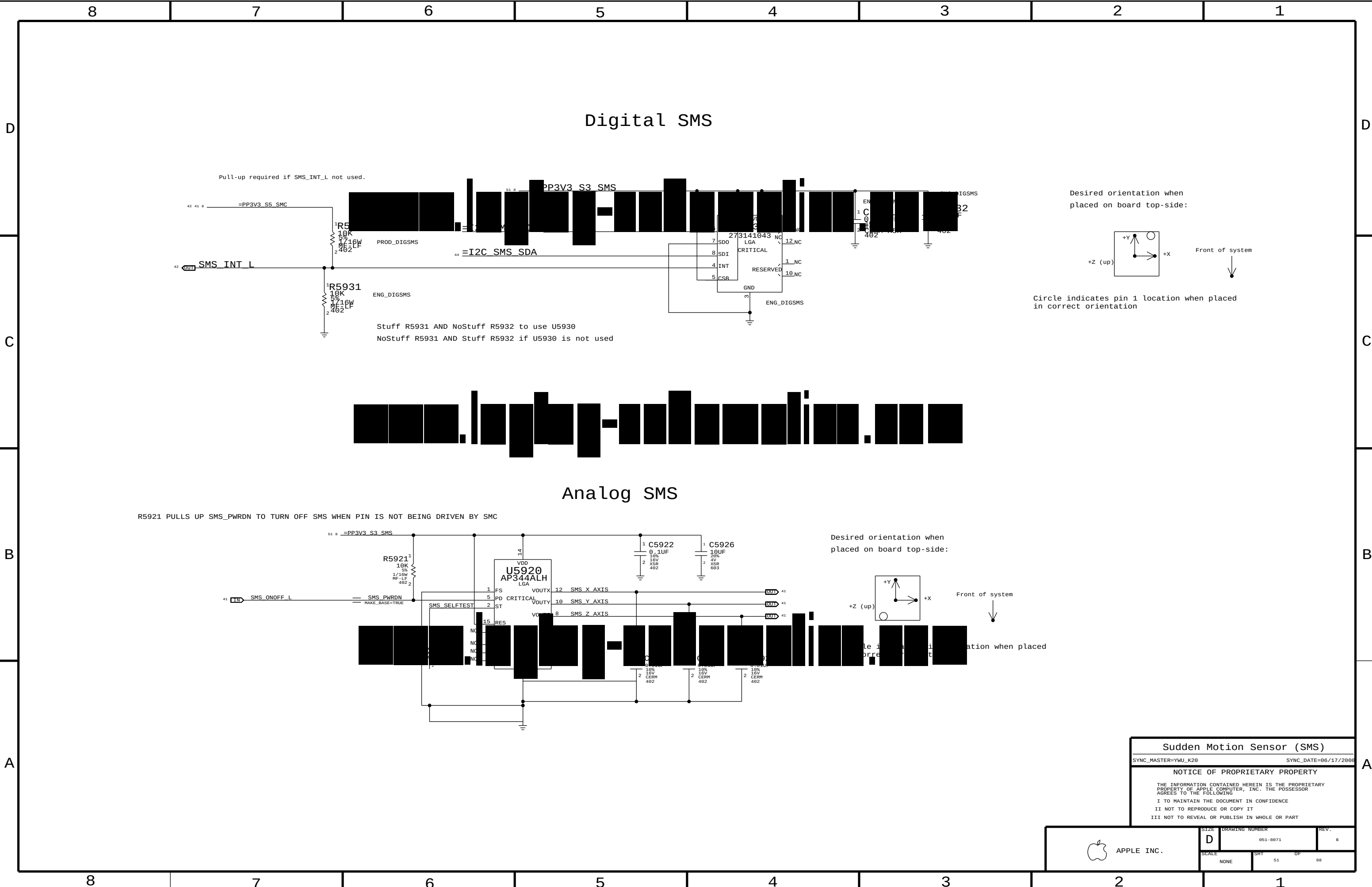
DRAWING NUMBER 051-8071

REV. B

SCALE NONE

SHT 50

OF 98



Desired orientation when placed on board top-side:

+Z (up)

+Y

+X

Front of system

Circle indicates pin 1 location when placed in correct orientation

R5921 PULLS UP SMS\_PWRDN TO TURN OFF SMS WHEN PIN IS NOT BEING DRIVEN BY SMC

Desired orientation when placed on board top-side:

+Y

+X

+Z (up)

Front of system

Circle indicates pin 1 location when placed in correct orientation

Sudden Motion Sensor (SMS)

SYNC\_MASTER=YWJ\_K20 SYNC\_DATE=06/17/2008

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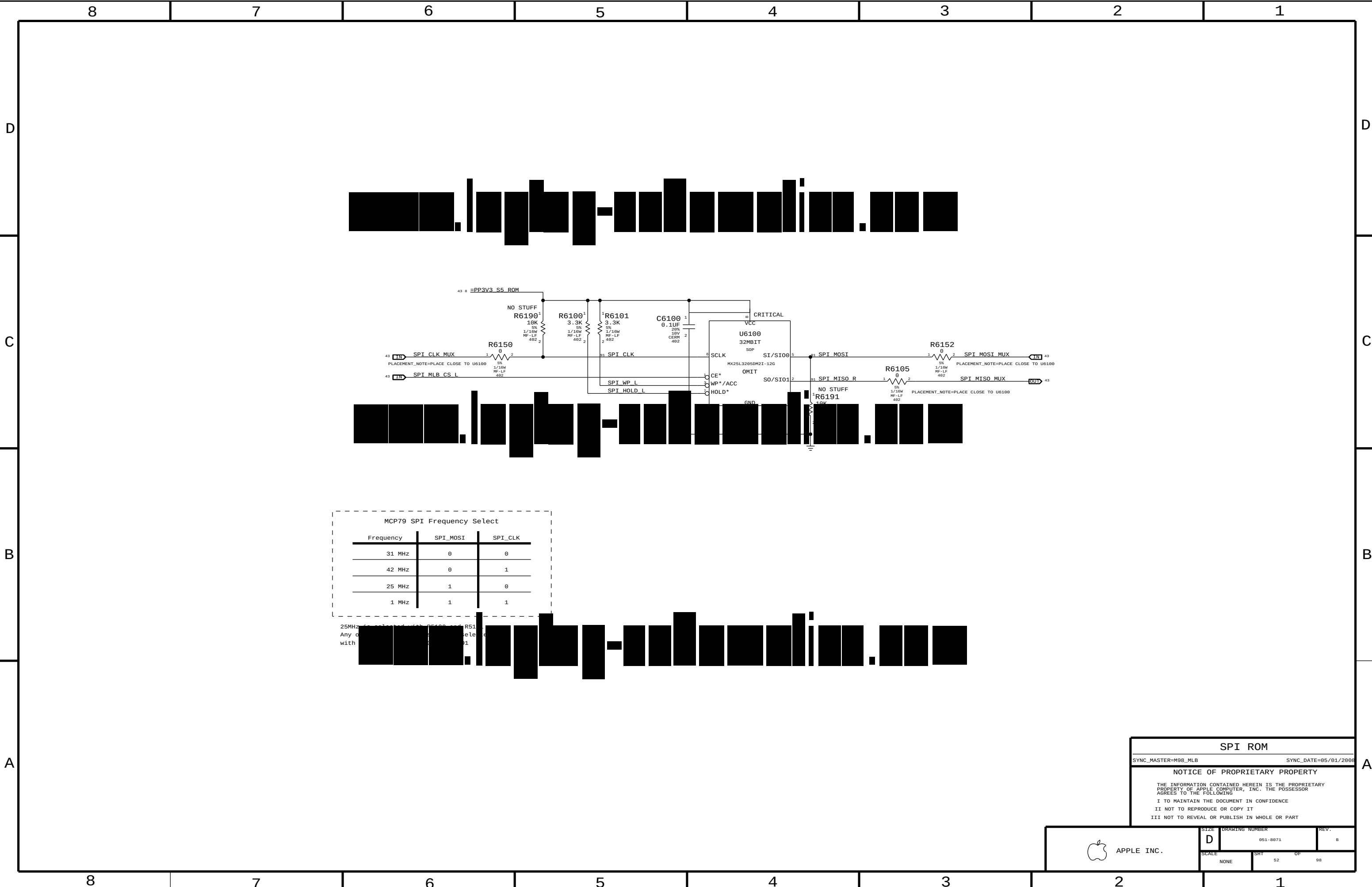
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| NONE       |                |              |      |



SPI ROM

SYNC\_MASTER=M98\_MLB SYNC\_DATE=05/01/2009

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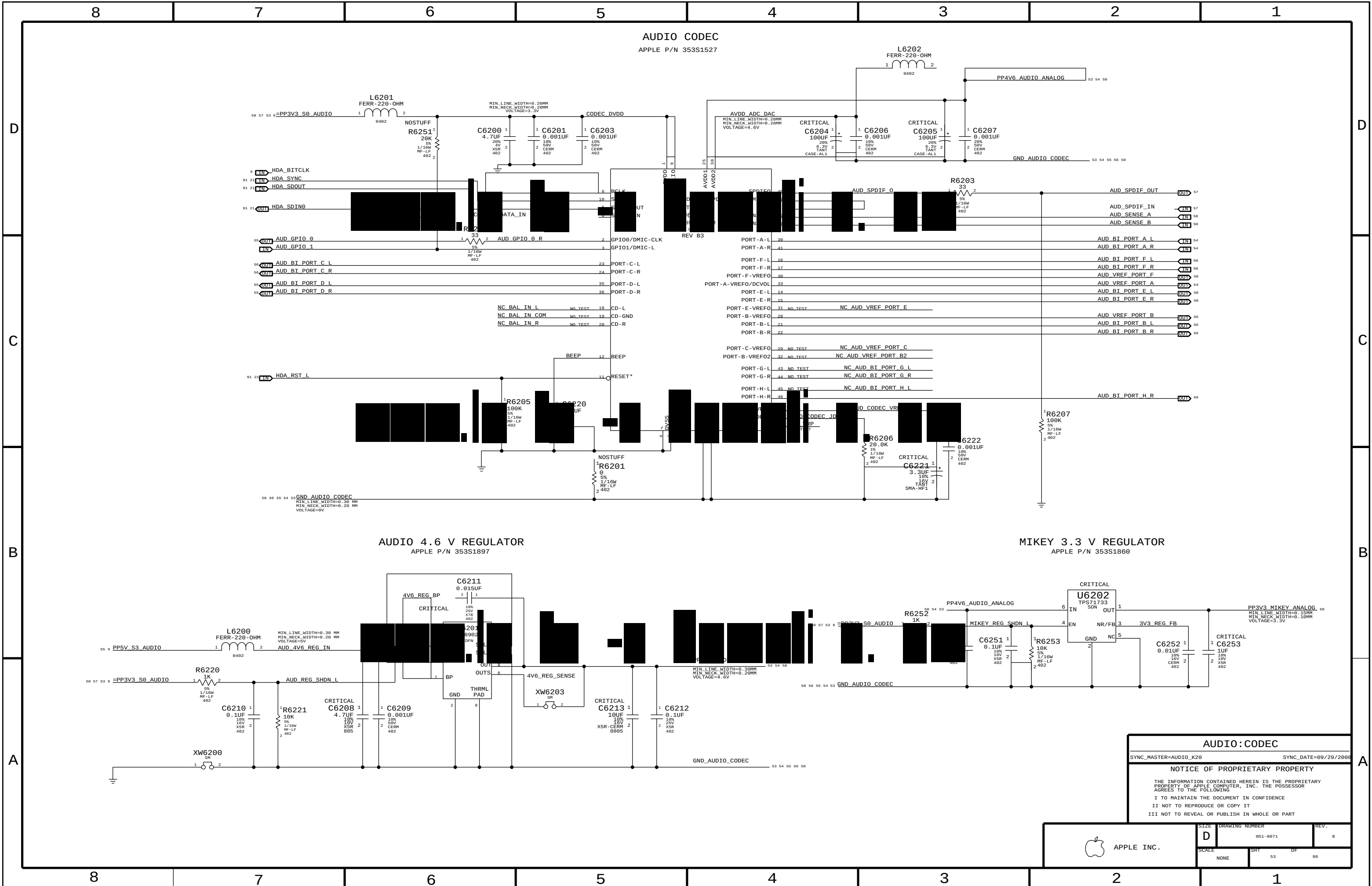
DRAWING NUMBER 051-8071

REV. B

SCALE NONE

SHT 52

OF 98



AUDIO:CODEC

SYNC\_MASTER=AUDIO\_K20 SYNC\_DATE=09/29/2009

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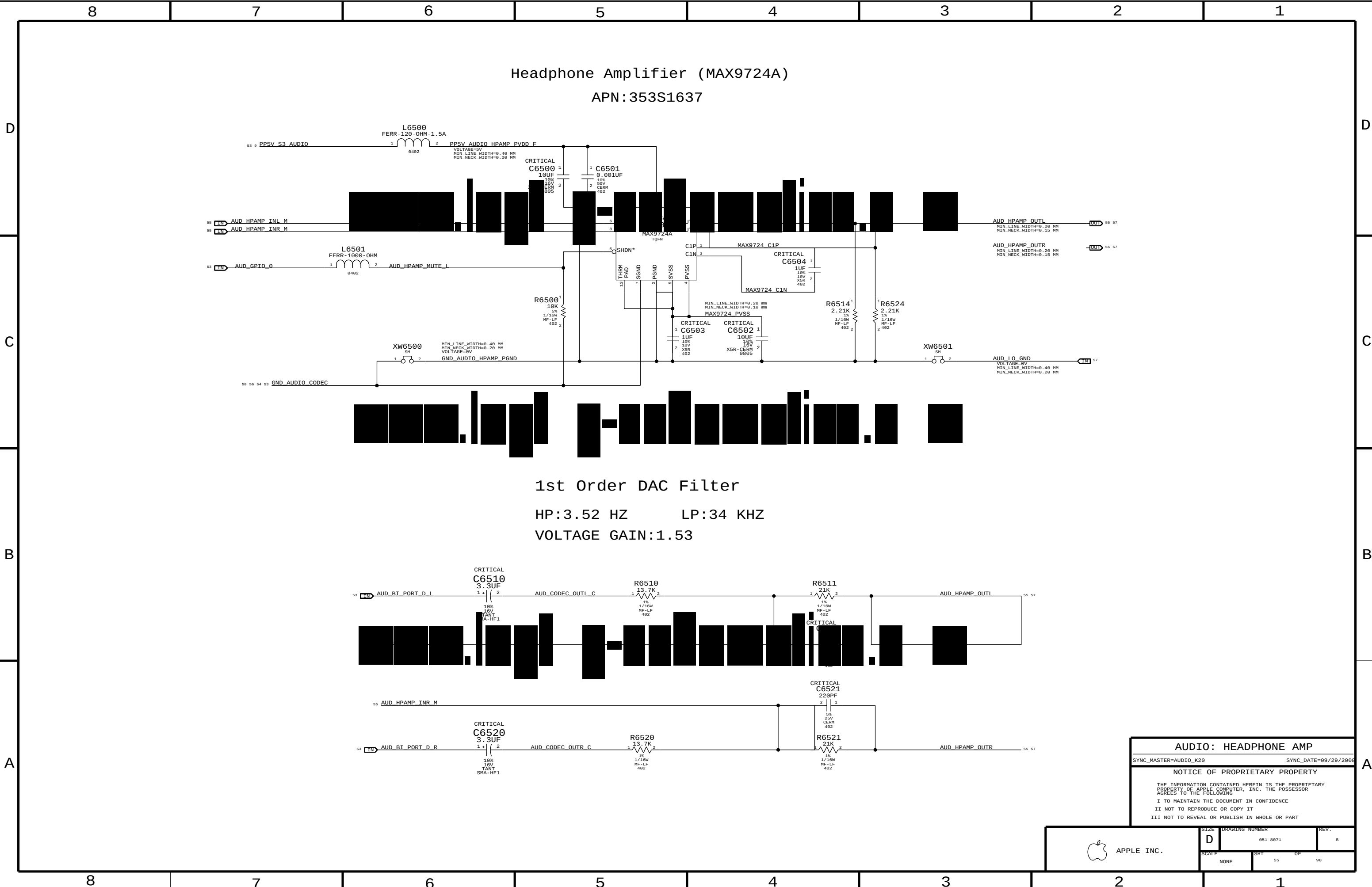
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Headphone Amplifier (MAX9724A)

APN:353S1637

1st Order DAC Filter

HP:3.52 HZ LP:34 KHZ

VOLTAGE GAIN:1.53

AUDIO: HEADPHONE AMP

SYNC\_MASTER=AUDIO\_K20SYNC\_DATE=09/29/2009

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|            | D              | 051-8071 |      |
| SCALE      |                | SHT      | OF   |
| NONE       |                | 55       | 98   |





AUDIO JACK 1 LO/HP JACK, SPDIF TX

MIC CONNECTOR  
APN: 518S0520

SPEAKER CONNECTORS  
APN: 518S0521

APN: 518S0672

AUDIO JACK 2 LINE IN JACK, SPDIF RX

AUDIO: JACKS

SYNC\_MASTER=AUDIO\_K20 SYNC\_DATE=09/29/2008

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SIZE DRAWING NUMBER

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051-8071

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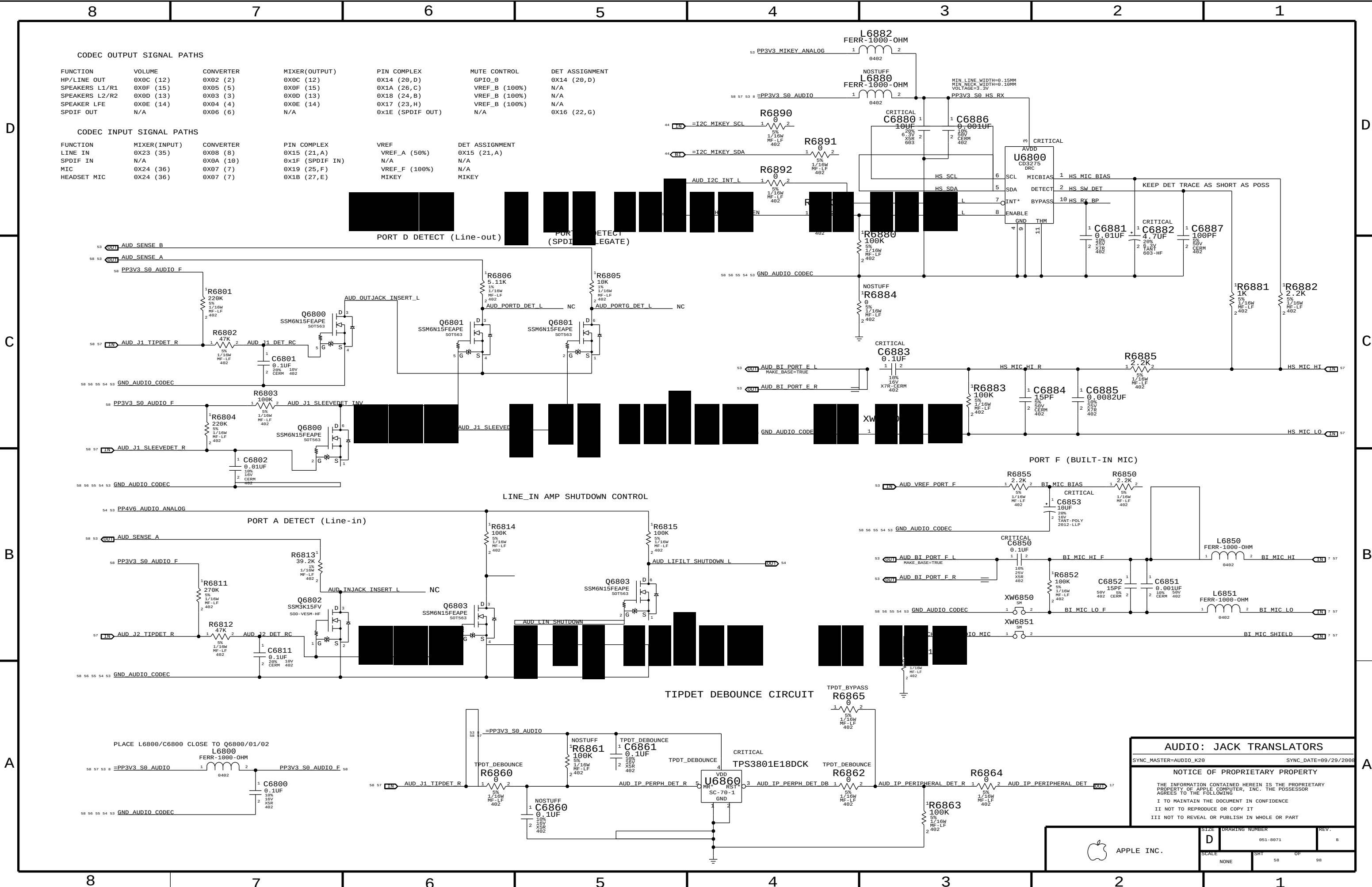
NONE

SHT

57

OF

98



CODEC OUTPUT SIGNAL PATHS

| FUNCTION       | VOLUME    | CONVERTER | MIXER(OUTPUT) | PIN COMPLEX      | MUTE CONTROL  | DET ASSIGNMENT |
|----------------|-----------|-----------|---------------|------------------|---------------|----------------|
| HP/LINE OUT    | 0X0C (12) | 0X0C (2)  | 0X0C (12)     | 0X14 (20,D)      | GPIO_0        | 0X14 (20,D)    |
| SPEAKERS L1/R1 | 0X0F (15) | 0X05 (5)  | 0X0F (15)     | 0X1A (26,C)      | VREF_B (100%) | N/A            |
| SPEAKERS L2/R2 | 0X0D (13) | 0X03 (3)  | 0X0D (13)     | 0X18 (24,B)      | VREF_B (100%) | N/A            |
| SPEAKER LFE    | 0X0E (14) | 0X04 (4)  | 0X0E (14)     | 0X17 (23,H)      | VREF_B (100%) | N/A            |
| SPDIF OUT      | N/A       | 0X06 (6)  | N/A           | 0X1E (SPDIF OUT) | N/A           | 0X16 (22,G)    |

CODEC INPUT SIGNAL PATHS

| FUNCTION    | MIXER(INPUT) | CONVERTER | PIN COMPLEX     | VREF          | DET ASSIGNMENT |
|-------------|--------------|-----------|-----------------|---------------|----------------|
| LINE IN     | 0X23 (35)    | 0X08 (8)  | 0X15 (21,A)     | VREF_A (50%)  | 0X15 (21,A)    |
| SPDIF IN    | N/A          | 0X0A (10) | 0X1F (SPDIF IN) | N/A           | N/A            |
| MIC         | 0X24 (36)    | 0X07 (7)  | 0X19 (25,F)     | VREF_F (100%) | N/A            |
| HEADSET MIC | 0X24 (36)    | 0X07 (7)  | 0X1B (27,E)     | MIKEY         | MIKEY          |

**AUDIO: JACK TRANSLATORS**

SYNC\_MASTER=AUDIO\_K20      SYNC\_DATE=09/29/2008

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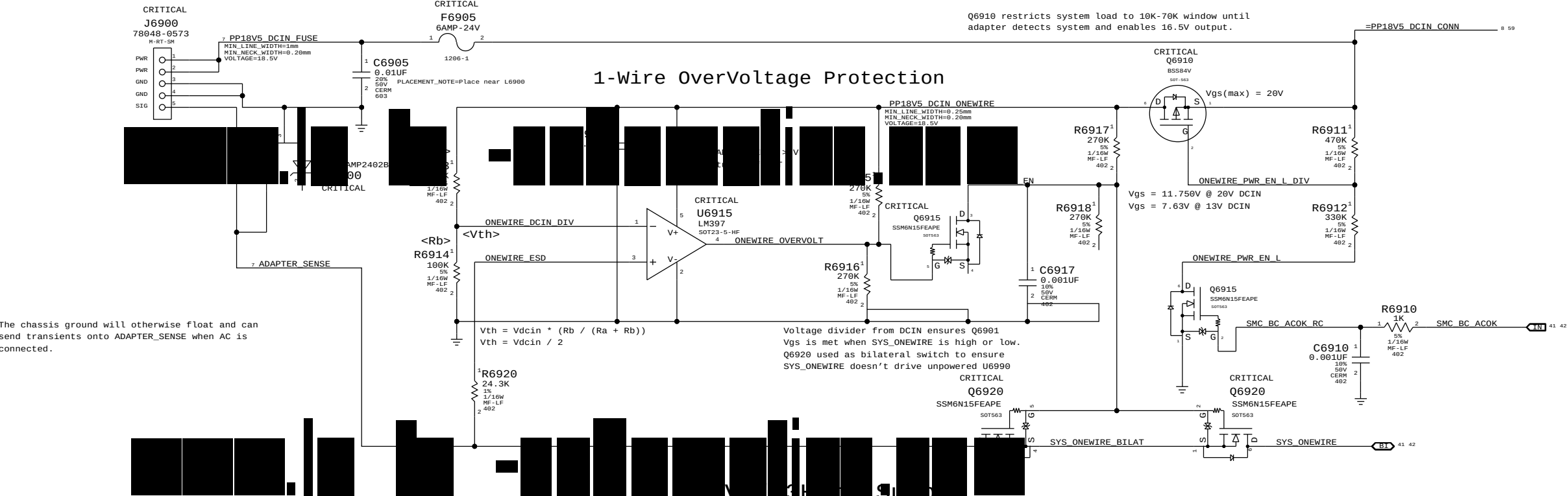
III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART

APPLE INC.

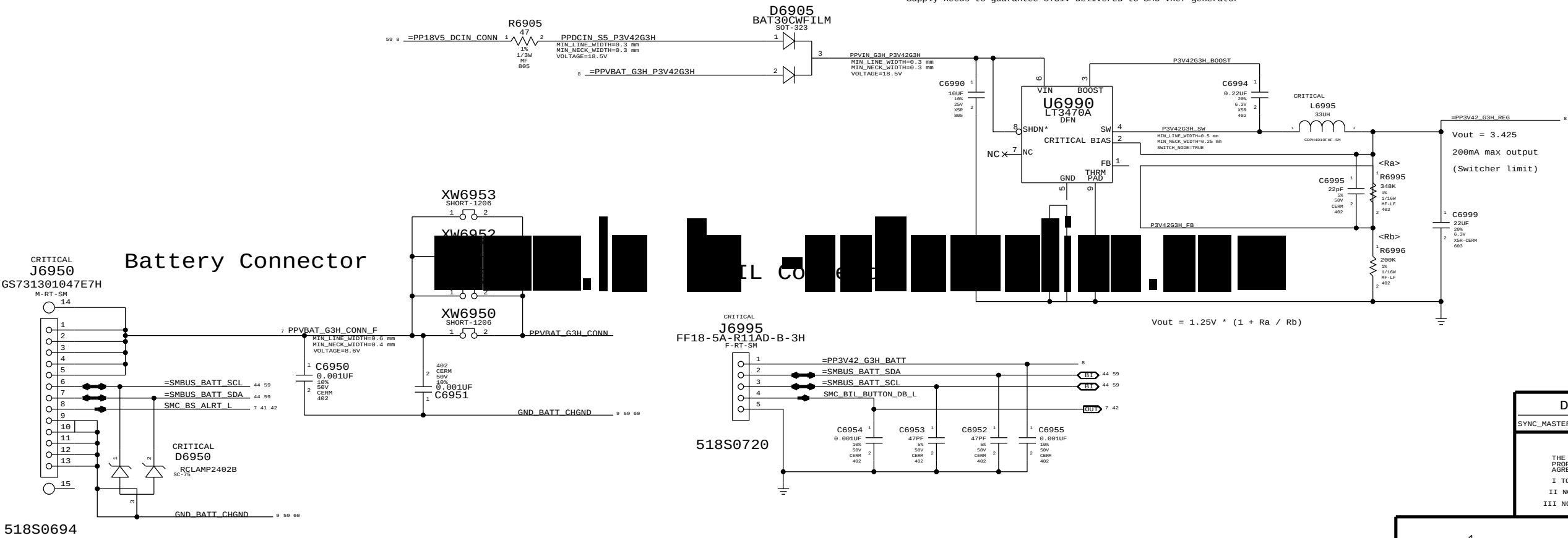
D      051-8071      B

SCALE      NONE      SRT      58      OF      98

MagSafe DC Power Jack



Battery Connector



DC-In & Battery Connectors

SYNC\_MASTER=RXU\_K20 SYNC\_DATE=05/21/2008

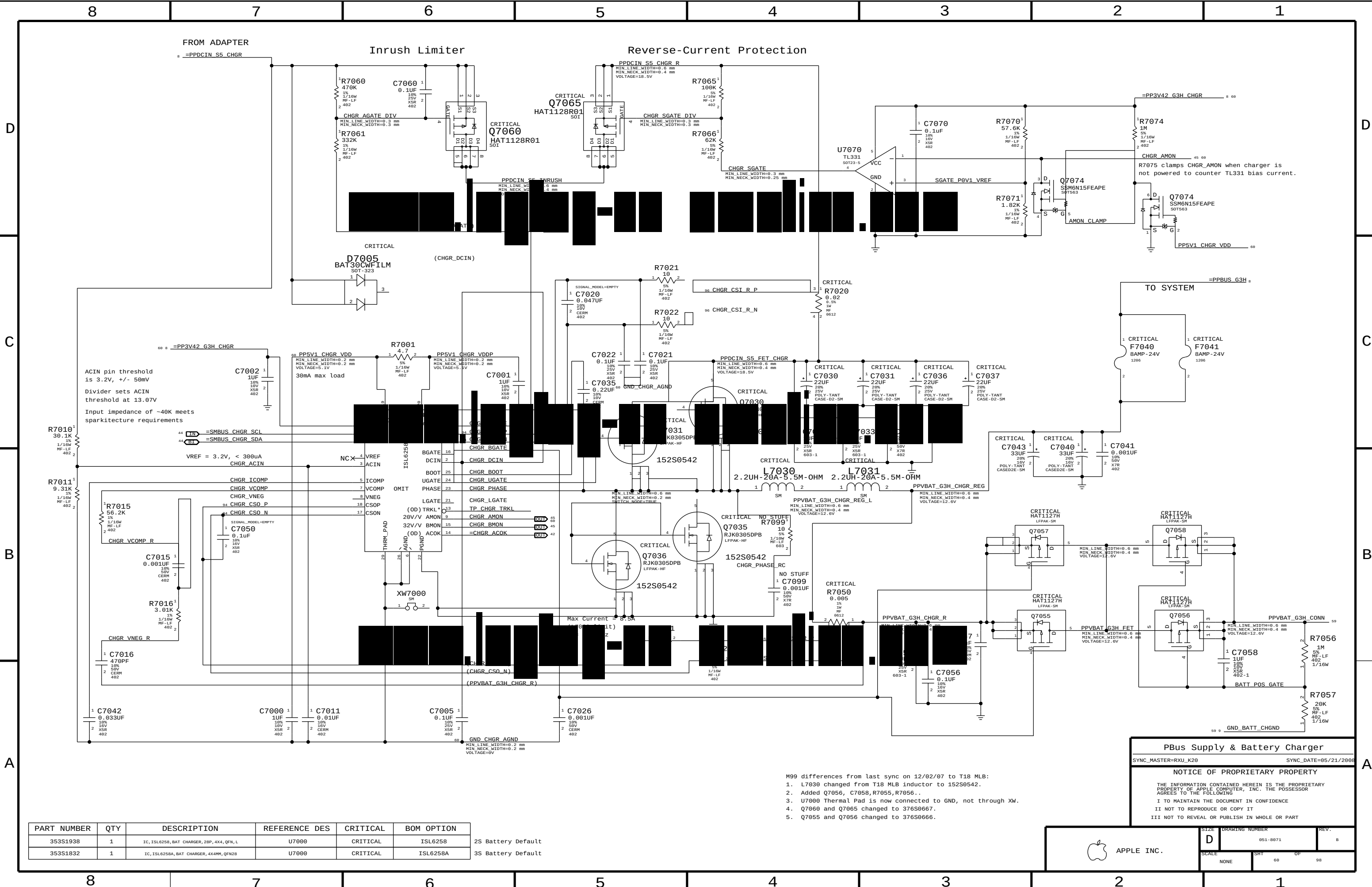
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| PART NUMBER | QTY | DESCRIPTION                                | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|--------------------------------------------|---------------|----------|------------|
| 353S1938    | 1   | IC, ISL6258, BAT CHARGER, 28P, 4X4, QFN, L | U7000         | CRITICAL | ISL6258    |
| 353S1832    | 1   | IC, ISL6258A, BAT CHARGER, 4X4MM, QFN28    | U7000         | CRITICAL | ISL6258A   |

2S Battery Default  
3S Battery Default

- M99 differences from last sync on 12/02/07 to T18 MLB:
- L7030 changed from T18 MLB inductor to 152S0542.
  - Added Q7056, C7058, R7055, R7056.
  - U7000 Thermal Pad is now connected to GND, not through XW.
  - Q7060 and Q7065 changed to 376S0667.
  - Q7055 and Q7056 changed to 376S0666.

PBUS Supply & Battery Charger

SYNC\_MASTER=RXU\_K20

SYNC\_DATE=05/21/2008

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DRAWING NUMBER

REV.

SCALE

SHT

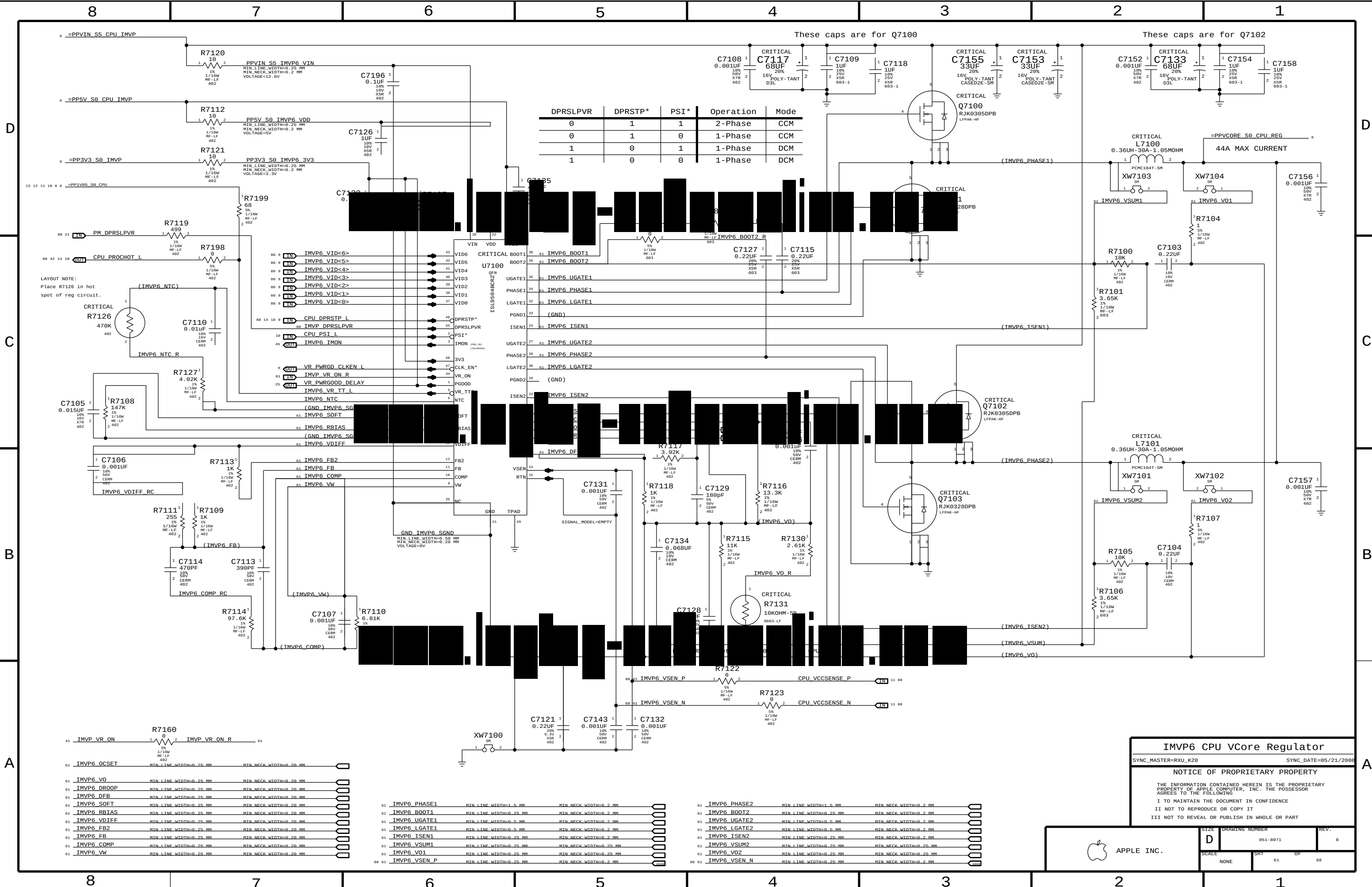
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051-8071

B



| DPRSLPVR | DPRSTP* | PSI* | Operation | Mode |
|----------|---------|------|-----------|------|
| 0        | 1       | 1    | 2-Phase   | CCM  |
| 0        | 1       | 0    | 1-Phase   | CCM  |
| 1        | 0       | 1    | 1-Phase   | DCM  |
| 1        | 0       | 0    | 1-Phase   | DCM  |

IMVP6 CPU VCore Regulator

SYNC\_MASTER=RXU\_K20

SYNC\_DATE=05/21/2008

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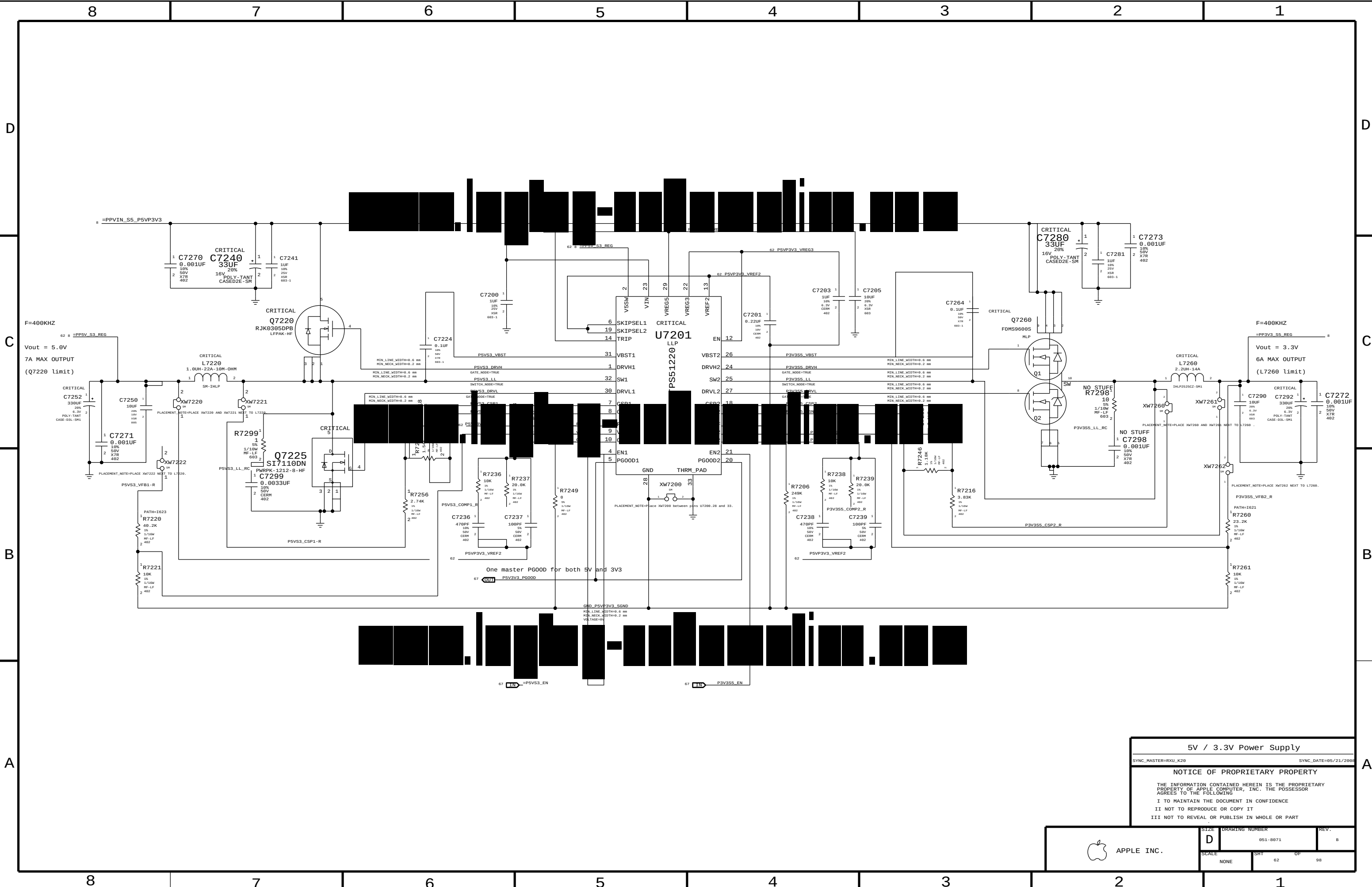
NONE

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61

OF

98



5V / 3.3V Power Supply

SYNC\_MASTER=RXU\_K20 SYNC\_DATE=05/21/2008

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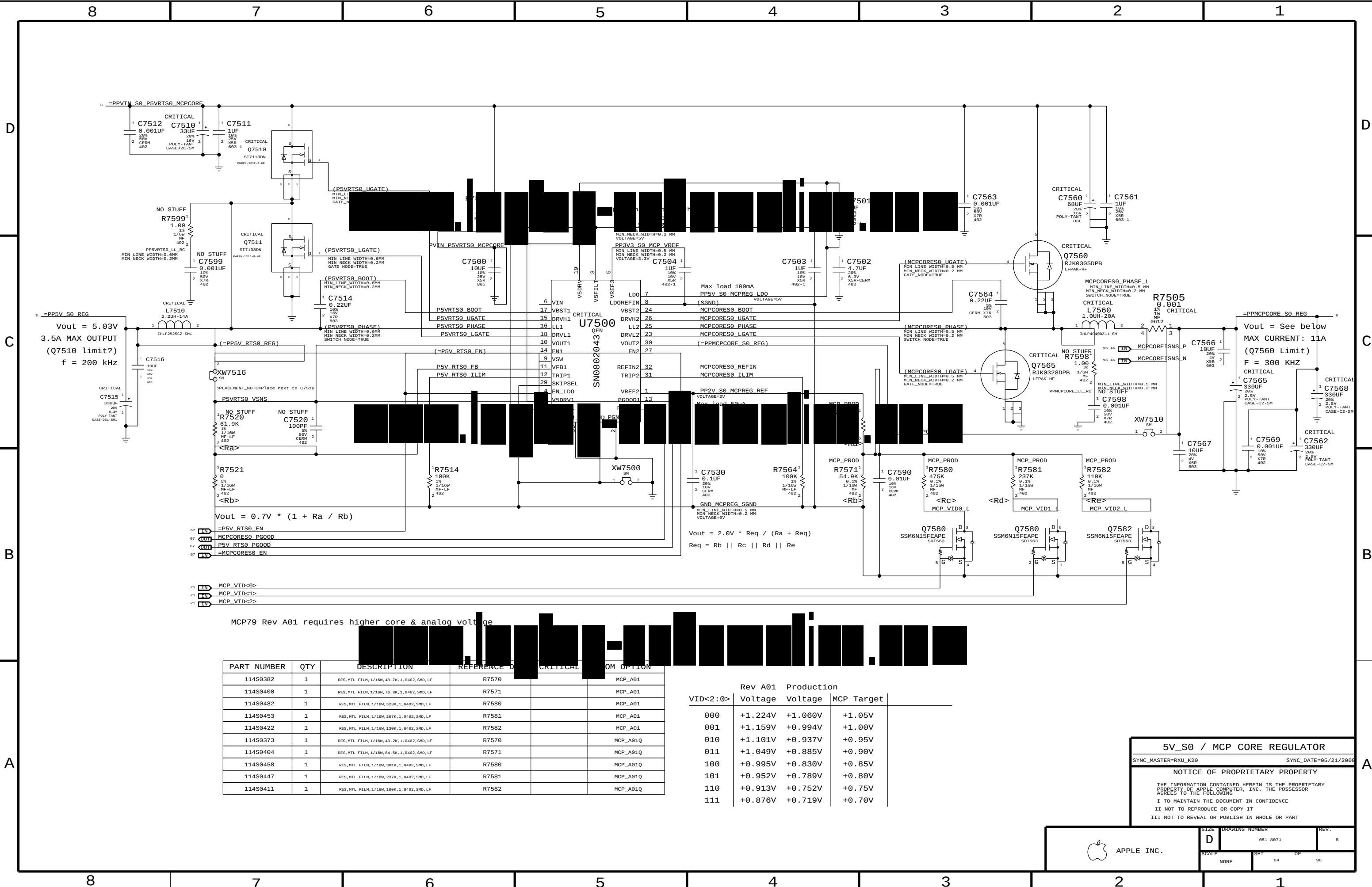
|                                                                                                  |       |                |      |
|--------------------------------------------------------------------------------------------------|-------|----------------|------|
|  APPLE INC. | SIZE  | DRAWING NUMBER | REV. |
|                                                                                                  | D     | 051-8071       | B    |
|                                                                                                  | SCALE | SHT OF         |      |
|                                                                                                  | NONE  | 62 98          |      |



APPLE INC.







MCP79 Rev A01 requires higher core & analog voltage

| PART NUMBER | QTY | DESCRIPTION                            | REFERENCE | CRITICAL | COM OPTION |
|-------------|-----|----------------------------------------|-----------|----------|------------|
| 114S0382    | 1   | RES,MTL FILM,1/16W,48.7K,1,0402,SMD,LF | R7570     |          | MCP_A01    |
| 114S0400    | 1   | RES,MTL FILM,1/16W,76.8K,1,0402,SMD,LF | R7571     |          | MCP_A01    |
| 114S0482    | 1   | RES,MTL FILM,1/16W,523K,1,0402,SMD,LF  | R7580     |          | MCP_A01    |
| 114S0453    | 1   | RES,MTL FILM,1/16W,267K,1,0402,SMD,LF  | R7581     |          | MCP_A01    |
| 114S0422    | 1   | RES,MTL FILM,1/16W,130K,1,0402,SMD,LF  | R7582     |          | MCP_A01    |
| 114S0373    | 1   | RES,MTL FILM,1/16W,40.2K,1,0402,SMD,LF | R7570     |          | MCP_A01Q   |
| 114S0404    | 1   | RES,MTL FILM,1/16W,84.5K,1,0402,SMD,LF | R7571     |          | MCP_A01Q   |
| 114S0458    | 1   | RES,MTL FILM,1/16W,301K,1,0402,SMD,LF  | R7580     |          | MCP_A01Q   |
| 114S0447    | 1   | RES,MTL FILM,1/16W,237K,1,0402,SMD,LF  | R7581     |          | MCP_A01Q   |
| 114S0411    | 1   | RES,MTL FILM,1/16W,100K,1,0402,SMD,LF  | R7582     |          | MCP_A01Q   |

| VID<2:0> | Rev A01 |            |            |
|----------|---------|------------|------------|
|          | Voltage | Production | MCP Target |
| 000      | +1.224V | +1.060V    | +1.05V     |
| 001      | +1.159V | +0.994V    | +1.00V     |
| 010      | +1.101V | +0.937V    | +0.95V     |
| 011      | +1.049V | +0.885V    | +0.90V     |
| 100      | +0.995V | +0.830V    | +0.85V     |
| 101      | +0.952V | +0.789V    | +0.80V     |
| 110      | +0.913V | +0.752V    | +0.75V     |
| 111      | +0.876V | +0.719V    | +0.70V     |

5V\_S0 / MCP CORE REGULATOR

SYNC\_MASTER=RXU\_K20      SYNC\_DATE=05/21/2008

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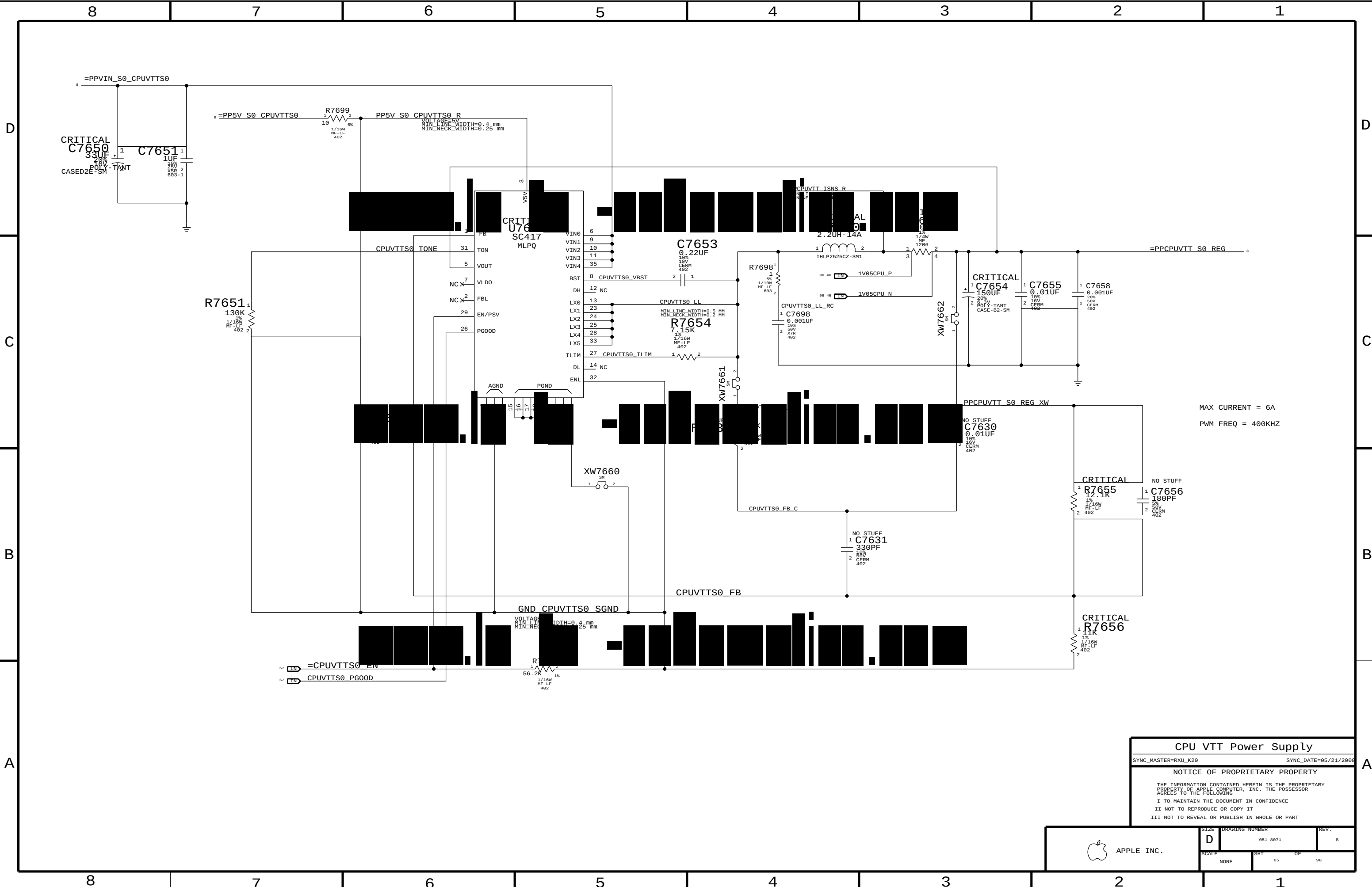
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NONE      64      98





CPU VTT Power Supply

SYNC\_MASTER=RXU\_K20

SYNC\_DATE=05/21/2008

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SCALE

NONE

D

DRAWING NUMBER

051-8071

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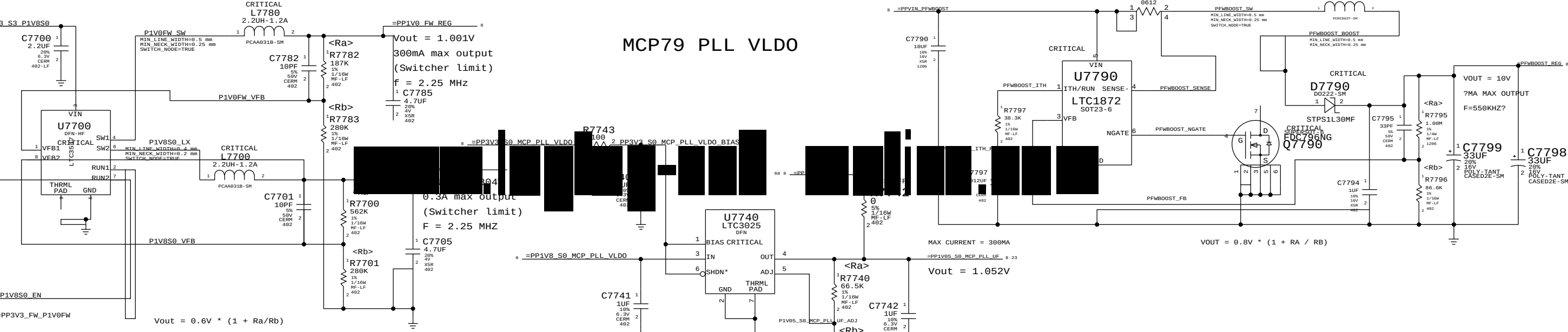
98

1.8V S0 Switcher / 1.0VFW SWITCHER

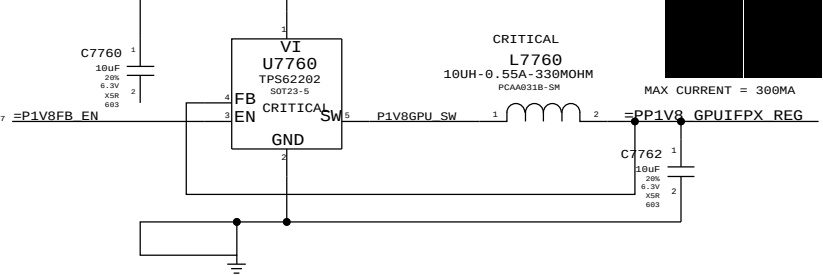
S5 power required for output discharge feature

MCP79 PLL VLD0

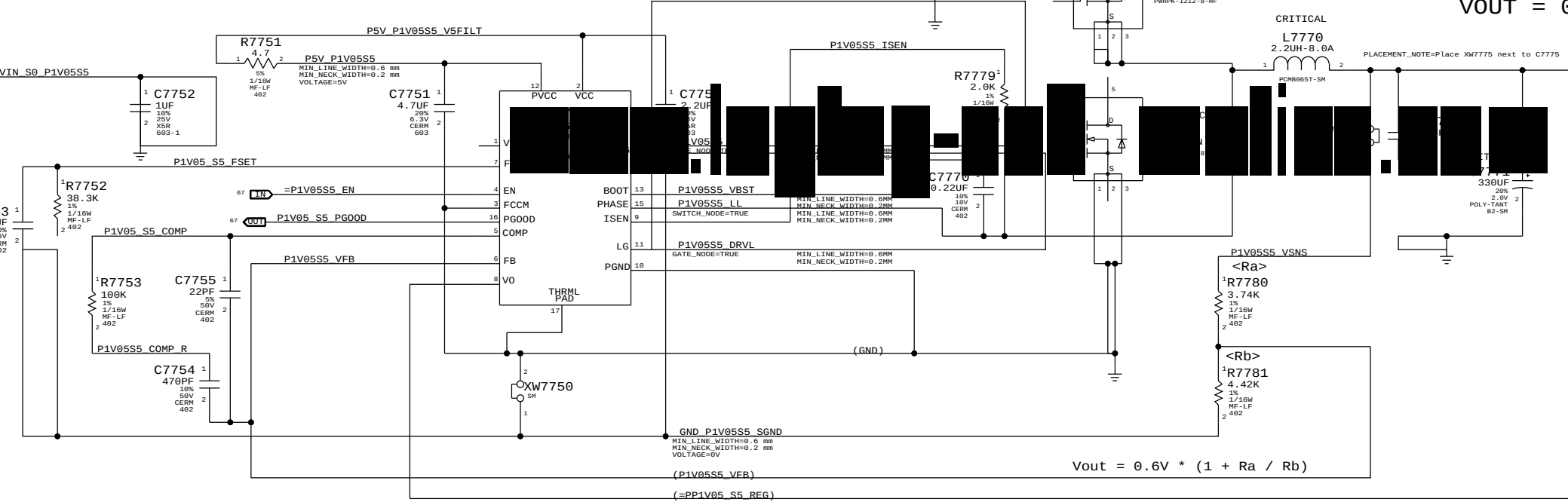
FW BOOST POWER



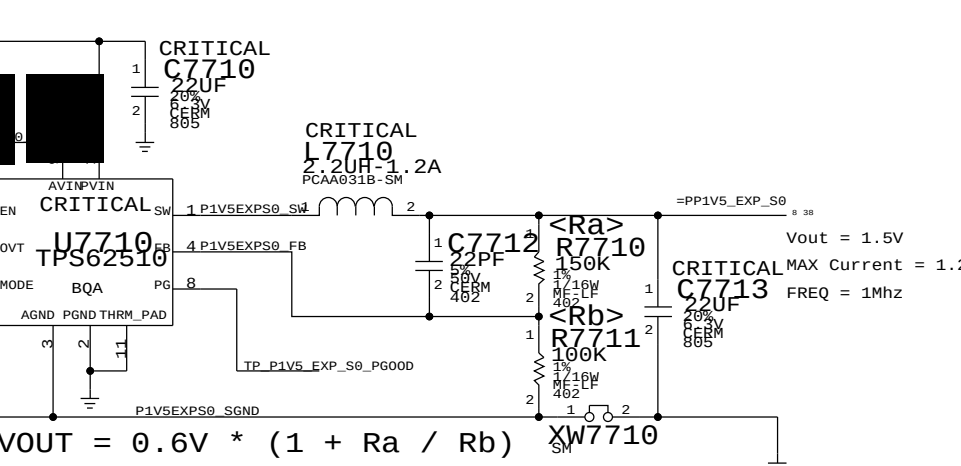
1.8V S0 Switcher



MCP 1.05V AUXC Supply



EXPRESSCARD 1.5V\_S0 SUPPLY



Misc Power Supplies

SYNC\_MASTER=RXU\_K20 SYNC\_DATE=05/21/2008

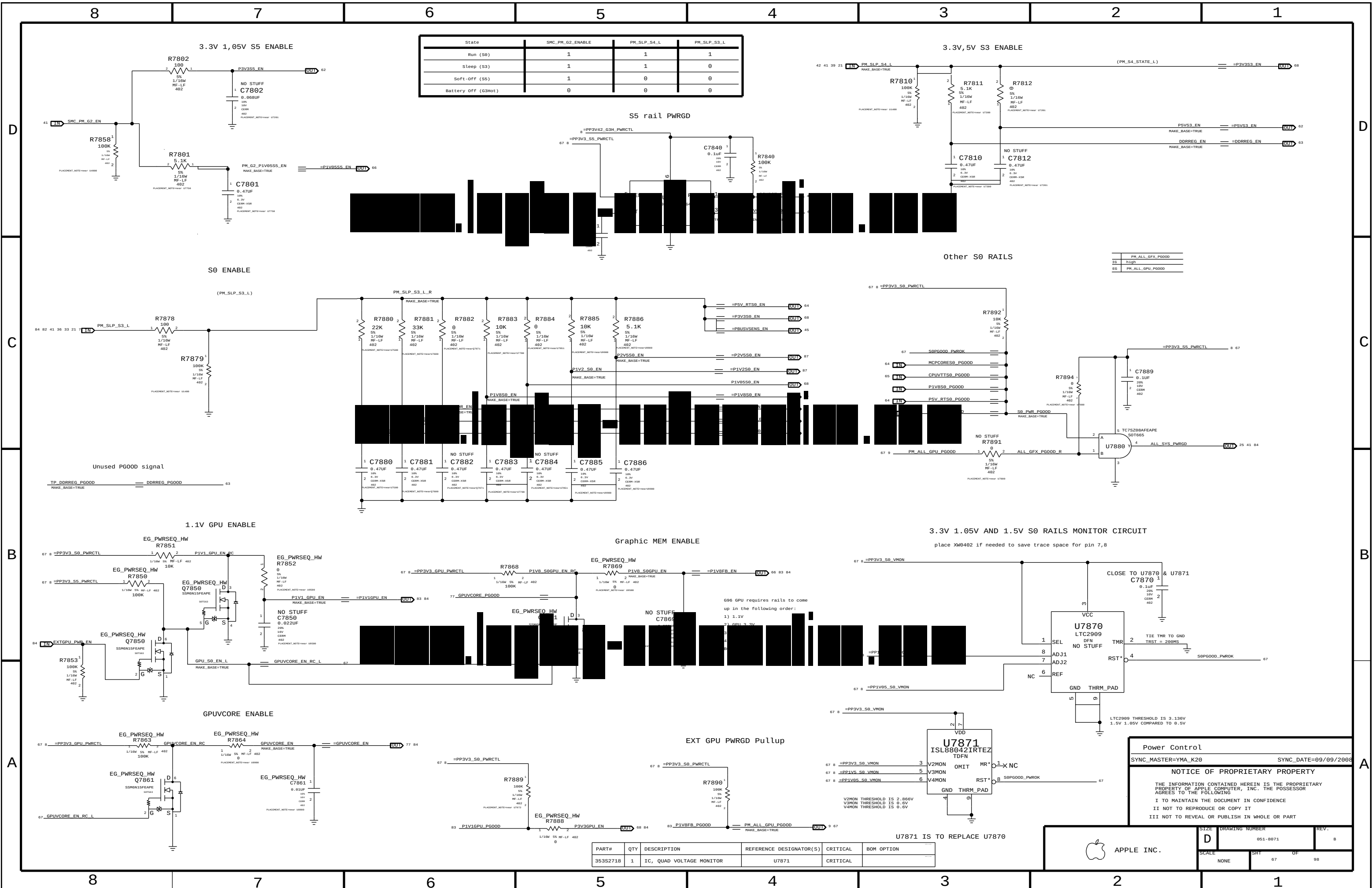
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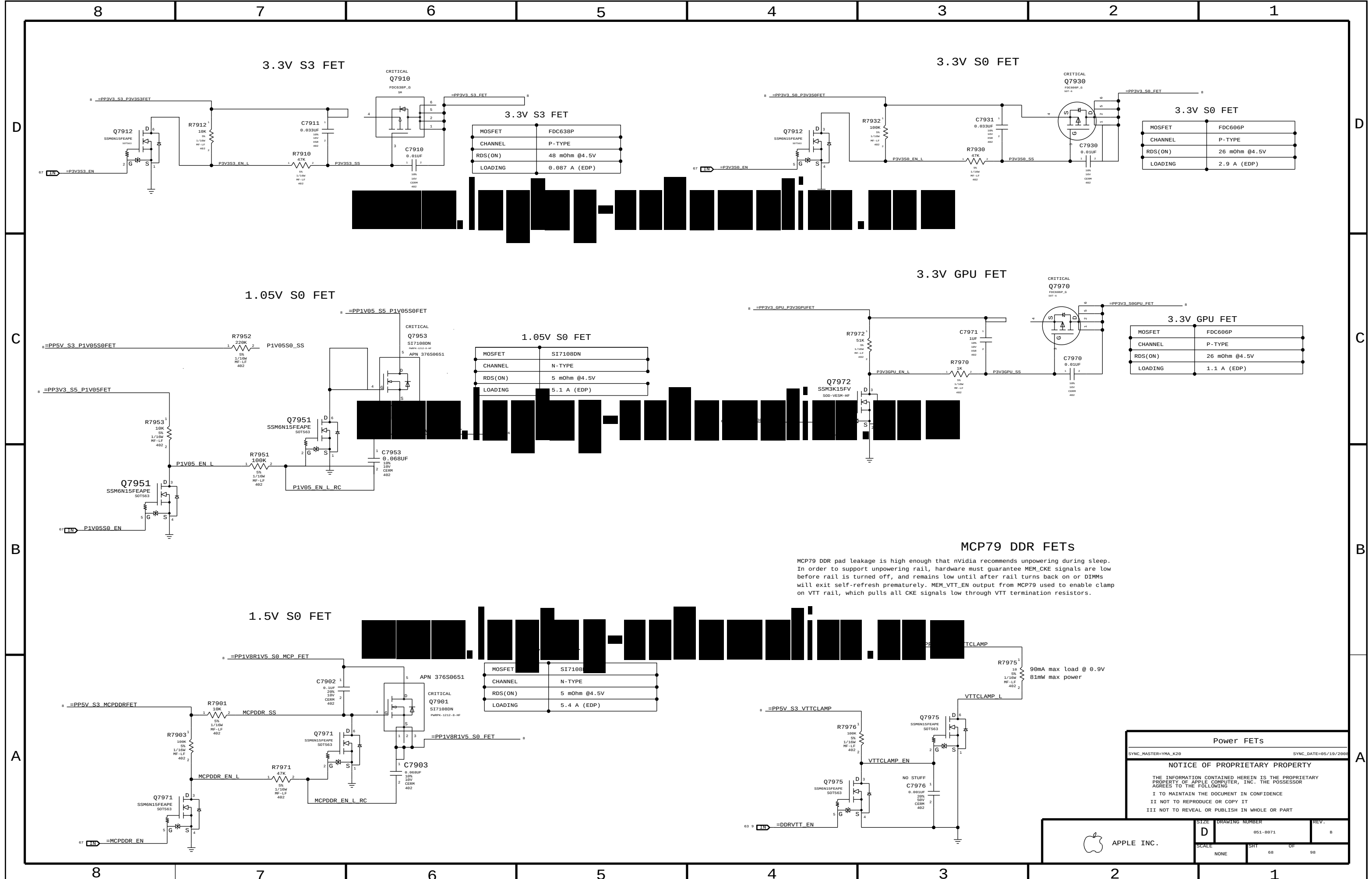
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Power aliases required by this page:

- @PP1V2\_GPU\_PEX\_PL1KVD0
- @PP1V2\_GPU\_PEX\_IOVD0Q
- @PP1V2\_GPU\_PEX\_IOVD0

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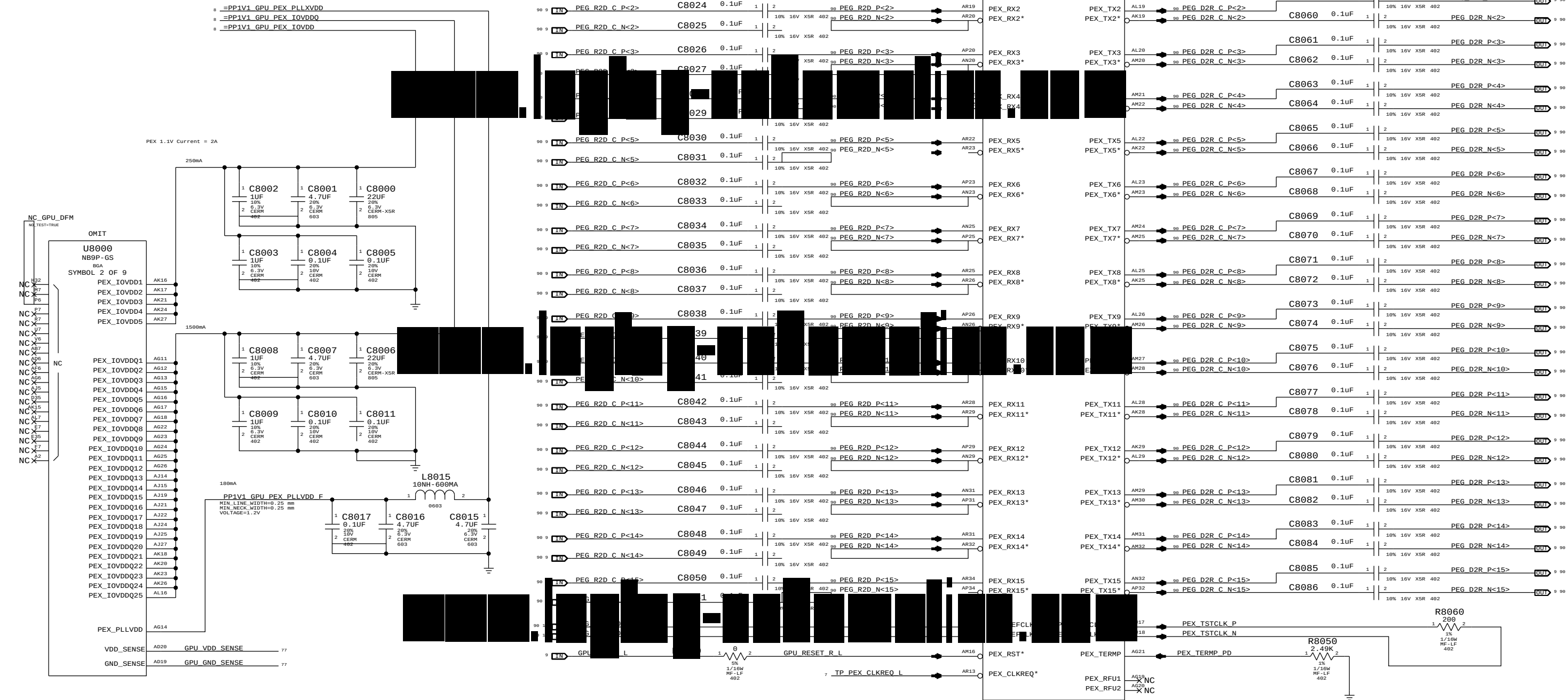
Signal aliases required by this page:

(NONE)

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BOM options provided by this page:

(NONE)



|                                                                                                                            |                      |
|----------------------------------------------------------------------------------------------------------------------------|----------------------|
| NV G96 PCI-E                                                                                                               |                      |
| SYNC_MASTER=M9B_MLB                                                                                                        | SYNC_DATE=04/03/2008 |
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| SCALE | INCH     | OF |

| SCALE | SHI | UP |
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| NONE  | 69  | 98 |

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|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|

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## Page Notes

Power aliases required by this page:

- =PPVCORE\_GPU  
- =PP1V8\_GPU\_FBVDDQ

Signal aliases required by this page:

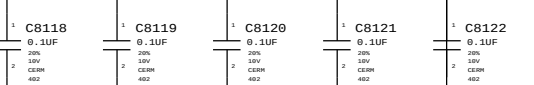
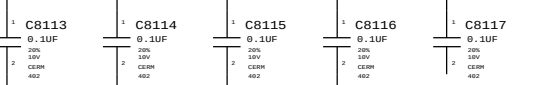
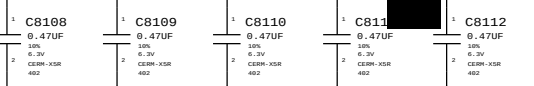
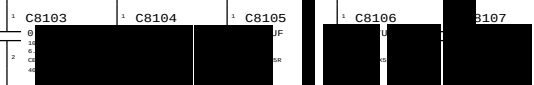
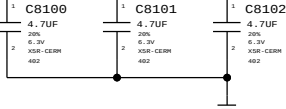
(NONE)

BOM options provided by this page:

(NONE)

=PPVCORE\_GPU

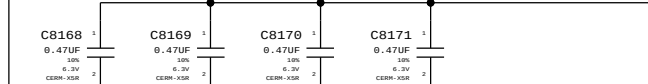
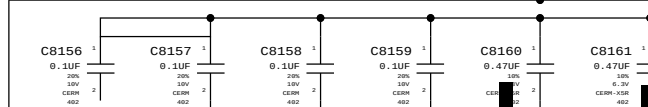
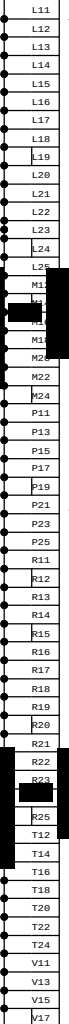
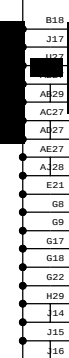
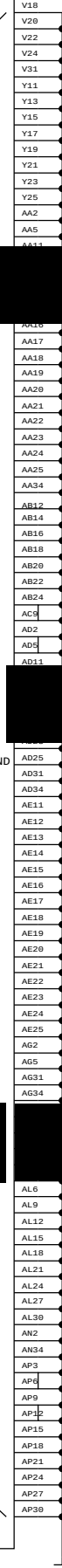
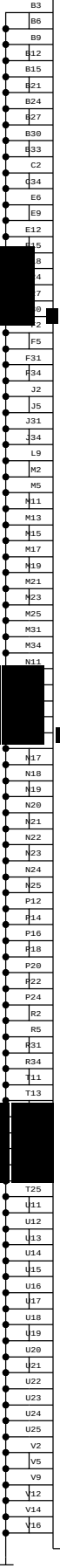
???A @ ???/???MHz Core/Mem Clk for VDD



=PP1V8\_GPU\_FBVDDQ

Nvidia PRD for GB-128 uses 4x4.7uF, 8x0.47uF, 16x0.1uF

???A @ ???MHz 1.8V GDDR3

U8000  
NB9P-GS  
BGA  
SYMBOL 9 OF 9U8000  
NB9P-GS  
BGA  
SYMBOL 7 OF 7U8000  
NB9P-GS  
BGA  
SYMBOL 8 OF 9

## NV G96 CORE/FB POWER

SYNC\_MASTER=H9B\_MLB

SYNC\_DATE=04/01/2008

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SIZE: DRAWING NUMBER

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051-8071

REV.

B

SCALE

NONE

SHT

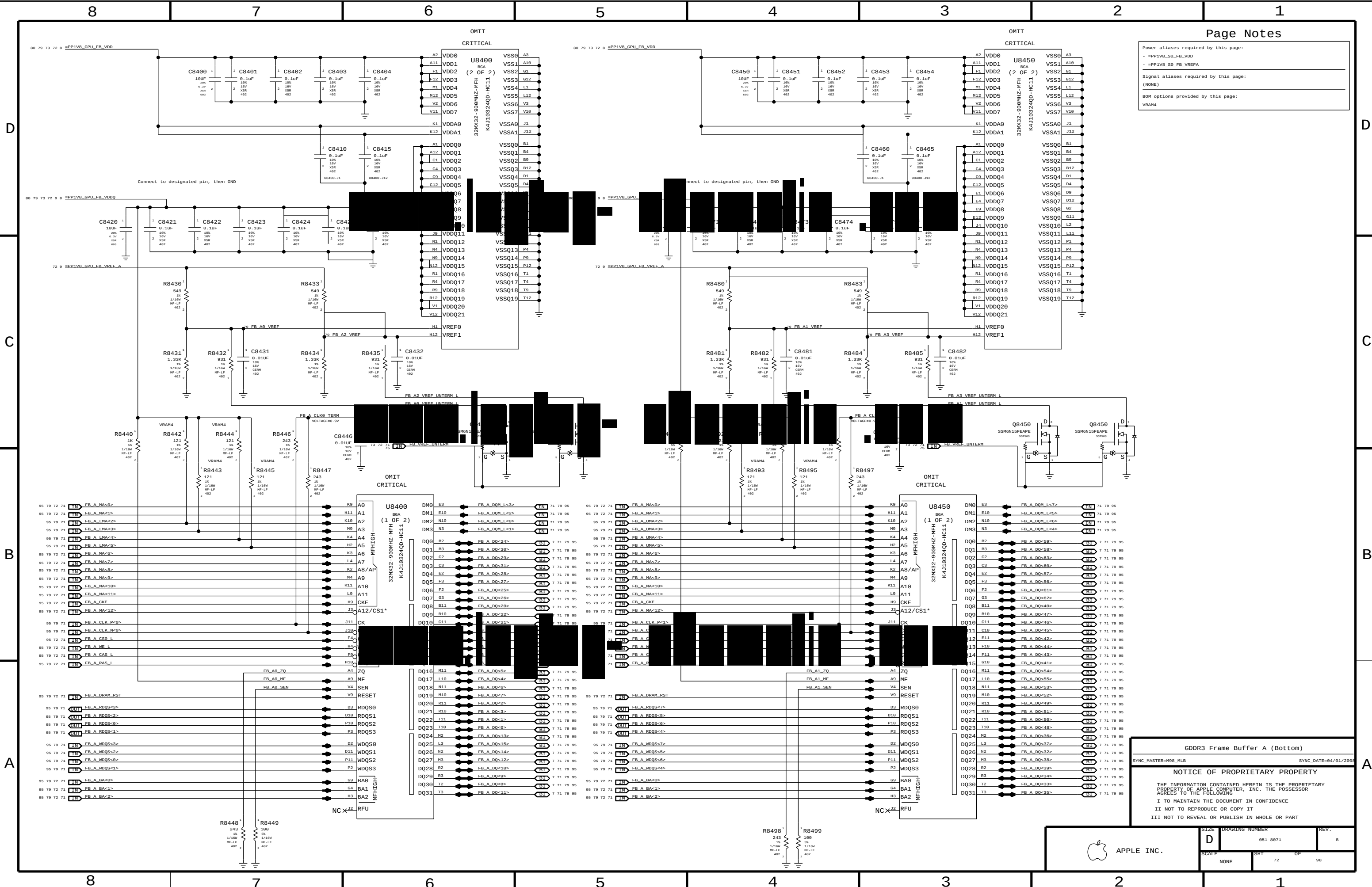
70

OF

98







Page Notes

Power aliases required by this page:

- =PP1VB\_S0\_FB\_VDD
- =PP1VB\_S0\_FB\_VREFA

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

VRAM4

GDDR3 Frame Buffer A (Bottom)

SYNC\_MASTER=M8B\_MLB SYNC\_DATE=04/01/2008

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## Page Notes

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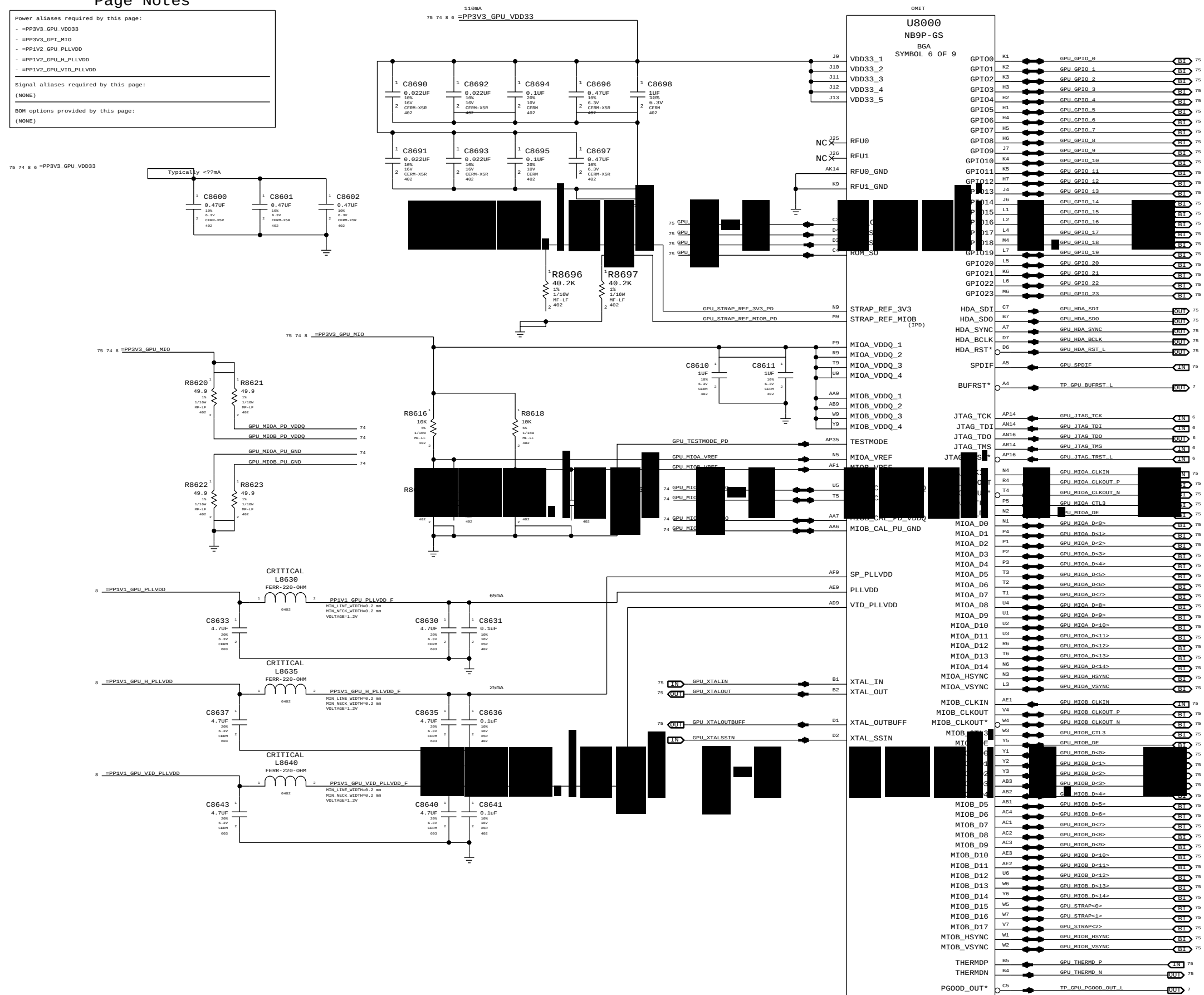
- =PP3V3\_GPU\_VDD33
- =PP3V3\_GPI\_MIO
- =PP1V2\_GPU\_PLLVDD
- =PP1V2\_GPU\_H\_PLLVDD
- =PP1V2\_GPU\_VID\_PLLVDD

Signal aliases required by this page:

( NONE )

BOM options provided by this page:

( NONE )



NV G96 GPIO/MIO/MISC

SYNC\_MASTER=K20\_MLB

SYNC\_DATE=09/24/2008

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| SIZE | DRAWING NUMBER |
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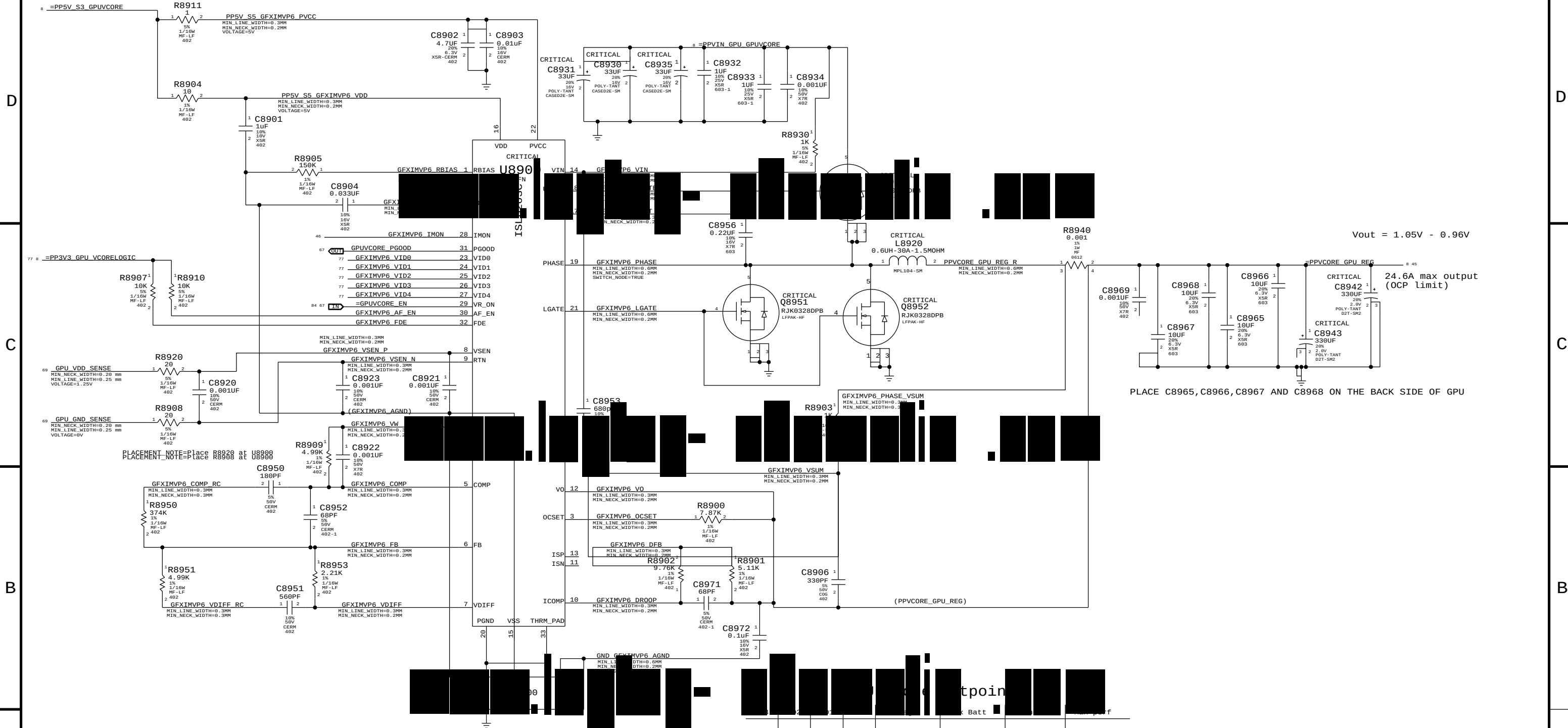
1

OF





GPU VCore Regulator



Vout = 1.05V - 0.96V

24.6A max output (OCP limit)

PLACE C8965,C8966,C8967 AND C8968 ON THE BACK SIDE OF GPU

Default Vcore Setpoints

| BOM GROUP    | BOM OPTIONS                     |
|--------------|---------------------------------|
| GPUVID_0P90V | GPUVID2_1, GPUVID1_1, GPUVID0_1 |
| GPUVID_1P00V | GPUVID2_0, GPUVID1_1, GPUVID0_1 |

GPU (G96) CORE SUPPLY

SYNC\_MASTER=RXU\_K20 SYNC\_DATE=05/21/2008

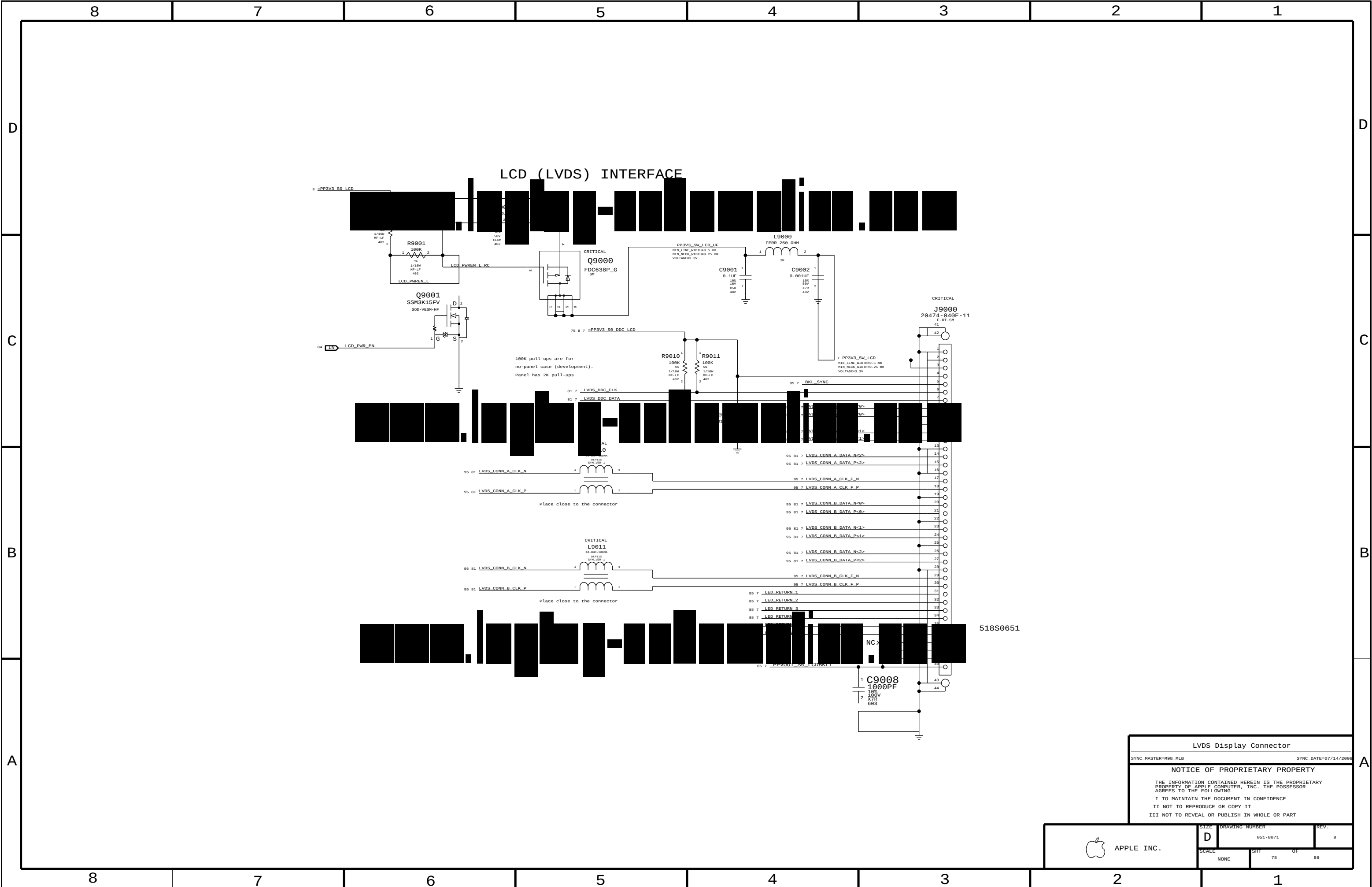
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|-------|----------------|------|
| D     | 051-8071       | B    |
| SCALE | SHT            | OF   |
| NONE  | 77             | 98   |



LVDS Display Connector

SYNC\_MASTER=M9B\_MLB SYNC\_DATE=07/14/2008

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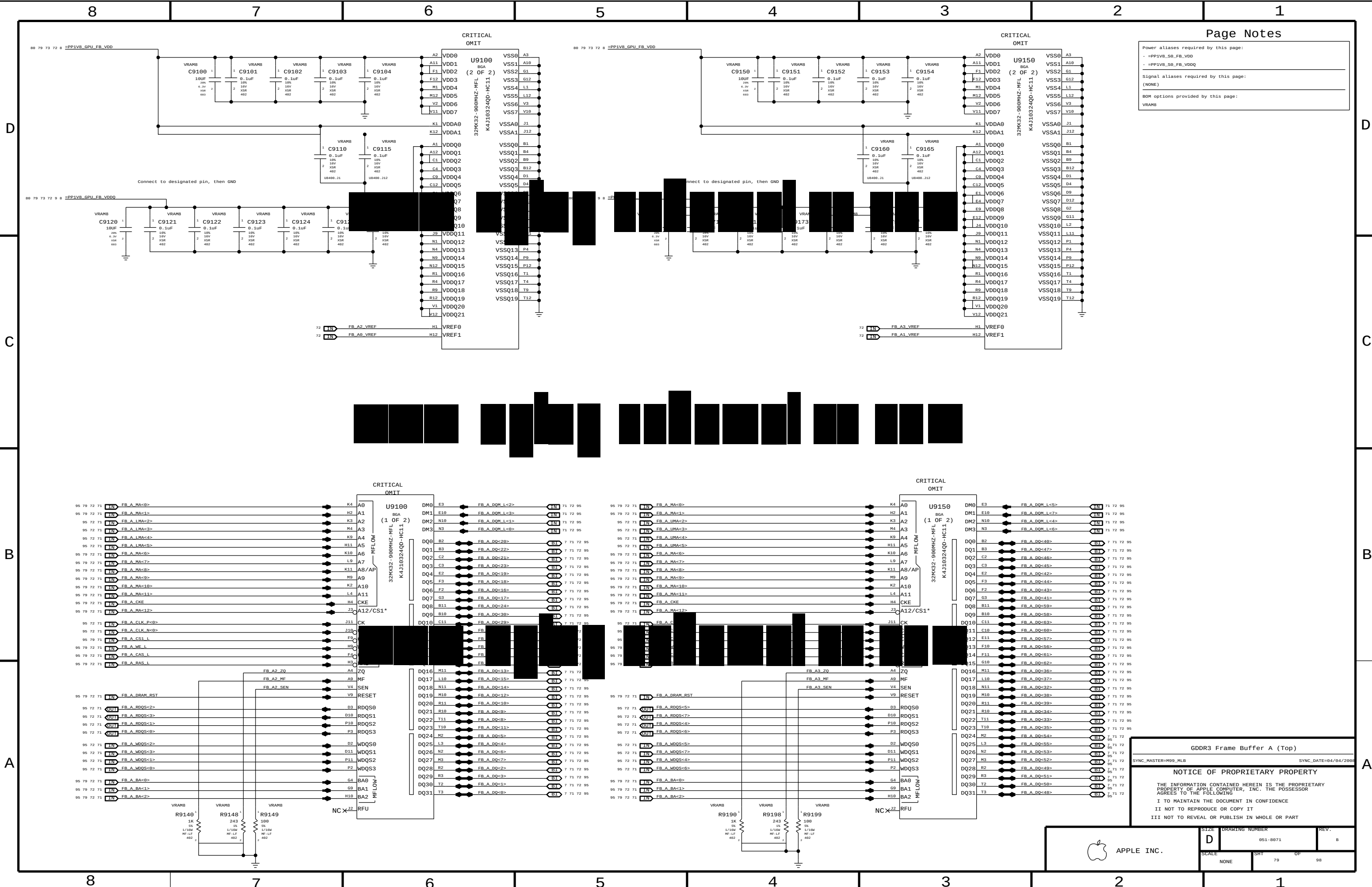
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APPLE INC.

SIZE D DRAWING NUMBER 051-8071 REV. B

SCALE NONE SHY OF 98





Page Notes

Power aliases required by this page:

- PP1VB\_S0\_FB\_VDD
- PP1VB\_S0\_FB\_VDDQ

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

VRAMB

GDDR3 Frame Buffer A (Top)

SYNC\_MASTER=M99\_MLB SYNC\_DATE=04/04/2008

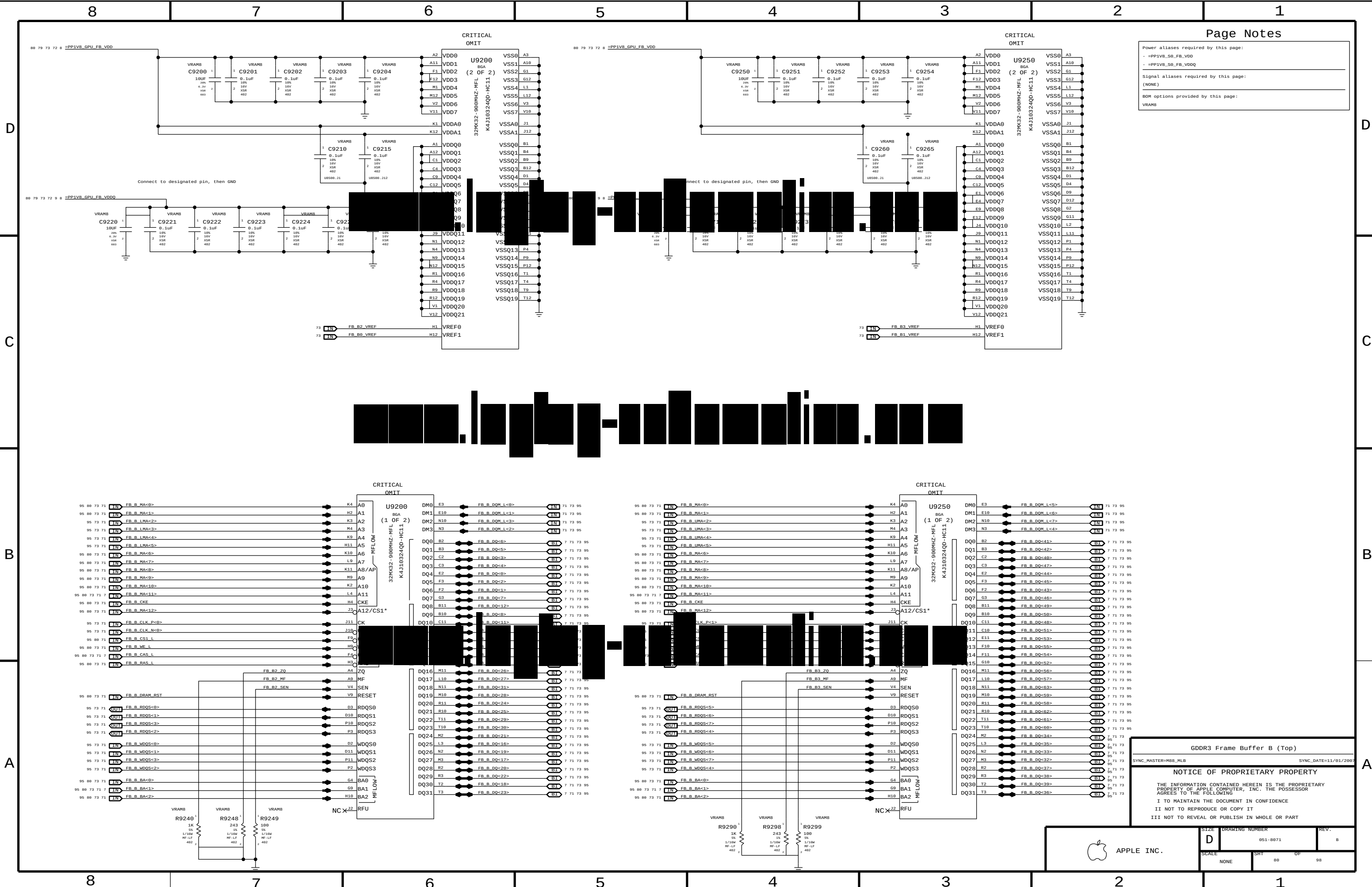
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Page Notes

Power aliases required by this page:

- =PP1V8\_GPU\_FB\_VDD
- =PP1V8\_S0\_FB\_VDDQ

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

VRAMB

GDDR3 Frame Buffer B (Top)

SYNC\_MASTER=M8B\_MLB SYNC\_DATE=11/01/2007

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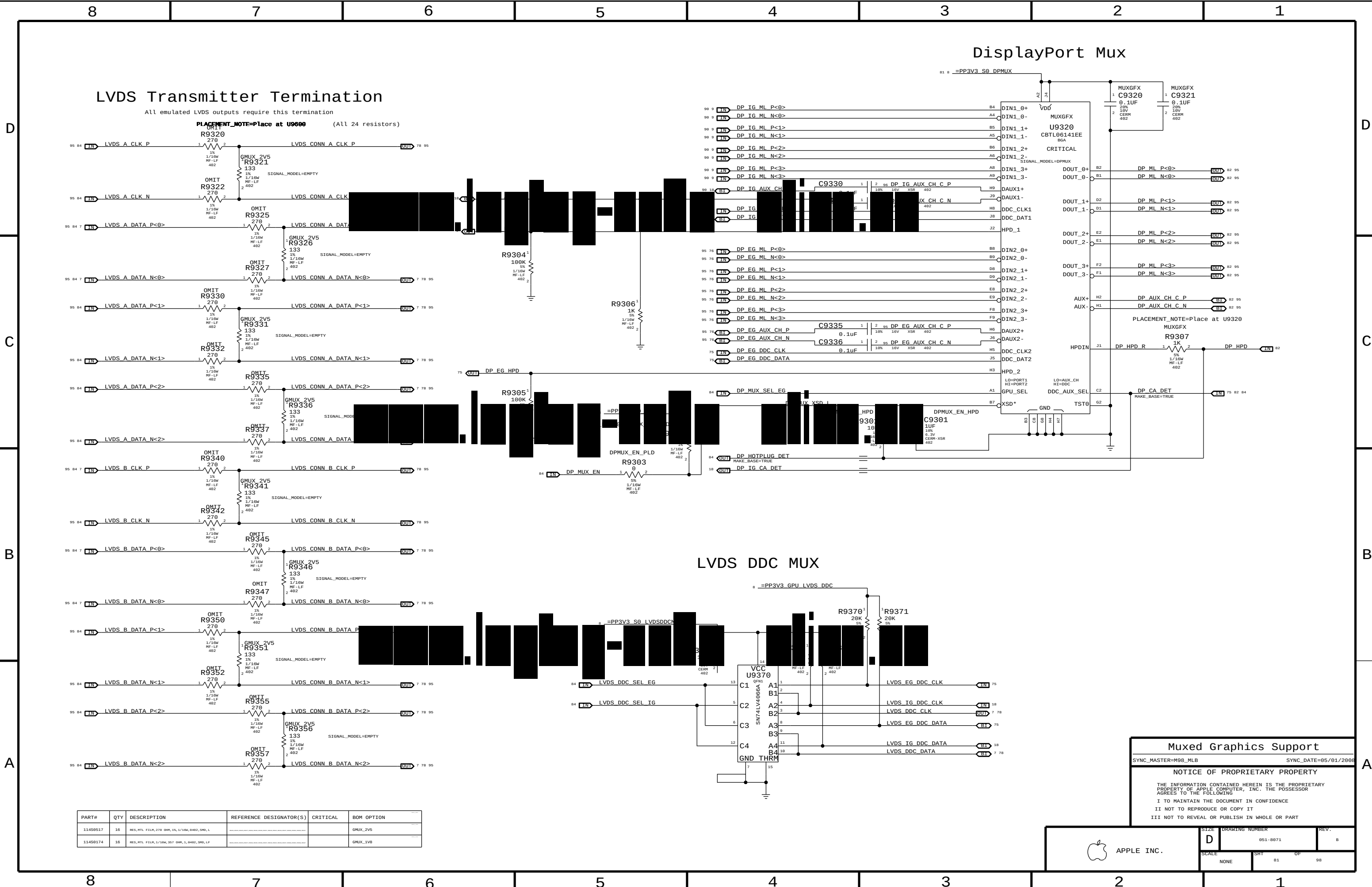
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Muxed Graphics Support

SYNC\_MASTER=M98\_MLB

SYNC\_DATE=05/01/2008

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| PART#    | QTY | DESCRIPTION                               | REFERENCE DESIGNATOR(S)                                                                                                                                                                                                                                                                                                                   | CRITICAL | BOM OPTION |
|----------|-----|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------|
| 11450517 | 15  | RES,MTL F1LM,270 OHM,1%,1/16W,0402,SMD,L  | R9320,R9321,R9322,R9323,R9324,R9325,R9326,R9327,R9328,R9329,R9330,R9331,R9332,R9333,R9334,R9335,R9336,R9337,R9338,R9339,R9340,R9341,R9342,R9343,R9344,R9345,R9346,R9347,R9348,R9349,R9350,R9351,R9352,R9353,R9354,R9355,R9356,R9357                                                                                                       |          | GMUX_2V5   |
| 11450174 | 15  | RES,MTL F1LM,1/16W,35T OHM,1%,0402,SMD,LF | R9303,R9304,R9305,R9306,R9307,R9308,R9309,R9310,R9311,R9312,R9313,R9314,R9315,R9316,R9317,R9318,R9319,R9320,R9321,R9322,R9323,R9324,R9325,R9326,R9327,R9328,R9329,R9330,R9331,R9332,R9333,R9334,R9335,R9336,R9337,R9338,R9339,R9340,R9341,R9342,R9343,R9344,R9345,R9346,R9347,R9348,R9349,R9350,R9351,R9352,R9353,R9354,R9355,R9356,R9357 |          | GMUX_1V8   |

APPLE INC.

D

DRAWING NUMBER

051-8071

REV.

B

SCALE

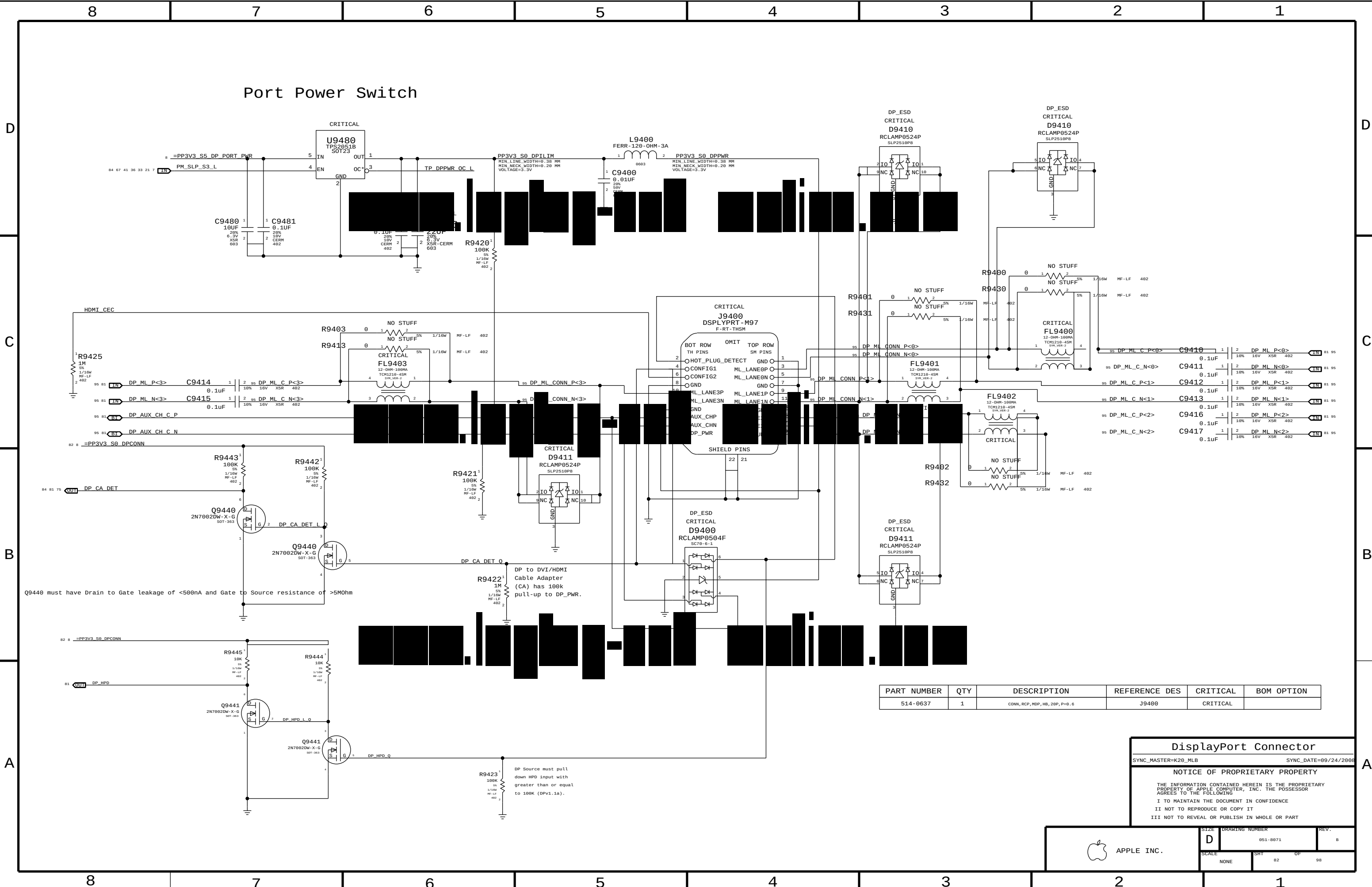
NONE

SHT

81

OF

98



| PART NUMBER | QTY | DESCRIPTION                    | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|--------------------------------|---------------|----------|------------|
| 514-0637    | 1   | CONN, RCP, MDP, HB, 28P, P=0.6 | J9400         | CRITICAL |            |

DisplayPort Connector

SYNC\_MASTER=K20\_MLB      SYNC\_DATE=09/24/2008

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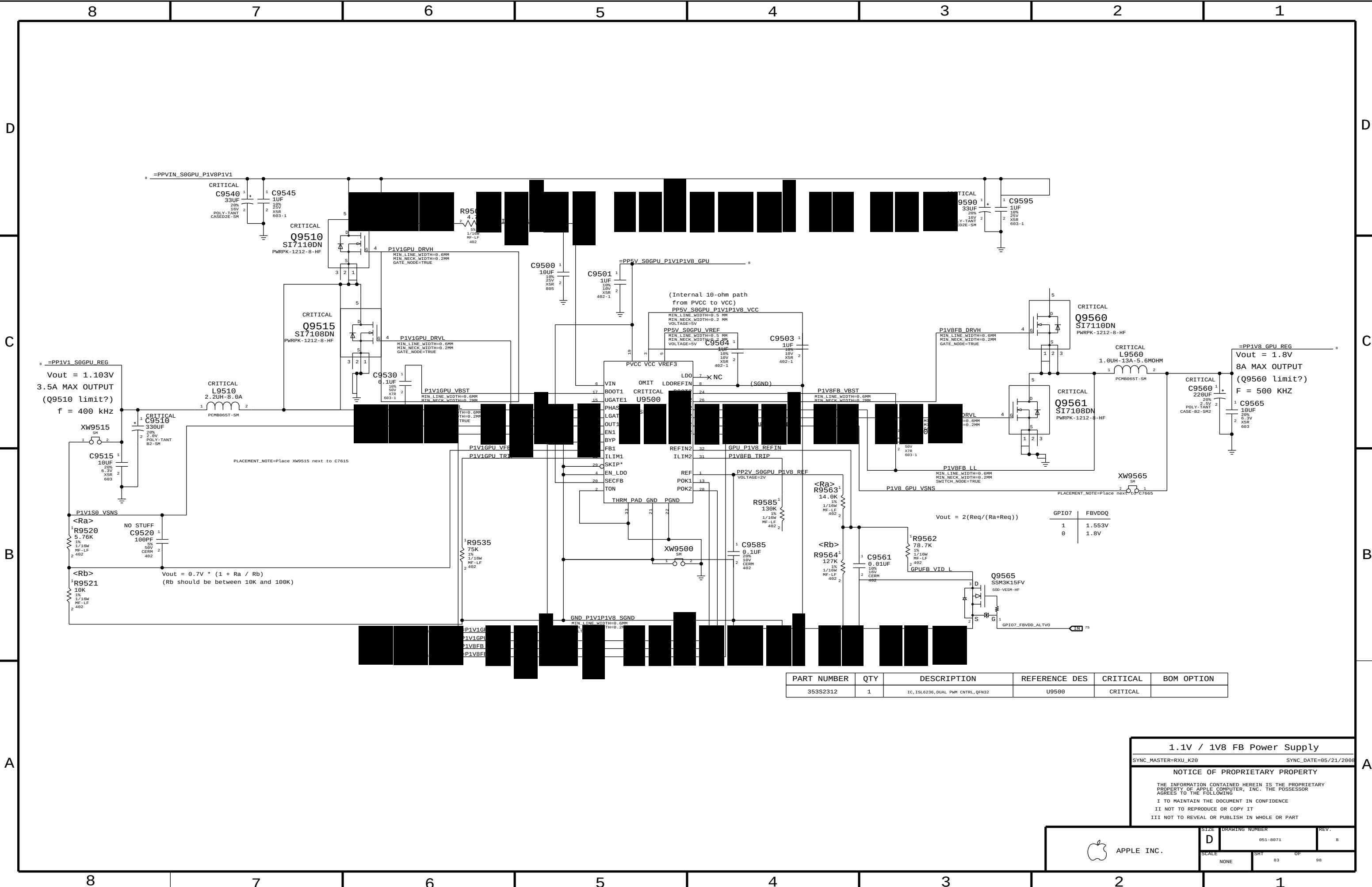
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APPLE INC.

D      051-8071      B

SCALE      NONE      SHT      82      OF      98



| PART NUMBER | QTY | DESCRIPTION                        | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|------------------------------------|---------------|----------|------------|
| 353S2312    | 1   | IC, ISL6236, DUAL PWM CNTRL, QFN32 | U9500         | CRITICAL |            |

1.1V / 1V8 FB Power Supply

SYNC\_MASTER=RXU\_K20      SYNC\_DATE=05/21/2008

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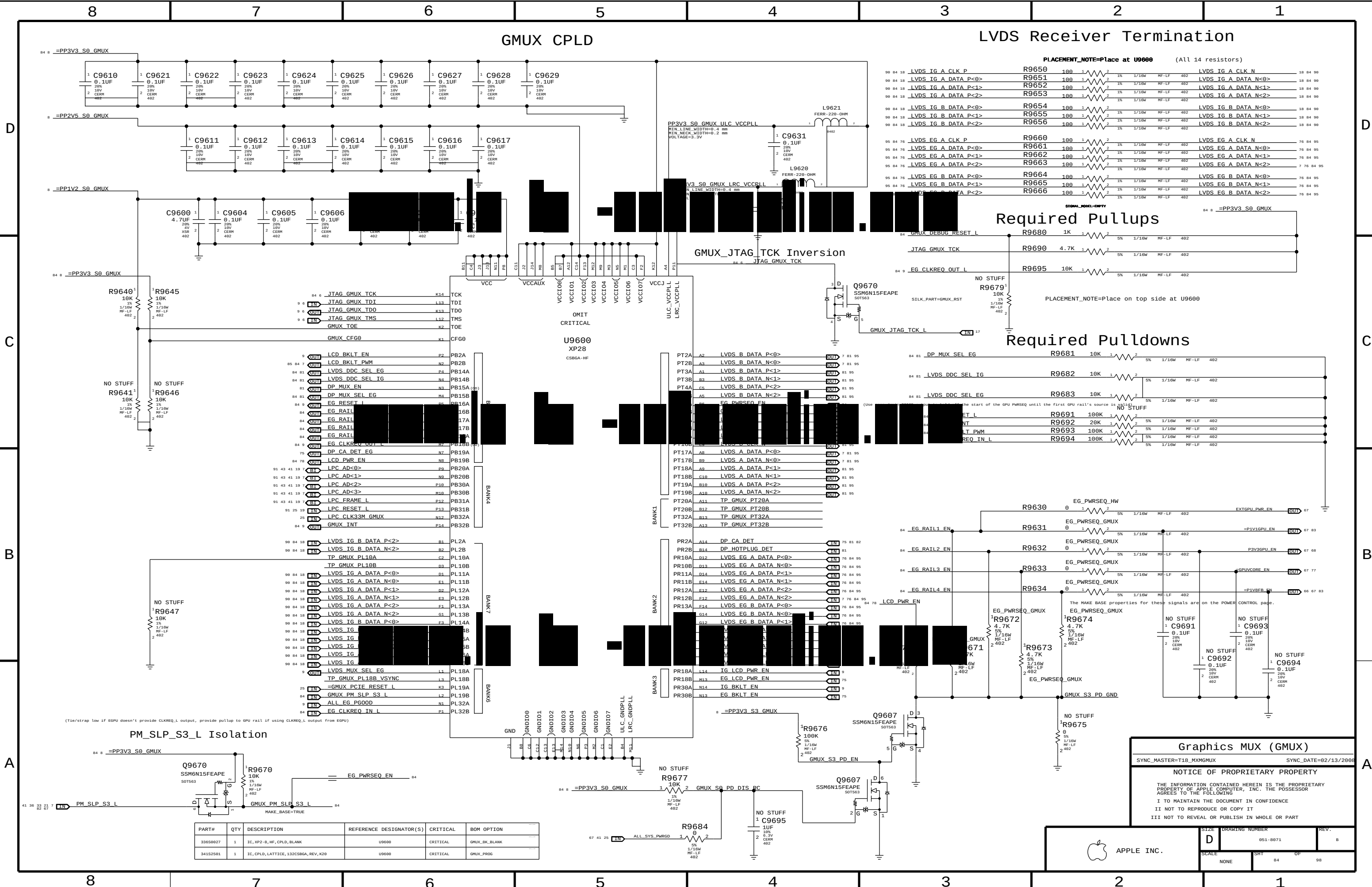
DRAWING NUMBER: 051-8071

REV.: B

SCALE: NONE

SHT: 83

OF: 98



| PART#    | QTY | DESCRIPTION                           | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION    |
|----------|-----|---------------------------------------|-------------------------|----------|---------------|
| 3358927  | 1   | IC, X2-8, HF, CPLD, BLANK             | U9600                   | CRITICAL | GMUX_SK_BLANK |
| 34152501 | 1   | IC, CPLD, LATTICE, 132CSBGA, REV, K20 | U9600                   | CRITICAL | GMUX_PROG     |

Graphics MUX (GMUX)

SYNC\_MASTER=T18\_MXGMUX

SYNC\_DATE=02/13/2008

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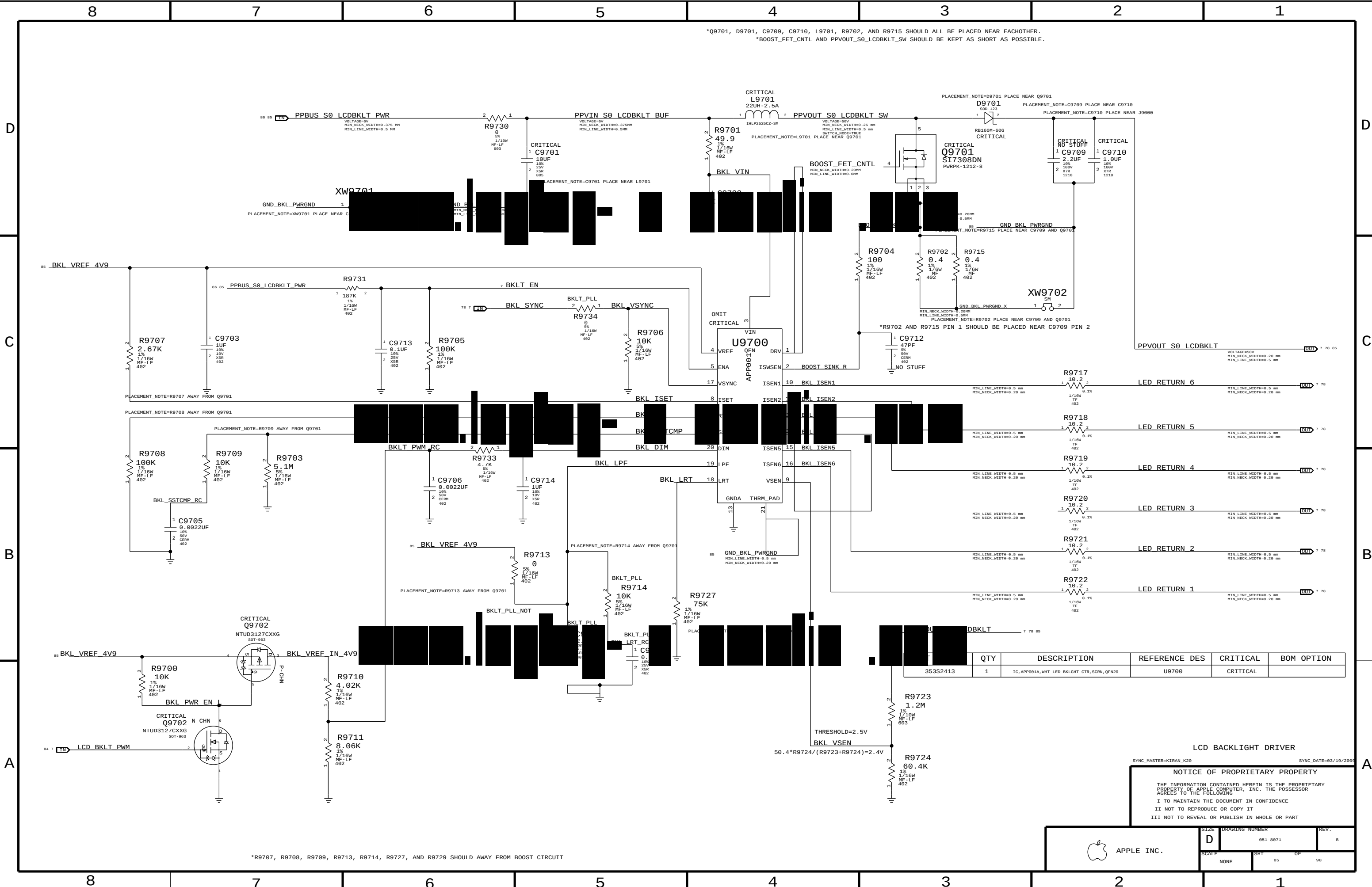
NONE

SHT

84

OF

98



| QTY | DESCRIPTION                                  | REFERENCE DES | CRITICAL | BOM OPTION |
|-----|----------------------------------------------|---------------|----------|------------|
| 1   | IC, APP001A, WHT LED BKLGHT CTR, SCRN, QFN20 | U9700         | CRITICAL |            |

LCD BACKLIGHT DRIVER

SYNC\_MASTER=KIRAN\_K20 SYNC\_DATE=03/19/2005

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| NONE  | 85             | 98   |

\*R9707, R9708, R9709, R9713, R9714, R9727, AND R9729 SHOULD AWAY FROM BOOST CIRCUIT





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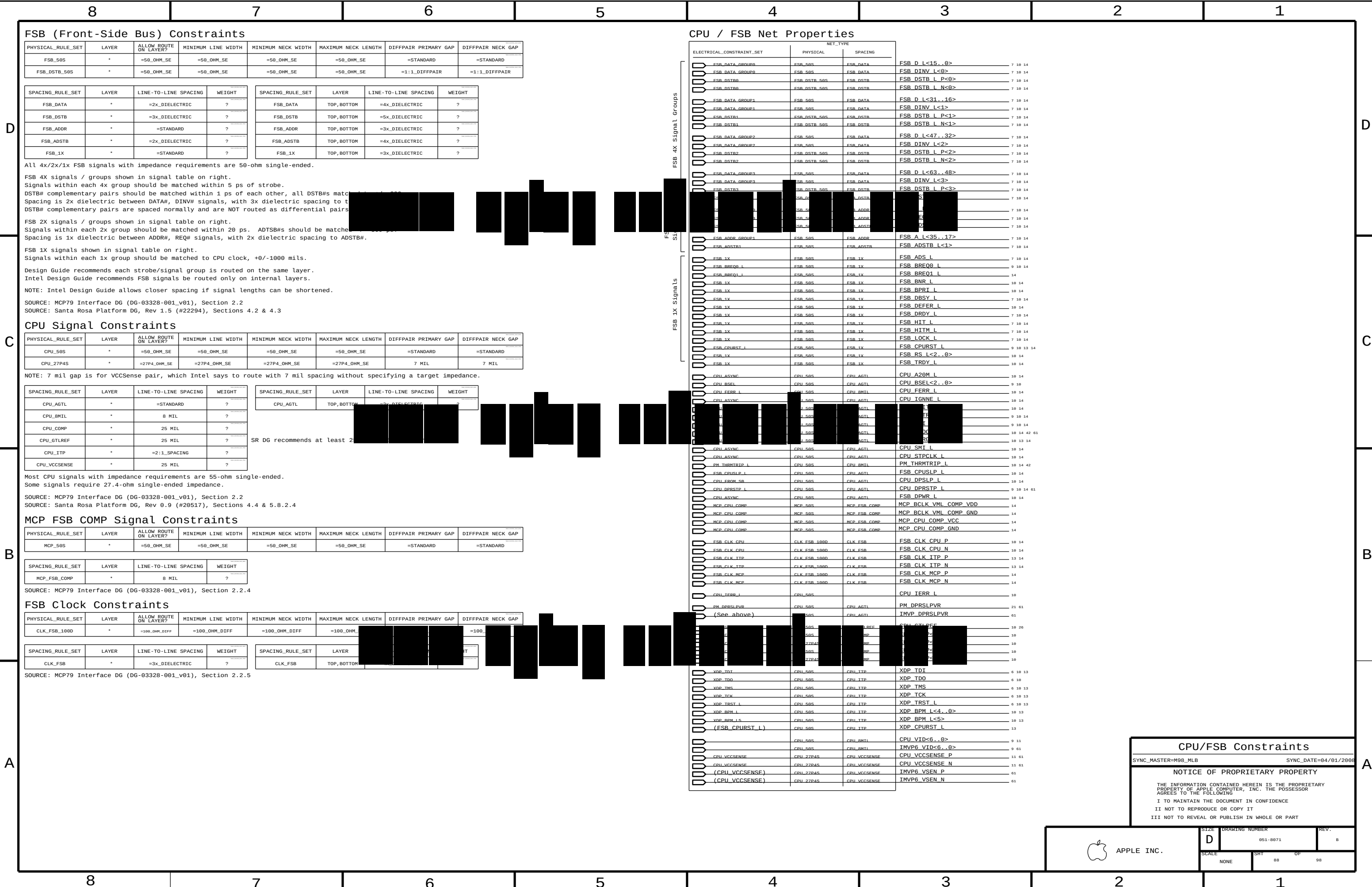


|                                                                                                  |       |                |  |      |
|--------------------------------------------------------------------------------------------------|-------|----------------|--|------|
|  APPLE INC. | SIZE  | DRAWING NUMBER |  | REV. |
|                                                                                                  | D     | 051-8071       |  | B    |
|                                                                                                  | SCALE | SHEET OF       |  |      |
|                                                                                                  | NONE  | 87 98          |  |      |

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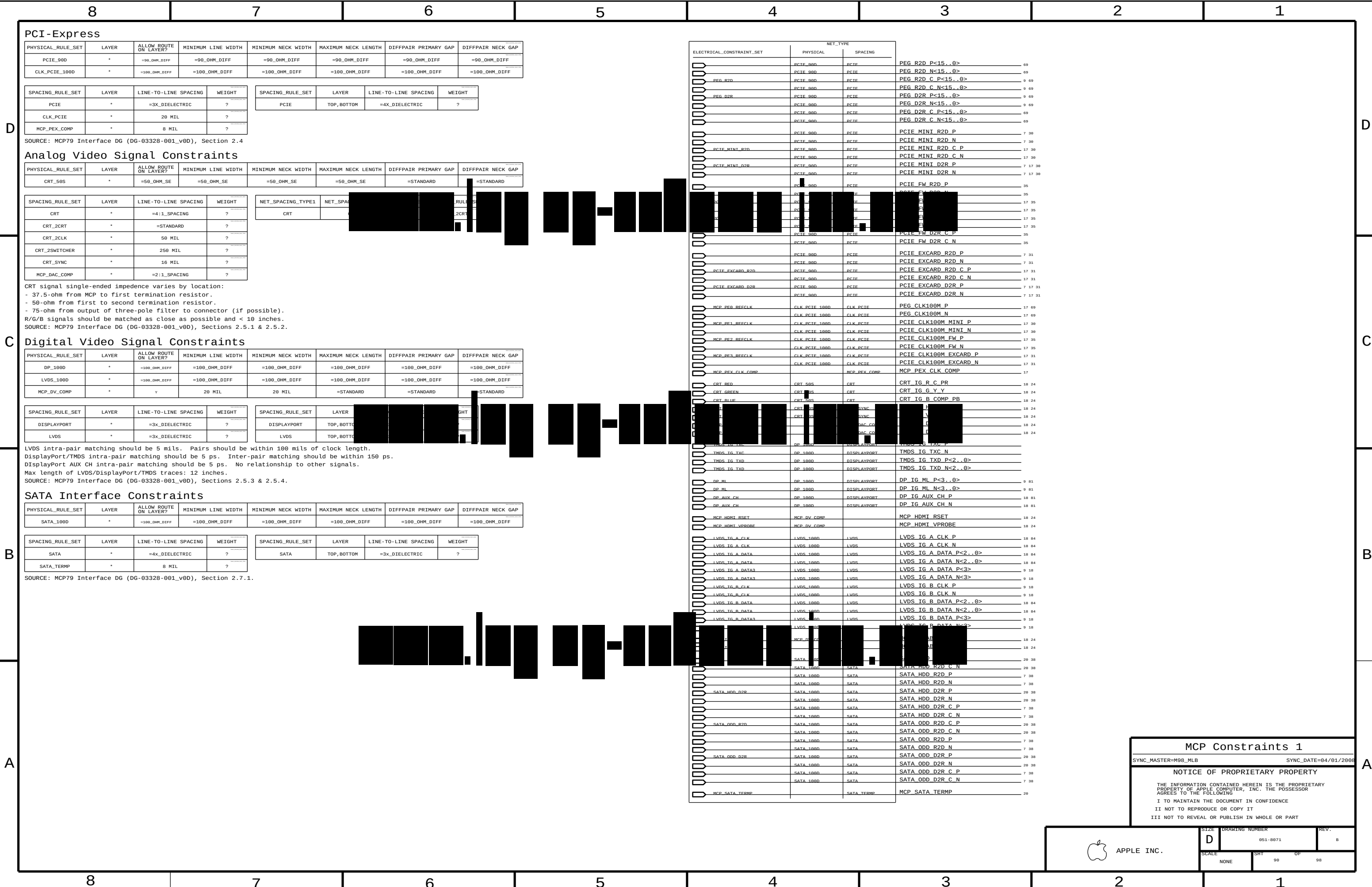
|       |     |    |
|-------|-----|----|
| SCALE | SHT | OF |
|       | 87  | 88 |

 APPLE INC.









PCI-Express

| PHYSICAL_RULE_SET | LAYER | ALLOW_ROUTE_ON_LAYER? | MINIMUM_LINE_WIDTH | MINIMUM_NECK_WIDTH | MAXIMUM_NECK_LENGTH | DIFFPAIR_PRIMARY_GAP | DIFFPAIR_NECK_GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| PCIE_960          | *     | =90_OHM_DIFF          | =90_OHM_DIFF       | =90_OHM_DIFF       | =90_OHM_DIFF        | =90_OHM_DIFF         | =90_OHM_DIFF      |
| CLK_PCIE_1080     | *     | =180_OHM_DIFF         | =180_OHM_DIFF      | =180_OHM_DIFF      | =180_OHM_DIFF       | =180_OHM_DIFF        | =180_OHM_DIFF     |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE_SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| PCIE             | *     | =3X_DIELECTRIC       | ?      |
| CLK_PCIE         | *     | 28_MIL               | ?      |
| MCP_PEX_COMP     | *     | 8_MIL                | ?      |

SOURCE: MCP79 Interface DG (DG-03328-001\_v0D), Section 2.4

Analog Video Signal Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW_ROUTE_ON_LAYER? | MINIMUM_LINE_WIDTH | MINIMUM_NECK_WIDTH | MAXIMUM_NECK_LENGTH | DIFFPAIR_PRIMARY_GAP | DIFFPAIR_NECK_GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| CRT_56S           | *     | =50_OHM_SE            | =50_OHM_SE         | =50_OHM_SE         | =50_OHM_SE          | =STANDARD            | =STANDARD         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE_SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| CRT              | *     | =4:1_SPACING         | ?      |
| CRT_2CRT         | *     | =STANDARD            | ?      |
| CRT_2CLK         | *     | 50_MIL               | ?      |
| CRT_2SWITCHER    | *     | 250_MIL              | ?      |
| CRT_SYNC         | *     | 16_MIL               | ?      |
| MCP_DAC_COMP     | *     | =2:1_SPACING         | ?      |

CRT signal single-ended impedance varies by location:  
- 37.5-ohm from MCP to first termination resistor.  
- 50-ohm from first to second termination resistor.  
- 75-ohm from output of three-pole filter to connector (if possible).  
R/G/B signals should be matched as close as possible and < 10 inches.  
SOURCE: MCP79 Interface DG (DG-03328-001\_v0D), Sections 2.5.1 & 2.5.2.

Digital Video Signal Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW_ROUTE_ON_LAYER? | MINIMUM_LINE_WIDTH | MINIMUM_NECK_WIDTH | MAXIMUM_NECK_LENGTH | DIFFPAIR_PRIMARY_GAP | DIFFPAIR_NECK_GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| DP_1080           | *     | =180_OHM_DIFF         | =180_OHM_DIFF      | =180_OHM_DIFF      | =180_OHM_DIFF       | =180_OHM_DIFF        | =180_OHM_DIFF     |
| LVDS_1080         | *     | =180_OHM_DIFF         | =180_OHM_DIFF      | =180_OHM_DIFF      | =180_OHM_DIFF       | =180_OHM_DIFF        | =180_OHM_DIFF     |
| MCP_DV_COMP       | *     | Y                     | 20_MIL             | 20_MIL             | =STANDARD           | =STANDARD            | =STANDARD         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE_SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| DISPLAYPORT      | *     | =3X_DIELECTRIC       | ?      |
| LVDS             | *     | =3X_DIELECTRIC       | ?      |

LVDS intra-pair matching should be 5 mils. Pairs should be within 100 mils of clock length.  
DisplayPort/TMDS intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.  
DisplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals.  
Max length of LVDS/DisplayPort/TMDS traces: 12 inches.  
SOURCE: MCP79 Interface DG (DG-03328-001\_v0D), Sections 2.5.3 & 2.5.4.

SATA Interface Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW_ROUTE_ON_LAYER? | MINIMUM_LINE_WIDTH | MINIMUM_NECK_WIDTH | MAXIMUM_NECK_LENGTH | DIFFPAIR_PRIMARY_GAP | DIFFPAIR_NECK_GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| SATA_1080         | *     | =180_OHM_DIFF         | =180_OHM_DIFF      | =180_OHM_DIFF      | =180_OHM_DIFF       | =180_OHM_DIFF        | =180_OHM_DIFF     |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE_SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| SATA             | *     | =4X_DIELECTRIC       | ?      |
| SATA_TERM        | *     | 8_MIL                | ?      |

SOURCE: MCP79 Interface DG (DG-03328-001\_v0D), Section 2.7.1.

| ELECTRICAL_CONSTRAINT_SET | NET_TYPE      |              |                        |
|---------------------------|---------------|--------------|------------------------|
|                           | PHYSICAL      | SPACING      |                        |
| PEG_R2D                   | PCIE_960      | PCIE         | PEG_R2D P<15..0>       |
| PEG_R2D                   | PCIE_960      | PCIE         | PEG_R2D N<15..0>       |
| PEG_R2D                   | PCIE_960      | PCIE         | PEG_R2D C P<15..0>     |
| PEG_R2D                   | PCIE_960      | PCIE         | PEG_R2D C N<15..0>     |
| PEG_D2R                   | PCIE_960      | PCIE         | PEG_D2R P<15..0>       |
| PEG_D2R                   | PCIE_960      | PCIE         | PEG_D2R N<15..0>       |
| PEG_D2R                   | PCIE_960      | PCIE         | PEG_D2R C P<15..0>     |
| PEG_D2R                   | PCIE_960      | PCIE         | PEG_D2R C N<15..0>     |
| PCIE_MINI_R2D_P           | PCIE_960      | PCIE         | PCIE_MINI_R2D_P        |
| PCIE_MINI_R2D_N           | PCIE_960      | PCIE         | PCIE_MINI_R2D_N        |
| PCIE_MINI_R2D_C_P         | PCIE_960      | PCIE         | PCIE_MINI_R2D_C_P      |
| PCIE_MINI_R2D_C_N         | PCIE_960      | PCIE         | PCIE_MINI_R2D_C_N      |
| PCIE_MINI_D2R_P           | PCIE_960      | PCIE         | PCIE_MINI_D2R_P        |
| PCIE_MINI_D2R_N           | PCIE_960      | PCIE         | PCIE_MINI_D2R_N        |
| PCIE_FW_R2D_P             | PCIE_960      | PCIE         | PCIE_FW_R2D_P          |
| PCIE_FW_R2D_N             | PCIE_960      | PCIE         | PCIE_FW_R2D_N          |
| PCIE_EXCARD_R2D_P         | PCIE_960      | PCIE         | PCIE_EXCARD_R2D_P      |
| PCIE_EXCARD_R2D_N         | PCIE_960      | PCIE         | PCIE_EXCARD_R2D_N      |
| PCIE_EXCARD_R2D_C_P       | PCIE_960      | PCIE         | PCIE_EXCARD_R2D_C_P    |
| PCIE_EXCARD_R2D_C_N       | PCIE_960      | PCIE         | PCIE_EXCARD_R2D_C_N    |
| PCIE_EXCARD_D2R_P         | PCIE_960      | PCIE         | PCIE_EXCARD_D2R_P      |
| PCIE_EXCARD_D2R_N         | PCIE_960      | PCIE         | PCIE_EXCARD_D2R_N      |
| PEG_CLK100M_P             | CLK_PCIE_1080 | CLK_PCIE     | PEG_CLK100M_P          |
| PEG_CLK100M_N             | CLK_PCIE_1080 | CLK_PCIE     | PEG_CLK100M_N          |
| PCIE_CLK100M_MINI_P       | CLK_PCIE_1080 | CLK_PCIE     | PCIE_CLK100M_MINI_P    |
| PCIE_CLK100M_MINI_N       | CLK_PCIE_1080 | CLK_PCIE     | PCIE_CLK100M_MINI_N    |
| PCIE_CLK100M_FW_P         | CLK_PCIE_1080 | CLK_PCIE     | PCIE_CLK100M_FW_P      |
| PCIE_CLK100M_FW_N         | CLK_PCIE_1080 | CLK_PCIE     | PCIE_CLK100M_FW_N      |
| PCIE_CLK100M_EXCARD_P     | CLK_PCIE_1080 | CLK_PCIE     | PCIE_CLK100M_EXCARD_P  |
| PCIE_CLK100M_EXCARD_N     | CLK_PCIE_1080 | CLK_PCIE     | PCIE_CLK100M_EXCARD_N  |
| MCP_PEX_CLK_COMP          |               | MCP_PEX_COMP | MCP_PEX_CLK_COMP       |
| CRT_RED                   | CRT_56S       | CRT          | CRT_IG_R_C_PR          |
| CRT_GREEN                 | CRT_56S       | CRT          | CRT_IG_G_Y_Y           |
| CRT_BLUE                  | CRT_56S       | CRT          | CRT_IG_B_COMP_PB       |
| CRT_SYNC                  | CRT_SYNC      | CRT          | CRT_SYNC               |
| CRT_SYNC                  | CRT_SYNC      | CRT          | CRT_SYNC               |
| CRT_SYNC                  | CRT_SYNC      | CRT          | CRT_SYNC               |
| CRT_SYNC                  | CRT_SYNC      | CRT          | CRT_SYNC               |
| DP_1080                   | DP_1080       | DISPLAYPORT  | DP_IG_TXC_P            |
| DP_1080                   | DP_1080       | DISPLAYPORT  | DP_IG_TXC_N            |
| DP_1080                   | DP_1080       | DISPLAYPORT  | DP_IG_TXD_P<2..0>      |
| DP_1080                   | DP_1080       | DISPLAYPORT  | DP_IG_TXD_N<2..0>      |
| DP_ML                     | DP_1080       | DISPLAYPORT  | DP_IG_ML_P<3..0>       |
| DP_ML                     | DP_1080       | DISPLAYPORT  | DP_IG_ML_N<3..0>       |
| DP_AUX_CH                 | DP_1080       | DISPLAYPORT  | DP_IG_AUX_CH_P         |
| DP_AUX_CH                 | DP_1080       | DISPLAYPORT  | DP_IG_AUX_CH_N         |
| MCP_HDMI_RSET             | MCP_DV_COMP   |              | MCP_HDMI_RSET          |
| MCP_HDMI_VPROBE           | MCP_DV_COMP   |              | MCP_HDMI_VPROBE        |
| LVDS_IG_A_CLK             | LVDS_1080     | LVDS         | LVDS_IG_A_CLK_P        |
| LVDS_IG_A_CLK             | LVDS_1080     | LVDS         | LVDS_IG_A_CLK_N        |
| LVDS_IG_A_DATA            | LVDS_1080     | LVDS         | LVDS_IG_A_DATA_P<2..0> |
| LVDS_IG_A_DATA            | LVDS_1080     | LVDS         | LVDS_IG_A_DATA_N<2..0> |
| LVDS_IG_A_DATA3           | LVDS_1080     | LVDS         | LVDS_IG_A_DATA_P<3>    |
| LVDS_IG_A_DATA3           | LVDS_1080     | LVDS         | LVDS_IG_A_DATA_N<3>    |
| LVDS_IG_B_CLK             | LVDS_1080     | LVDS         | LVDS_IG_B_CLK_P        |
| LVDS_IG_B_CLK             | LVDS_1080     | LVDS         | LVDS_IG_B_CLK_N        |
| LVDS_IG_B_DATA            | LVDS_1080     | LVDS         | LVDS_IG_B_DATA_P<2..0> |
| LVDS_IG_B_DATA            | LVDS_1080     | LVDS         | LVDS_IG_B_DATA_N<2..0> |
| LVDS_IG_B_DATA3           | LVDS_1080     | LVDS         | LVDS_IG_B_DATA_P<3>    |
| LVDS_IG_B_DATA3           | LVDS_1080     | LVDS         | LVDS_IG_B_DATA_N<3>    |
| MCP_SATA_TERM             | MCP_SATA_TERM |              | MCP_SATA_TERM          |
| SATA_HDD_R2D_C_N          | SATA_1080     | SATA         | SATA_HDD_R2D_C_N       |
| SATA_HDD_R2D_P            | SATA_1080     | SATA         | SATA_HDD_R2D_P         |
| SATA_HDD_R2D_N            | SATA_1080     | SATA         | SATA_HDD_R2D_N         |
| SATA_HDD_D2R_P            | SATA_1080     | SATA         | SATA_HDD_D2R_P         |
| SATA_HDD_D2R_N            | SATA_1080     | SATA         | SATA_HDD_D2R_N         |
| SATA_HDD_D2R_C_P          | SATA_1080     | SATA         | SATA_HDD_D2R_C_P       |
| SATA_HDD_D2R_C_N          | SATA_1080     | SATA         | SATA_HDD_D2R_C_N       |
| SATA_ODD_R2D_C_P          | SATA_1080     | SATA         | SATA_ODD_R2D_C_P       |
| SATA_ODD_R2D_C_N          | SATA_1080     | SATA         | SATA_ODD_R2D_C_N       |
| SATA_ODD_R2D_P            | SATA_1080     | SATA         | SATA_ODD_R2D_P         |
| SATA_ODD_R2D_N            | SATA_1080     | SATA         | SATA_ODD_R2D_N         |
| SATA_ODD_D2R_P            | SATA_1080     | SATA         | SATA_ODD_D2R_P         |
| SATA_ODD_D2R_N            | SATA_1080     | SATA         | SATA_ODD_D2R_N         |
| SATA_ODD_D2R_C_P          | SATA_1080     | SATA         | SATA_ODD_D2R_C_P       |
| SATA_ODD_D2R_C_N          | SATA_1080     | SATA         | SATA_ODD_D2R_C_N       |
| MCP_SATA_TERM             | MCP_SATA_TERM |              | MCP_SATA_TERM          |

MCP Constraints 1

SYNC\_MASTER=M98\_MLB SYNC\_DATE=04/01/2008

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98





MCP RGMII (Ethernet) Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW_ROUTE_ON_LAYER? | MINIMUM_LINE_WIDTH | MINIMUM_NECK_WIDTH | MAXIMUM_NECK_LENGTH | DIFFPAIR_PRIMARY_GAP | DIFFPAIR_NECK_GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| MCP_MII_COMP      | *     | =STANDARD             | 7.5 MIL            | 7.5 MIL            | =STANDARD           | =STANDARD            | =STANDARD         |
| ENET_MII_55S      | *     | =55_OHM_SE            | =55_OHM_SE         | =55_OHM_SE         | =55_OHM_SE          | =STANDARD            | =STANDARD         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE_SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| MCP_BUF0_CLK     | *     | =3:1_SPACING         | ?      |
| ENET_MII         | *     | 12 MIL               | ?      |

SOURCE: MCP73 Interface DG (DG-02974-001\_v01), Sections 2.7.2 & 2.7.4

88E1116R (Ethernet PHY) Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW_ROUTE_ON_LAYER? | MINIMUM_LINE_WIDTH | MINIMUM_NECK_WIDTH | MAXIMUM_NECK_LENGTH | DIFFPAIR_PRIMARY_GAP | DIFFPAIR_NECK_GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| ENET_MDI_100D     | *     | =100_OHM_DIFF         | =100_OHM_DIFF      | =100_OHM_DIFF      | =100_OHM_DIFF       | =100_OHM_DIFF        | =100_OHM_DIFF     |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE_SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| ENET_MDI         | *     | 25 MIL               | ?      |

SOURCE: MCP73 Interface DG (DG-02974-001\_v01), Section 2.7.4

| NET_TYPE                  |               |              |                              |
|---------------------------|---------------|--------------|------------------------------|
| ELECTRICAL_CONSTRAINT_SET | PHYSICAL      | SPACING      |                              |
| MCP_MII_COMP              | MCP_MII_COMP  |              | MCP_MII_COMP_VDD 18          |
| MCP_MII_COMP              | MCP_MII_COMP  |              | MCP_MII_COMP_GND 18          |
| MCP_CLK25M_BUF0           | ENET_MII_55S  | MCP_BUF0_CLK | MCP_CLK25M_BUF0_R 18 33      |
|                           | ENET_MII_55S  | MCP_BUF0_CLK | RTL8211_CLK25M_CKXTAL1 32 33 |
| ENET_INTR_I               | ENET_MII_55S  | ENET_MII     | ENET_INTR_L 18 32            |
| ENET_MDIO                 | ENET_MII_55S  | ENET_MII     | ENET_MDIO 18 32              |
| ENET_MDC                  | ENET_MII_55S  | ENET_MII     | ENET_MDC 18 32               |
| ENET_PWDOWN_I             | ENET_MII_55S  | ENET_MII     | ENET_PWDOWN_L 18 32          |
|                           | ENET_MII_55S  | ENET_MII     | ENET_CLK125M_RXCLK_R 32      |
| ENET_RXCLK                | ENET_MII_55S  | ENET_MII     | ENET_CLK125M_RXCLK 18 32     |
|                           | ENET_MII_55S  | ENET_MII     | ENET_RXD_R<3..0> 32          |
| ENET_RXD                  | ENET_MII_55S  | ENET_MII     | ENET_RXD<0> 18 32            |
| ENET_RXD_STRAP            | ENET_MII_55S  | ENET_MII     | ENET_RXD<3..1> 18 32         |
| ENET_RXD                  | ENET_MII_55S  | ENET_MII     | ENET_RX_CTRL 18 32           |
|                           | ENET_MII_55S  | ENET_MII     | ENET_CLK125M_TXCLK 18 32     |
|                           | ENET_MII_55S  | ENET_MII     | ENET_TXD_I<0> 18 32          |
|                           | ENET_MII_55S  | ENET_MII     | ENET_TXD_I<1> 18 32          |
|                           | ENET_MII_55S  | ENET_MII     | ENET_TXD_I<2> 18 32          |
|                           | ENET_MII_55S  | ENET_MII     | ENET_TXD_I<3> 18 32          |
|                           | ENET_MII_55S  | ENET_MII     | ENET_TXD_I<4> 18 32          |
|                           | ENET_MII_55S  | ENET_MII     | ENET_TXD_I<5> 18 32          |
|                           | ENET_MII_55S  | ENET_MII     | ENET_TXD_I<6> 18 32          |
|                           | ENET_MII_55S  | ENET_MII     | ENET_TXD_I<7> 18 32          |
| ENET_MDI                  | ENET_MDI_100D | ENET_MDI     | ENET_MDI_P<3..0> 32 34       |
|                           | ENET_MDI_100D | ENET_MDI     | ENET_MDI_N<3..0> 32 34       |

Ethernet Constraints

SYNC\_MASTER=M9B\_MLB

SYNC\_DATE=04/01/2009

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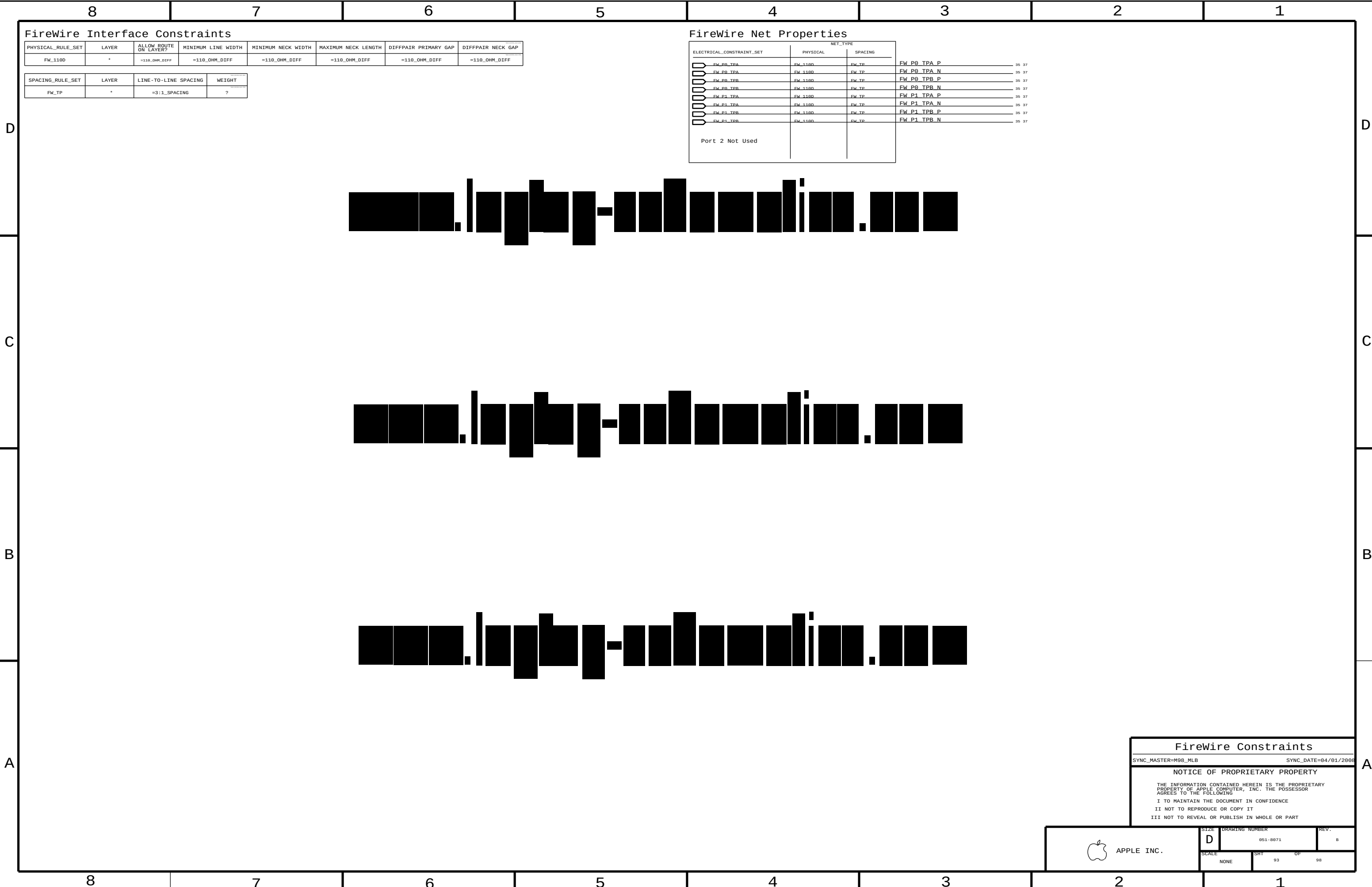
DRAWING NUMBER: 051-8071

REV.: B

SCALE: NONE

SHT: 92

OF: 98



FireWire Interface Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW_ROUTE_ON_LAYER? | MINIMUM_LINE_WIDTH | MINIMUM_NECK_WIDTH | MAXIMUM_NECK_LENGTH | DIFFPAIR_PRIMARY_GAP | DIFFPAIR_NECK_GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| FW_1100           | *     | =110_OHM_DIFF         | =110_OHM_DIFF      | =110_OHM_DIFF      | =110_OHM_DIFF       | =110_OHM_DIFF        | =110_OHM_DIFF     |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE_SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| FW_TP            | *     | =3:1_SPACING         | ?      |

FireWire Net Properties

| NET_TYPE                  |          |         |                   |
|---------------------------|----------|---------|-------------------|
| ELECTRICAL_CONSTRAINT_SET | PHYSICAL | SPACING |                   |
| FW_P0_TPA                 | FW_1100  | FW_TP   | FW_P0_TPA_P 35 37 |
| FW_P0_TPA                 | FW_1100  | FW_TP   | FW_P0_TPA_N 35 37 |
| FW_P0_TPB                 | FW_1100  | FW_TP   | FW_P0_TPB_P 35 37 |
| FW_P0_TPB                 | FW_1100  | FW_TP   | FW_P0_TPB_N 35 37 |
| FW_P1_TPA                 | FW_1100  | FW_TP   | FW_P1_TPA_P 35 37 |
| FW_P1_TPA                 | FW_1100  | FW_TP   | FW_P1_TPA_N 35 37 |
| FW_P1_TPB                 | FW_1100  | FW_TP   | FW_P1_TPB_P 35 37 |
| FW_P1_TPB                 | FW_1100  | FW_TP   | FW_P1_TPB_N 35 37 |
| Port 2 Not Used           |          |         |                   |

FireWire Constraints

SYNC\_MASTER=M98\_MLB

SYNC\_DATE=04/01/2009

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| SIZE  | DRAWING NUMBER | REV. |
| D     | 051-8071       | B    |
| SCALE | SHT            | OF   |
| NONE  | 93             | 98   |



| PHYSICAL_RULE_SET | LAYER | ALLOW_ROUTE_ON_LAYER? | MINIMUM_LINE_WIDTH | MINIMUM_NECK_WIDTH | MAXIMUM_NECK_LENGTH | DIFFPAIR_PRIMARY_GAP | DIFFPAIR_NECK_GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 1T01_DIFFPAIR     | *     | =STANDARD             | =STANDARD          | =STANDARD          | =STANDARD           | 0.1 MM               | 0.1 MM            |

| SMC SMBus Net Properties                                                            |                    |          |         |                         |
|-------------------------------------------------------------------------------------|--------------------|----------|---------|-------------------------|
|                                                                                     |                    | NET_TYPE |         |                         |
| ELECTRICAL_CONSTRAINT_SET                                                           |                    | PHYSICAL | SPACING |                         |
|  | SMBUS_SMC_A_S3_SCL | SMB_55G  | SMB     | SMBUS_SMC_A_S3_SCL 7 44 |
|  | SMBUS_SMC_A_S3_SDA | SMB_55G  | SMB     | SMBUS_SMC_A_S3_SDA 7 44 |
|  | SMBUS_SMC_B_S0_SCL | SMB_55G  | SMB     | SMBUS_SMC_B_S0_SCL 44   |
|  | SMBUS_SMC_B_S0_SDA | SMB_55G  | SMB     | SMBUS_SMC_B_S0_SDA 44   |
|  | SMBUS_SMC_0_S0_SCL | SMB_55G  | SMB     | SMBUS_SMC_0_S0_SCL 44   |
|  | SMBUS_SMC_0_S0_SDA | SMB_55G  | SMB     | SMBUS_SMC_0_S0_SDA 44   |
|  | SMBUS_SMC_BSA_SCL  | SMB_55G  | SMB     | SMBUS_SMC_BSA_SCL 7 44  |
|  | SMBUS_SMC_BSA_SDA  | SMB_55G  | SMB     | SMBUS_SMC_BSA_SDA 7 44  |
|  | SMBUS_SMC_MGMT_SCL | SMB_55G  | SMB     | SMBUS_SMC_MGMT_SCL 44   |
|  | SMBUS_SMC_MGMT_SDA | SMB_55G  | SMB     | SMBUS_SMC_MGMT_SDA 44   |

| SMBus Charger Net Properties |  |              |            |    |
|------------------------------|--|--------------|------------|----|
| ELECTRICAL_CONSTRAINT_SET    |  | NET_TYPE     |            |    |
|                              |  | PHYSICAL     | SPACING    |    |
| CHGR_CST                     |  | T01_DIFFPAIR | CHGR_CST_P | 60 |
|                              |  |              |            | 60 |
|                              |  |              |            | 60 |
|                              |  |              |            | 60 |

SMC Constraints

SYNC\_MASTER=M9B\_MLB

SYNC\_DATE=04/01/2009

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SIZE: D

DRAWING NUMBER: 051-8071

REV.: B

SCALE: NONE

SHT: 94

OF: 98



## D

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From T18 MXM:

## B

A

LVDS intra-pair matching should be 5 mils. Pairs should be within 100 mils of clock length.  
DisplayPort/TMD5 intra-pair matching should be 5 ps. Inter-pair matching should be within 10 ps.  
DisplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals.  
Max length of LVDS/DisplayPort/TMD5 traces: 12 inches.  
SOURCE: MCP79 Interface DG (DG-03328-001\_v0D), Sections 2.5.3 & 2.5.4.

## B

A

## 996 Net Properties

| ELECTRICAL_CONSTRAINT_SET                                                             |                | NET_TYPE     |             |
|---------------------------------------------------------------------------------------|----------------|--------------|-------------|
|                                                                                       |                | PHYSICAL     | SPACING     |
|  | (CK565_P0735)  | CLK_SLOW_555 | CLK_SLOW    |
|  | CK565_CLK275SS | CLK_SLOW_555 | CLK_SLOW    |
|  | LVDS_EG_A_CLK  | LVDS_1800    | LVDS        |
|  | LVDS_EG_A_CLK  | LVDS_1800    | LVDS        |
|  | LVDS_EG_A_DATA | LVDS_1800    | LVDS        |
|  | LVDS_EG_A_DATA | LVDS_1800    | LVDS        |
|  | LVDS_EG_B_DATA | LVDS_1800    | LVDS        |
|  | LVDS_EG_B_DATA | LVDS_1800    | LVDS        |
|  | DP_1800        | DP_1800      | DTSP1AYP0RT |

## 996 Net Properties

| ELECTRICAL_CONSTRAINT_SET                                                             |                | NET_TYPE     |          |
|---------------------------------------------------------------------------------------|----------------|--------------|----------|
|                                                                                       |                | PHYSICAL     | SPACING  |
|  | (CK565_P0735)  | CLK_SLOW_555 | CLK_SLOW |
|  | CK565_CLK2555  | CLK_SLOW_555 | CLK_SLOW |
|  | LVDS_EG_A_CLK  | LVDS_1800    | LVDS     |
|  | LVDS_EG_A_CLK  | LVDS_1800    | LVDS     |
|  | LVDS_EG_A_DATA | LVDS_1800    | LVDS     |
|  | LVDS_EG_A_DATA | LVDS_1800    | LVDS     |
|  | LVDS_EG_B_DATA | LVDS_1800    | LVDS     |
|  | LVDS_EG_B_DATA | LVDS_1800    | LVDS     |
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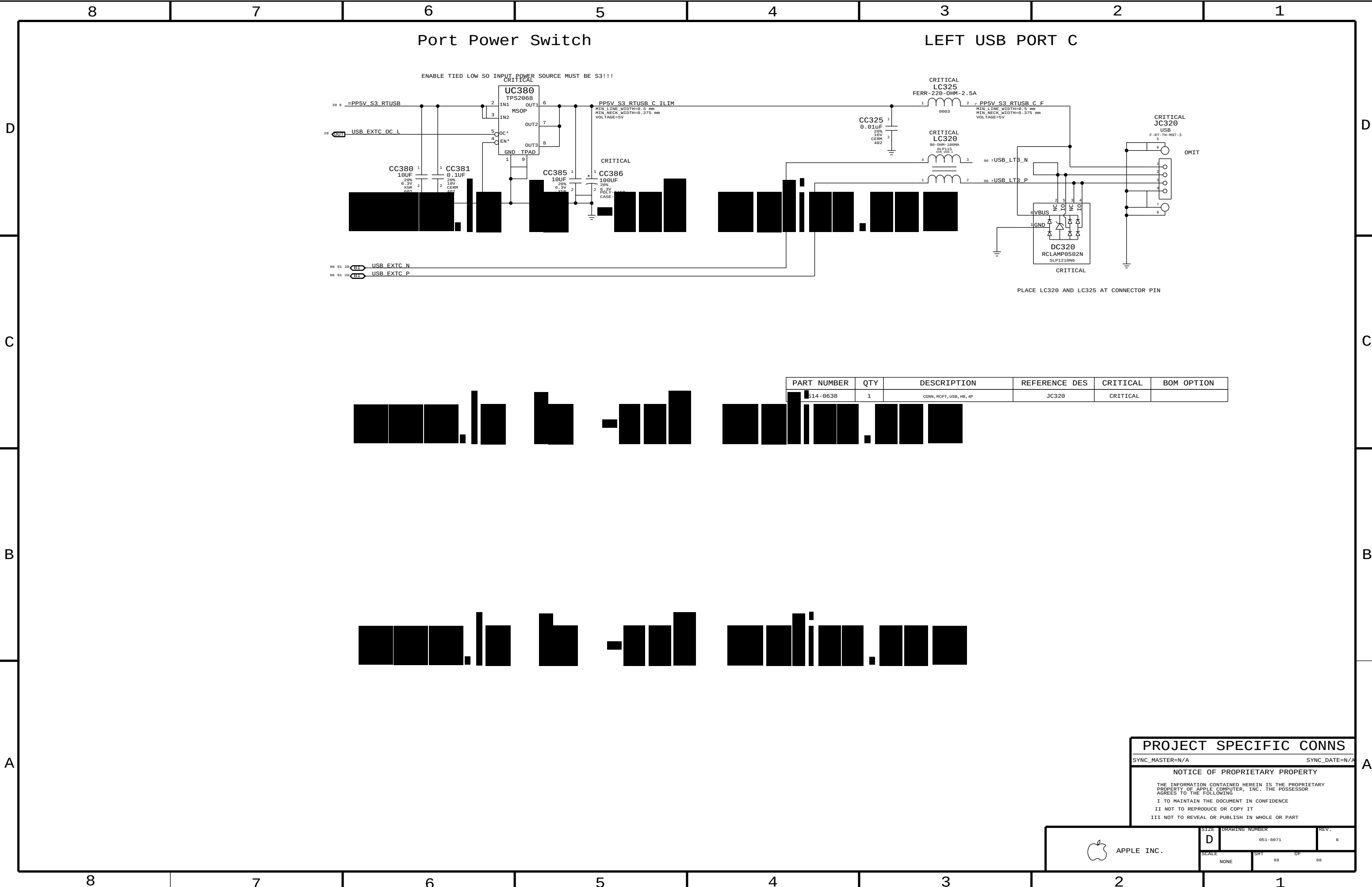
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|----------------------------------------------------------------------------------------------------------------------------|-------------------|-----------------------|--------------------|-----------------------------------------------------|-------------------------|----------------------|-------------------|
| 8                                                                                                                          | 7                 | 6                     | 5                  | 4                                                   | 3                       | 2                    | 1                 |
| M99 Board-Specific Spacing & Physical Constraints                                                                          |                   |                       |                    |                                                     |                         |                      |                   |
| BOARD LAYERS                                                                                                               |                   |                       | BOARD AREAS        |                                                     | BOARD UNITS (MIL OR MM) | ALLEGRO VERSION      |                   |
| TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM                                                  |                   |                       | NO_TYPE, BGA       |                                                     | MM                      | 15.5.1               |                   |
| PHYSICAL_RULE_SET                                                                                                          | LAYER             | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH                                  | MAXIMUM NECK LENGTH     | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
| DEFAULT                                                                                                                    | *                 | Y                     | =50_OHM_SE         | =50_OHM_SE                                          | 10 MM                   | 0 MM                 | 0 MM              |
| STANDARD                                                                                                                   | *                 | Y                     | =DEFAULT           | =DEFAULT                                            | 10 MM                   | =DEFAULT             | =DEFAULT          |
| PHYSICAL_RULE_SET                                                                                                          | LAYER             | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH                                  | MAXIMUM NECK LENGTH     | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
| 55_OHM_SE                                                                                                                  | TOP, BOTTOM       | Y                     | 0.090 MM           | 0.090 MM                                            |                         |                      |                   |
| 55_OHM_SE                                                                                                                  | *                 | Y                     | 0.076 MM           | 0.076 MM                                            | =STANDARD               | =STANDARD            | =STANDARD         |
| PHYSICAL_RULE_SET                                                                                                          | LAYER             | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH                                  | MAXIMUM NECK LENGTH     | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
| 50_OHM_SE                                                                                                                  | TOP, BOTTOM       | Y                     | 0.110 MM           | 0.095 MM                                            |                         |                      |                   |
| 50_OHM_SE                                                                                                                  | *                 | Y                     | 0.090 MM           | 0.090 MM                                            | =STANDARD               | =STANDARD            | =STANDARD         |
| PHYSICAL_RULE_SET                                                                                                          | LAYER             | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH                                  | MAXIMUM NECK LENGTH     | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
| 40_OHM_SE                                                                                                                  | TOP, BOTTOM       | Y                     | 0.165 MM           | 0.095 MM                                            |                         |                      |                   |
| 40_OHM_SE                                                                                                                  | *                 | Y                     | 0.135 MM           | 0.135 MM                                            | =STANDARD               | =STANDARD            | =STANDARD         |
| PHYSICAL_RULE_SET                                                                                                          | LAYER             | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH                                  | MAXIMUM NECK LENGTH     | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
| 27P4_OHM_SE                                                                                                                | TOP, BOTTOM       | Y                     | 0.310 MM           | 0.095 MM                                            |                         |                      |                   |
| 27P4_OHM_SE                                                                                                                | *                 | Y                     | 0.250 MM           | 0.250 MM                                            | =STANDARD               | =STANDARD            | =STANDARD         |
| PHYSICAL_RULE_SET                                                                                                          | LAYER             | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH                                  | MAXIMUM NECK LENGTH     | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
| 70_OHM_DIFF                                                                                                                | *                 | N                     | =STANDARD          | =STANDARD                                           | =STANDARD               | =STANDARD            | =STANDARD         |
| 70_OHM_DIFF                                                                                                                | ISL3, ISL4        | Y                     | 0.160 MM           | 0.160 MM                                            |                         |                      |                   |
| 70_OHM_DIFF                                                                                                                | ISL9, ISL10       | Y                     | 0.160 MM           | 0.160 MM                                            |                         |                      |                   |
| 70_OHM_DIFF                                                                                                                | ISL2, ISL11       | Y                     | 0.170 MM           | 0.170 MM                                            |                         |                      |                   |
| 70_OHM_DIFF                                                                                                                | TOP, BOTTOM       | Y                     | 0.170 MM           | 0.095 MM                                            |                         | 0.150 MM             | 0.150 MM          |
| PHYSICAL_RULE_SET                                                                                                          | LAYER             | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH                                  | MAXIMUM NECK LENGTH     | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
| 80_OHM_DIFF                                                                                                                | *                 | N                     | =STANDARD          | =STANDARD                                           | =STANDARD               | =STANDARD            | =STANDARD         |
| 80_OHM_DIFF                                                                                                                | ISL3, ISL4        | Y                     | 0.125 MM           | 0.125 MM                                            |                         | 0.180 MM             | 0.180 MM          |
| 80_OHM_DIFF                                                                                                                | ISL9, ISL10       | Y                     | 0.125 MM           | 0.125 MM                                            |                         | 0.180 MM             | 0.180 MM          |
| 80_OHM_DIFF                                                                                                                | ISL2, ISL11       | Y                     | 0.140 MM           | 0.140 MM                                            |                         | 0.190 MM             | 0.190 MM          |
| 80_OHM_DIFF                                                                                                                | TOP, BOTTOM       | Y                     | 0.140 MM           | 0.095 MM                                            |                         | 0.190 MM             | 0.190 MM          |
| PHYSICAL_RULE_SET                                                                                                          | LAYER             | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH                                  | MAXIMUM NECK LENGTH     | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
| 90_OHM_DIFF                                                                                                                | *                 | N                     | =STANDARD          | =STANDARD                                           | =STANDARD               | =STANDARD            | =STANDARD         |
| 90_OHM_DIFF                                                                                                                | ISL3, ISL4        | Y                     | 0.102 MM           | 0.102 MM                                            |                         |                      |                   |
| 90_OHM_DIFF                                                                                                                | ISL9, ISL10       | Y                     | 0.102 MM           | 0.102 MM                                            |                         | 0.220 MM             | 0.220 MM          |
| 90_OHM_DIFF                                                                                                                | ISL2, ISL11       | Y                     | 0.115 MM           | 0.115 MM                                            |                         | 0.230 MM             | 0.230 MM          |
| 90_OHM_DIFF                                                                                                                | TOP, BOTTOM       | Y                     | 0.115 MM           | 0.095 MM                                            |                         | 0.230 MM             | 0.230 MM          |
| PHYSICAL_RULE_SET                                                                                                          | LAYER             | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH                                  | MAXIMUM NECK LENGTH     | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
| 100_OHM_DIFF                                                                                                               | *                 | N                     | =STANDARD          | =STANDARD                                           | =STANDARD               | =STANDARD            | =STANDARD         |
| 100_OHM_DIFF                                                                                                               | ISL3, ISL4        | Y                     | 0.080 MM           | 0.080 MM                                            |                         | 0.200 MM             | 0.200 MM          |
| 100_OHM_DIFF                                                                                                               | ISL9, ISL10       | Y                     | 0.080 MM           | 0.080 MM                                            |                         | 0.200 MM             | 0.200 MM          |
| 100_OHM_DIFF                                                                                                               | ISL2, ISL11       | Y                     | 0.080 MM           | 0.080 MM                                            |                         | 0.220 MM             | 0.220 MM          |
| 100_OHM_DIFF                                                                                                               | TOP, BOTTOM       | Y                     | 0.080 MM           | 0.080 MM                                            |                         | 0.220 MM             | 0.220 MM          |
| PHYSICAL_RULE_SET                                                                                                          | LAYER             | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH                                  | MAXIMUM NECK LENGTH     | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
| 110_OHM_DIFF                                                                                                               | *                 | N                     | =STANDARD          | =STANDARD                                           | =STANDARD               | =STANDARD            | =STANDARD         |
| 110_OHM_DIFF                                                                                                               | ISL3, ISL4        | Y                     | 0.077 MM           | 0.077 MM                                            |                         | 0.330 MM             | 0.330 MM          |
| 110_OHM_DIFF                                                                                                               | ISL9, ISL10       | Y                     | 0.077 MM           | 0.077 MM                                            |                         | 0.330 MM             | 0.330 MM          |
| 110_OHM_DIFF                                                                                                               | ISL2, ISL11       | Y                     | 0.077 MM           | 0.077 MM                                            |                         | 0.330 MM             | 0.330 MM          |
| 110_OHM_DIFF                                                                                                               | TOP, BOTTOM       | Y                     | 0.077 MM           | 0.077 MM                                            |                         | 0.330 MM             | 0.330 MM          |
| SPACING_RULE_SET                                                                                                           | LAYER             | LINE-TO-LINE SPACING  | WEIGHT             |                                                     |                         |                      |                   |
| DEFAULT                                                                                                                    | *                 | 0.1 MM                | ?                  |                                                     |                         |                      |                   |
| STANDARD                                                                                                                   | *                 | =DEFAULT              | ?                  |                                                     |                         |                      |                   |
| BGA_P1MM                                                                                                                   | *                 | =DEFAULT              | ?                  |                                                     |                         |                      |                   |
| BGA_P2MM                                                                                                                   | *                 | =DEFAULT              | ?                  |                                                     |                         |                      |                   |
| BGA_P3MM                                                                                                                   | *                 | =DEFAULT              | ?                  |                                                     |                         |                      |                   |
| SPACING_RULE_SET                                                                                                           | LAYER             | LINE-TO-LINE SPACING  | WEIGHT             |                                                     |                         |                      |                   |
| 1:0:1_SPACING                                                                                                              | *                 | 0.15 MM               | ?                  |                                                     |                         |                      |                   |
| 1:0:1_SPACING                                                                                                              | *                 |                       | ?                  |                                                     |                         |                      |                   |
| 2:1_SPACING                                                                                                                | *                 |                       |                    |                                                     |                         |                      |                   |
| 2:5:1_SPACING                                                                                                              | *                 |                       |                    |                                                     |                         |                      |                   |
| 3:1_SPACING                                                                                                                | *                 |                       |                    |                                                     |                         |                      |                   |
| 4:1_SPACING                                                                                                                | *                 | 0.4 MM                | ?                  |                                                     |                         |                      |                   |
| NET_SPACING_TYPE1                                                                                                          | NET_SPACING_TYPE2 | AREA_TYPE             | SPACING_RULE_SET   |                                                     |                         |                      |                   |
| *                                                                                                                          | *                 | BGA                   | BGA_P1MM           |                                                     |                         |                      |                   |
| NEP_CLK                                                                                                                    | *                 | BGA                   | BGA_P2MM           |                                                     |                         |                      |                   |
| CLK_FSB                                                                                                                    | *                 | BGA                   | BGA_P3MM           |                                                     |                         |                      |                   |
| CLK_PCIE                                                                                                                   | *                 | BGA                   | BGA_P2MM           |                                                     |                         |                      |                   |
| CLK_SLOW                                                                                                                   | *                 | BGA                   | BGA_P2MM           |                                                     |                         |                      |                   |
| FSB_DSTB                                                                                                                   | FSB_DSTB          | BGA                   | BGA_P3MM           | NOTE: From T18 MLB, changed to reflect M99 stackup. |                         |                      |                   |
|                                                                                                                            |                   |                       |                    |                                                     |                         |                      |                   |
| SPACING_RULE_SET                                                                                                           | LAYER             | LINE-TO-LINE SPACING  | WEIGHT             |                                                     |                         |                      |                   |
| 5X_DIELECTRIC                                                                                                              | *                 | 0.350 MM              | ?                  |                                                     |                         |                      |                   |
| PHYSICAL_RULE_SET                                                                                                          | LAYER             | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH                                  | MAXIMUM NECK LENGTH     | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
| 1:1_DIFFPAIR                                                                                                               | *                 | N                     | =STANDARD          | =STANDARD                                           | =STANDARD               | 0.150 MM             | 0.1 MM            |
| PCB Rule Definitions                                                                                                       |                   |                       |                    |                                                     |                         |                      |                   |
| SYNC_MASTER=M98_MLB                                                                                                        |                   |                       |                    | SYNC_DATE=04/01/2008                                |                         |                      |                   |
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| NONE                                                                                                                       |                   | 97                    |                    | 98                                                  |                         |                      |                   |



PROJECT SPECIFIC CONNS

SYNC\_MASTER=N/A

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